

Independent Surreal Copies and Arithmetic Intersections

Prescribed Hahn cores, algebraic independence,
two transcendental descent obstructions, composita,
and maximal transcendence of Hahn joins

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Built from manuscripts of batches 31, 34 and 35 of the collection, pinned to `bcac55a`, `7b256e0`, `f388f95`

Abstract

Let H be any set-sized divisible additive subgroup of the surreal numbers, and let $L = \mathcal{K}_{\mathbb{R}}(H)$ be the full Conway–Hahn field with exponents in H . We construct an ordinal-indexed proper class of strong, real-fixing, monomial-preserving self-embeddings of $\mathbf{No}$ whose images are jointly linearly disjoint over L and have pairwise intersection exactly L . All embeddings fix L pointwise and preserve and reflect the omnific integers. The same construction gives complex-fixing, conjugation-compatible copies of $\mathbf{No}(i)$ and its Gaussian omnific ring. Every set of parameters can be fixed by choosing a suitable H .

The proof combines a cut-realization lemma that avoids an arbitrary set of previously chosen values with a coset-slice criterion for linear disjointness of full Hahn fields. It yields exact Boolean meet diagrams and descent of all polynomial relations across their squares. Two consequences distinguish this construction from mere universality. First, the fraction field of the intersection of two full omnific copies is a proper subfield of their common field, with a transcendental gap for every set-sized H . For $H = \mathbb{Q}$ a witness is $1 + \sum_{n \geq 1} \omega^{-n!}$; real constants already belong to the smaller fraction field. Second, explicit independent copies with intersection $\mathbb{R}$ have a full Hahn join containing an explicitly described series transcendental over their ordinary compositum. Both obstructions persist on complexification.

Status and scope. The article provides complete mathematical arguments relative to standard Conway normal form and Hahn-field foundations. The proposed new contribution is the combined prescribed-intersection theorem and its arithmetic and compositum applications. The ingredients and the bounded-support transcendence mechanism are not claimed as new. This is an unrefereed, non-formalized draft, not a certified priority claim or a claimed solution of a named longstanding conjecture. [write] Independent proof review and formalization are pending. Labels carry the prefix `isc:`. Conventions, the place in the collection, the non-claims and the provenance are in [sections 1.5, 1.6 and 14](#) and [Appendix A](#); added text is marked [write]. [merge] Sections 15–27 and 28–41 add two more manuscripts (`isc:cp:`, `isc:hj:`).

Repository comparison is pinned to commit

`bcac55ae6fd3f2b354e568ba1d2e94d496a2cc59`.

Sources, limitations, and a concrete research program are included.

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1 Three questions and their answers

The existence of an embedding into the surreal numbers says little about how two embedded copies meet. An intersection question asks for much more: which elements must be shared, which algebraic relations can pass from one copy to another, and whether their distinguished integer rings reconstruct the same common field. This article treats these questions for *full proper-class copies* of the surreal field, rather than only for small subfields.

The ambient presentation matters. A surreal has a unique Conway normal form

$$x = \sum_{g \in S} r_g \omega^g,$$

where $S \subseteq \mathbf{No}$ is a reverse-well-ordered *set*. For an additive subgroup $G \subseteq \mathbf{No}$, put

$$\mathcal{K}_{\mathbb{R}}(G) = \{x \in \mathbf{No} : \text{supp}(x) \subseteq G\}, \quad \mathcal{K}_{\mathbb{C}}(G) = \mathcal{K}_{\mathbb{R}}(G)(i).$$

These are full Hahn fields: there is no additional bound on the cardinality or order type of a set-sized support. If G is divisible, the real field is real closed and the complex field is algebraically closed. These foundations are classical; see [1, 2, 3].

1.1 Prescribing the common field

Theorem A (Independent full copies over a prescribed Hahn core). *Let $H \subseteq \mathbf{No}$ be a set-sized divisible additive subgroup and write $L_k = \mathcal{K}_k(H)$ for $k = \mathbb{R}, \mathbb{C}$. There are class maps*

$$\Phi_{\alpha} : \mathbf{No} \longrightarrow \mathbf{No} \quad (\alpha \in \mathbf{On})$$

with the following properties.

(i) *Each Φ_{α} is a proper ordered field embedding, fixes $L_{\mathbb{R}}$ pointwise, sends Conway monomials to Conway monomials, and preserves and reflects Hahn summability of set-indexed families.*

(ii) *With $K_{\alpha} = \Phi_{\alpha}(\mathbf{No})$,*

$$K_{\alpha} \cap K_{\beta} = L_{\mathbb{R}} \quad (\alpha \neq \beta).$$

Every finite subfamily is jointly linearly disjoint over $L_{\mathbb{R}}$.

(iii) *For $R_{\alpha} = \Phi_{\alpha}(\mathbf{Oz})$,*

$$R_{\alpha} = K_{\alpha} \cap \mathbf{Oz}, \quad R_{\alpha} \cap R_{\beta} = L_{\mathbb{R}} \cap \mathbf{Oz}.$$

Moreover $\Phi_{\alpha}^{-1}(\mathbf{Oz}) = \mathbf{Oz}$.

(iv) *The coefficientwise complexifications have the same assertions over $L_{\mathbb{C}}$, commute with conjugation, and preserve and reflect $\mathbf{Oz}[i]$.*

The proof is given in [sections 3 to 6](#). “Jointly linearly disjoint” is stronger than pairwise intersection being small: finitely many independent choices from different copies satisfy no new polynomial relations over the common core. Proper-class terminology is made precise in [section 2](#); no tensor product indexed by a proper class is required.

A useful immediate specialization is $H = \{0\}$. It produces full copies of $\mathbf{No}$ intersecting exactly in $\mathbb{R}$, while their omnific rings intersect exactly in $\mathbb{Z}$. A second specialization, $H = \mathbb{Q}$, fixes not just ω and its rational powers but every Hahn series with rational exponents. In either case each copy is as large as the whole surreal field.

1.2 Arithmetic intersection is not field reconstruction

Theorem B (Transcendental failure of fraction-field descent). *Choose two copies in Theorem A. Then*

$$\text{Frac}(R_\alpha) = K_\alpha, \quad \text{Frac}(R_\alpha \cap R_\beta) \subsetneq K_\alpha \cap K_\beta = L_{\mathbb{R}}.$$

The latter extension is transcendental for every set-sized H , including $H = \{0\}$. For $H \neq 0$, the smaller field already contains every real constant. For $H = \mathbb{Q}$, the element

$$F = 1 + \sum_{n=1}^{\infty} \omega^{-n!}$$

is transcendental over $\text{Frac}(R_\alpha \cap R_\beta)$. The corresponding assertions hold for the Gaussian omnific copies inside $\mathbf{No}(i)$.

This is a failure of a particular descent operation, not a failure of $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$. Indeed that classical identity is indispensable to the argument. Each copy separately supplies denominators for every element of its field, but the *shared* denominators need not suffice for their shared field. The phenomenon remains after the missing-real-constants explanation has been removed. Sections 9 and 10 prove the result.

1.3 An ordinary compositum is smaller than a full Hahn join

For full Hahn fields in the same normal-form presentation, their *Hahn join* is the full Hahn field on the sum of their exponent groups. Their *ordinary compositum* is the field generated using finitely many field operations. These should not be conflated.

Theorem C (A transcendental gap above the compositum). *There are explicit strong embeddings Φ_0, Φ_1 as in Theorem A with $H = 0$ for which, writing*

$$K_j = \Phi_j(\mathbf{No}), \quad E = \mathcal{K}_{\mathbb{R}}(G_0 + G_1), \quad M = K_0 K_1,$$

the field E contains an explicit countably supported element S transcendental over M . In particular even the relative real algebraic closure of the ordinary compositum is a proper subfield of the Hahn join. The same S is transcendental over the compositum of the complexified copies inside $\mathcal{K}_{\mathbb{C}}(G_0 + G_1)$.

The two-level normal-form construction is explicit in section 7. The formula for S and the coefficient-field obstruction are proved in section 8. The latter argument is independent of the lacunary-series argument used for Theorem B.

1.4 What is new here, and what is not

The classical inputs are the set-cut property of $\mathbf{No}$, its normal-form presentation, Hahn multiplication and inversion, and the real/algebraic closedness criteria. The group extension argument is a concrete saturation construction, not a new saturation principle. The monomial lifting argument is standard Hahn functoriality. The coset-slice proof is presented as an elementary reusable lemma rather than as a certified new theorem of Hahn field theory.

The relevant repository comparison used the documentation catalogue and the reports on omnific-preserving automorphisms and bounded-support Hahn arithmetic at the pinned snapshot [6, 7, 8]. The former concerns stabilizers and rigidity; the latter already proposes much larger algebraically

independent families over bounded-support fraction fields. We neither repackage those results as discoveries nor use their unrefereed main theorems as black boxes. A self-contained one-witness gap argument is supplied here. In the published literature, the failure of fraction-field equality at set-sized cofinality already appears in L’Innocente–Mantova, Proposition 2.4.5 [3].

The proposed contribution is instead the *simultaneous arrangement*: full copies, a prescribed set-sized full Hahn intersection, strong arithmetic compatibility, independence of every finite subfamily, exact meet diagrams, and the two transcendental descent obstructions inside those arrangements. Targeted literature searches do not certify that this combined formulation is absent from all published or unpublished work. No claim of having audited every report or every later repository revision is made.

[write] This subsection is the source’s, written at its pin. Its description of the omnific-preserving report as concerning stabilizers and rigidity was accurate then: at the pin that report had no part on embeddings. Its Part III, written afterwards, classifies the strong exact-monomial embeddings that preserve $\mathbf{Oz}$, and the lifts used here are the real-fixing case of that classification. The bounded-support report was unchanged between the pin and batch 31; batch 32 added its Galois part (its Sections 11–19). The attribution to L’Innocente–Mantova was checked (Appendix A.3). The relation to the collection as it now stands, with exact labels, is in [section 1.6](#).

1.5 Notation and conventions in this report

[write] This subsection and the next were written when the manuscript joined the collection. No symbol of the source was renamed and no normalization was changed. The meanings below are fixed once for the whole report, and the collisions with other reports are disclosed here rather than removed. Paths such as `surreal/omnific-preserving-automorphisms` are relative to `docs/`, and labels of other reports are cited by name.

- *Two exponent conventions.* [Section 1](#) to [section 8](#), [section 11](#) and the beginning of [section 9](#) use Conway’s decreasing large- ω convention $x = \sum r_g \omega^g$. There $\mathbf{Oz}$ consists of the series supported in $\mathbf{No}^{>0} \cup \{0\}$ with integral constant coefficient: the purely infinite part sits at *positive* exponents. From [section 9.1](#) to the end of [section 10](#) the source switches to $t^h = \omega^{-h}$, the convention of the omnific-preserving report’s Part III (`opa:as:sub:notation`) and of the bounded-support report (`bst:sec:setup`); there the purely infinite part has *negative* t -exponents and v is the least t -exponent. The tempting false reading of $\mathcal{B}_k(H)$ is “support bounded above in the Conway exponents”. The true reading is “bounded above in the t -exponents”, that is, Conway exponents bounded *below*.
- *Fields.* $\mathcal{K}_k(G)$ is the *full* Hahn field on G , support-closed and not generated by monomials; for $k \neq \mathbb{R}, \mathbb{C}$ it is formal. $L_k = \mathcal{K}_k(H)$, with H a set-sized divisible subgroup of $\mathbf{No}$. This H need not be convex. It is not a convex subgroup H_g of the omnific-preserving report (`opa:def:convex`), and not the large-cardinal report’s embedding H_j .
- *Rings.* $\mathcal{A}_k(H) = D_k \oplus \Pi_k(H)$ with $D_{\mathbb{R}} = \mathbb{Z}$, $D_{\mathbb{C}} = \mathbb{Z}[i]$ restricts the constant coefficient to D_k . It is the ring $\mathcal{R}_{\Gamma}$ (or $\mathcal{R}_{\Gamma}^{\mathbb{C}}$) of the omnific-preserving report’s Part III and the ring $\mathcal{R}_{D_k}(k, \Gamma)$ of the Diophantine report’s Section 11. It is *not* that Part’s $\mathcal{A}_{\Gamma} = \mathbb{R} \oplus \Pi_{\Gamma}$. Nor is it the Diophantine report’s $\mathcal{A}_k(\Gamma)$, whose constant coefficient ranges over all of k . D_k is a ring of integers and never a derivation; the omnific-preserving report reserves d and D for derivations. $\mathcal{B}_k(H)$ is the bounded-support report’s $\mathcal{B}_H(k)$, with the two arguments in the opposite order, and $\text{Frac } \mathcal{B}_k(H)$ is its $\mathcal{F}_H(k)$.

- *Maps.* Φ_h of (4.1) is the monomial map $M_{1,h} = J_{h,\text{id}}$ of `opa:as:eq:classification`. It is not the Taylor automorphism Φ_d of the surcomplex-automorphisms report (the omnific-preserving report's $\mathcal{T}_{d,\delta}$), and not the large-cardinal report's sign-sequence embedding J . The letters h_α , G_α , Φ_α , K_α denote the family of [theorem 3.4](#) in [sections 3 to 6](#) and the explicit family in [sections 7 and 8](#). There h_α is the *inner* map (7.1), which is the inner lift ι_{σ_α} of `opa:as:eq:innerlift`. The maps h_α fix H as a set of exponents, and Φ_α fixes $\mathcal{K}_{\mathbb{R}}(H)$ ([remark 3.6](#)). “Monomial-preserving” is the omnific-preserving report's “exact-monomial”: a Conway monomial goes to a Conway monomial with coefficient one. It does not mean commuting with the omega-map. The order isomorphism ψ of [lemma 7.1](#) is not that report's μ_{12} (its source 12's ψ), nor μ_{11} or μ_{13} . The slice $\lambda_{g,H}$ of (5.1) is the bounded-support report's shifted coset projection π_g (`bst:eq:projection`); it is not the class functional $\lambda_{\mathbf{On}}$.
- *Local letters.* L, R are the sides of a cut in (2.1) and [lemmas 3.1 and 3.2](#); L is also a generic base field in the definition of linear disjointness and in [lemma 5.3](#). $M = K_0 K_1$ is an ordinary compositum, not a monomial map $M_{\chi,\tau}$. E is a span in the proof of [theorem 3.4](#), the field E_J of (6.1), and the Hahn join of [Theorem C](#). F is the witness of [Theorem B](#) and (10.2), and a polynomial in [corollary 5.6](#). It is not the image field F of `opa:as:prop:overlap`, and not the fraction field $\mathcal{F}_G(K)$. A is a set of parameters in [corollary 6.1](#), a matrix in [lemma 5.3](#) and a subgroup in [lemma 8.1](#), where B is a subgroup too; the calligraphic $\mathcal{A}, \mathcal{B}$ are the rings. D is a domain or subspace in [section 3](#) and a coefficient field in [section 8](#). S is a support set and the witness (8.3). I is an index set and I_α an interval.
- *Cardinals.* In [section 10](#), $\kappa = \text{cf}(H)$. In [section 11](#), κ is an arbitrary uncountable regular cardinal. Neither is the critical point κ of the large-cardinal report.
- *Strong* means preserving Hahn summability of set-indexed families; it is not convergence of partial sums.

1.6 Place in the collection

[write] The source compared itself with the catalogue, the omnific-preserving report and the bounded-support report at its pin ([Appendix A.2](#)). Checked against the collection at the placement commit, the relations are these.

Omnific-preserving automorphisms (`surreal/omnific-preserving-automorphisms`, prefix `opa:`). Its Part III (sub-prefix `opa:as:`) was written after the pin, so the source did not see it. Its bottom-gap classification `opa:as:thm:classification` lists the strongly additive field embeddings J with $J(t^\gamma) = t^{\tau(\gamma)}$ and $J(\mathcal{R}_\Gamma) \subseteq \mathcal{R}_\Delta$ as the maps $J_{\tau,\rho}$. Here ρ ranges over the ordered embeddings of $\mathbb{R}$ into the Hahn field of the bottom gap of τ .

1. *Lifts.* With $\Gamma = \Delta = \mathbf{No}$, $t^\gamma = \omega^{-\gamma}$ and $\tau = h$, the lift Φ_h of [theorem 4.1](#) is $J_{h,\text{id}}$, the case $\rho = \text{id}$. The same exponent map serves in both conventions because h is additive. The preservation and reflection of $\mathbf{Oz}$ in [proposition 4.2](#) is Step 5 of that proof and the display `opa:as:eq:classprops`. [Proposition 4.4](#) is the uniqueness clause of Step 2, for real-fixing maps; here the order of h is derived from real fixing. The floor, standard-part and modulus clauses of [proposition 4.2](#) are not stated there.
2. *Real fixing is a choice.* The explicit ranges G_α of [section 7](#) have a nonzero bottom gap: $\omega^{3\alpha}$ lies below every positive element of G_α , and for $\alpha = 0$ the gap is the group of finite surreals. By `opa:as:thm:taylorcriterion`, the composite of a Taylor automorphism $\mathcal{T}_{d,\delta}$ with Φ_α , for

$0 < \delta$ in that gap, therefore also preserves and reflects **Oz** and moves a transcendental real. The source takes $\rho = \text{id}$ throughout, which is consistent. For the family of [theorem 3.4](#) the construction does not determine the gap. [Theorem 6.2](#) depends on the real-fixing hypothesis: in the larger exact-monomial class, images meet $\mathbb{R}$ in the fixed field of ρ , for example $\ker d$ (`opa:as:prop:overlap`). No classification of intersections in that class is claimed.

3. *Inner and outer levels.* The inner map (7.1) is the inner lift ι_{σ_α} of `opa:as:eq:innerlift` and `opa:as:lem:inner`, and $\Phi_\alpha = M_{1, \iota_{\sigma_\alpha}}$. The order isomorphisms of `opa:as:eq:mus` have unbounded images; the explicit embeddings `opa:as:thm:explicit` combine them with Taylor sections in order to move a real. The maps σ_α here have bounded, pairwise disjoint images, which is what gives independence. The failure of omega-map commutation, $\Phi_0(\omega^1) \neq \omega^{\Phi_0(1)}$, is the same observation as the one recorded after `opa:as:cor:homogeneity`.
4. *Parameters.* [Corollary 6.1](#) extends `opa:as:thm:parameters`, which gives two proper strong exact-monomial copies fixing $A \cup \mathbb{R}$ and preserving and reflecting **Oz**. Here there is an **On**-indexed family whose pairwise intersection is exactly the full Hahn field $\mathcal{K}_{\mathbb{R}}(H) \supseteq A \cup \mathbb{R}$, jointly linearly disjoint over it. The mechanisms differ. That theorem fixes the inner supports through an order embedding (`opa:as:lem:parameters`); here the outer exponent group H is fixed by h_α . The topological assertions of that theorem (closed, discrete or nondiscrete images) are not claimed here, and that report does not study intersections.
5. *Complex version.* The complex clauses of [Theorem A](#) and [theorem 7.2](#) are the case $\varepsilon = 1$, $\rho = \text{id}$ of `opa:as:thm:complexembed`, parallel to `opa:as:cor:complexcopies`.
6. *Elementarity.* The ordered-field statement of [section 6.2](#) is the first assertion of `opa:as:thm:elementary`. That theorem's nonelementarity criterion needs a moved real, and the maps here fix $\mathbb{R}$.
7. *Questions.* None is answered. Research question 2 is the pair version, at a fixed Hahn core, of `opa:as:q:conjugacy`. Research question 9 asks `opa:as:q:elementary` for the maps here, with a prescribed intersection added. Research questions 6 and 7 lie inside `opa:as:q:exp`. [Theorem 6.3](#) decides image inclusion and intersection inside one independent family with $\rho = \text{id}$. That touches `opa:as:q:composition`, which asks for image inclusion of classified maps in general, but does not answer it. Research question 1 is related to `opa:as:q:units`, which asks to remove exact monomial preservation, but it asks about intersections and is new.

Bounded-support transcendence (`surreal/transcendence-over-bounded-support`, prefix `bst:`). The source cites it and says that [section 10](#) duplicates its one-witness case. That report was unchanged between the pin and batch 31; batch 32 added its Galois part (its Sections 11–19), which is not used here.

1. [Proposition 9.3](#) is the localization of `bst:prop:localdensity`, with the constant coefficient restricted to D_k . Since batch 32 that report also prints the general form, for every field K and every unital subring $D \subseteq K$, as `bst:gr:prop:fractions`.
2. [Theorem 10.2](#) is `bst:cor:onegap` when $\text{cf}(H) = \aleph_0$; that corollary's hypothesis forces a cofinal support of countable cofinality. In characteristic zero and at every cofinality it is the one-series case of `bst:thm:mainintro` (`bst:thm:independent`), which gives 2^κ independent series on one support of order type κ and proves that κ is optimal (`bst:cor:optimal`). The single-series form at uncountable cofinality over an arbitrary coefficient field, finite fields included, is not stated there; that report says that finite coefficient fields are not covered by its independence theorem.

Its proof is that report’s tail estimate and root count. [Lemma 10.1](#) is the second inequality of a separated exponent system (`bst:lem:skeleton`).

3. The factorial witness of [corollary 10.3](#) is `bst:cor:onegap` with $a_n = n!$ and all coefficients one; the constant 1 changes nothing. `bst:thm:actualreal` uses the same exponents, with integer codes, for a continuum of independent actual infinitesimals over the complex bounded-support fraction field. In the omnific-preserving report, `opa:as:ex:smallfrac` uses $\sum t^{n!}$ at $\Gamma = \mathbb{Z}$ to show that $\sum t^{n!}$ is not in $\text{Frac } \mathcal{R}_{\mathbb{Z}} = \mathbb{R}(t^{-1})$: non-membership, not transcendence.
4. Research question 8 is that report’s programme (`bst:cor:exactcard`, `bst:cor:realgroup`), as the source says. Only its uniformity over realized diagrams is new.
5. The coset slice $\lambda_{g,H}$ is that report’s projection π_g , and the linearity in [lemma 5.2](#) is `bst:lem:projection`. There it serves descent over a cofinal subgroup (`bst:thm:descent`). [Theorem 5.4](#) is not stated there: two full Hahn fields are linearly disjoint over the full Hahn field of the intersection of their groups, with no cofinality hypothesis. The source presents it as an elementary reusable lemma, not as a certified new theorem.

Omnific Diophantine geometry (`surreal/omnific-diophantine-geometry`, prefix `odg`). [Lemma 9.1](#) is the global monomial clearing `odg:thm:fractions`, with the same proof.

Large-cardinal embeddings (`foundations-and-computation/large-cardinal-embedding-s-and-normal-forms`, title “Critical-Point Defects in Surreal Arithmetic”). It was placed after the pin, so the source did not see it. Its write phase is in progress, so it is cited here by name and question title, not by label. An elementary embedding with critical point κ gives there two ordered-field embeddings J and H_j of **No**. The strong one, H_j , is the lift Φ_h of [theorem 4.1](#) with h the restriction of J to $(\mathbf{No}, +, <)$, since $H_j(\omega^a) = \omega^{J(a)}$. The map J is not strongly additive, and $J(\mathbf{No})$ is not a full Hahn field on a subgroup. The common field there (surreals with fewer than κ terms and exponents in $J(\mathbf{No})$) is a proper class, not a full Hahn field. That report’s question “Joint algebraic dependence of the images” asks whether $J(\mathbf{No})$ and $H_j(\mathbf{No})$ are linearly disjoint over that common field. It is related to [theorem 5.4](#) but *not answered* by it: the slice theorem needs full Hahn fields on subgroups on both sides.

Formalization. No `isc` label has a Lean mapping in `FORMALIZATION.md`. No theorem of this report is Lean-verified, and the identifiers proposed in [section 13.2](#) do not exist in the collection’s Lean library.

Status of the research questions. None of the collection’s named questions is answered. Research questions 1, 3, 4, 5 and 10 are new. Research question 2 is a pair version of `opa:as:q:conjugacy`, 6 and 7 lie inside `opa:as:q:exp`, 8 is the bounded-support programme, and 9 is an instance of `opa:as:q:elementary`. Research question 10 overlaps `opa:as:q:formal` in its Hahn-field kernel. [merge] *Status (batch 34)*. Research question 3 is now answered in the separated case with trivial common core, the case of [Theorem C](#), and in the natural Laurent-series cardinal cases, by the second manuscript printed in Sections 15–27 ([section 15.4](#)); the non-separated case and prescribed nonzero common cores stay open. Research questions 11–22 come from that manuscript and are new. The relations of the addition to the three-duals, tail-spans, bounded-support and dilation reports are in [section 15.6](#). [merge] *Status (batch 35)*. A third manuscript, printed in Sections 28–41, answers the transcendence-degree part of Research questions 3 and 11 for explicit

non-separated examples (interleaved groups at every infinite cardinal, dense Archimedean groups, and a class example); the description of the compositum in the non-separated case, and prescribed nonzero cores, stay open (section 28.4). Research questions 23–32 come from that manuscript and are new. Its relations to the tail-spans, Diophantine, set-sized-quotient, bounded-support and omnific-preserving reports are in section 28.6.

2 Foundations, support, and size

2.1 A class framework with set-sized constructions

We work in von Neumann–Bernays–Gödel class theory with Global Choice, with sets satisfying ZFC, as in the foundational convention of [2]. Classes are not members of other classes. Thus an ordinal-indexed family of class functions means one class relation whose sections are the functions. A field with carrier **No** is a *class field*; its elements, finite tuples, polynomial coefficient lists, and the supports of individual elements are sets.

The one surreal-specific order principle needed for the construction is:

$$L, R \subseteq \mathbf{No} \text{ sets, } L < R \implies \text{there is } y \in \mathbf{No} \text{ with } L < y < R. \quad (2.1)$$

Empty sides are permitted. The simplest such y , denoted $\{L \mid R\}$, is uniquely determined. We never apply (2.1) to proper-class sides. This restriction is central, not cosmetic.

Global Choice supplies a set-like class well-order of the surreals: order first by birthday and well-order each set-sized birthday level. It similarly supplies a set-like enumeration of $\mathbf{On} \times \mathbf{No}$. For the latter, group pairs by a bound on both their ordinal coordinate and the birthday of the surreal coordinate, then well-order each new set layer. Each initial segment is a set, and the resulting class order has order type **On**.

The recursion below has a *set as its entire state at every ordinal stage*. For any fixed ordinal, ordinary set transfinite recursion with the fixed class parameters constructs that initial segment. Uniqueness makes these initial segments coherent, and elementary class comprehension collects their stages. This is not an appeal to recursion with a proper-class-valued state at each stage, nor to an unmentioned class truth predicate.

2.2 Full Hahn fields in decreasing-exponent convention

If G is an ordered additive subgroup of **No** and k is a set-sized field, let $\mathcal{K}_k(G)$ denote the formal series

$$\sum_{g \in S} a_g \omega^g, \quad a_g \in k,$$

with reverse-well-ordered set support $S \subseteq G$. For $k = \mathbb{R}$ these are actual surreals. For $k = \mathbb{C}$, taking real and imaginary coefficients identifies them with actual elements of $\mathbf{No}(i)$. The union of the two real supports is again reverse well ordered, so this complexification is exact. For other k the notation is purely formal.

A nonzero series has a largest exponent, and its leading coefficient is nonzero. Addition is coefficientwise and multiplication is convolution. The Hahn support lemma guarantees that the support of a product is allowable and that each product coefficient is a finite sum. Normalizing a nonzero series as $a\omega^g(1 + \varepsilon)$ with all exponents of ε negative gives its inverse by a summable

geometric family. Consequently $\mathcal{K}_k(G)$ is a field. These statements are understood locally in set-sized subgroups when G is a proper class.

The standard closedness theorem says that a Hahn field of characteristic zero with divisible exponent group is real closed when its coefficients are real closed, and algebraically closed when its coefficients are algebraically closed [3]. The same conclusion holds for our class fields: the supports of finitely many coefficients of a polynomial generate a set-sized divisible subgroup, and the standard set-sized theorem provides the required roots there. In particular,

$$\mathcal{K}_{\mathbb{R}}(G) \text{ is real closed,} \quad \mathcal{K}_{\mathbb{C}}(G) \text{ is algebraically closed} \quad \text{if } G \text{ is divisible.}$$

If H is a set, then $\mathcal{K}_k(H)$ is a set: it is a subclass of the set of all functions $H \rightarrow k$. In contrast, if G is class-isomorphic to $(\mathbf{No}, +, <)$, then $\mathcal{K}_k(G)$ is a proper class, because it contains the proper class of monomials ω^g .

2.3 Strong summation and finite algebraic independence

Definition 2.1 (Hahn summability). A set-indexed family $(x_j)_{j \in J}$ of Hahn series is summable if the union of its supports is reverse well ordered and every exponent belongs to the support of only finitely many x_j . Its sum is the resulting coefficientwise finite sum at every exponent. A map is *strongly additive* if it sends such families to summable families and commutes with their sums.

The sum of a normal form is strong in this sense. Ordinary analytic convergence of a sequence of real constants is a different notion; it is not used in any proof. In particular strong additivity cannot be replaced by an unqualified assertion of continuity.

Definition 2.2 (Linear disjointness for class fields). Subfields P, Q of one class field are linearly disjoint over $L \subseteq P \cap Q$ if every finite L -linearly independent tuple from P remains linearly independent over Q . A finite family is jointly linearly disjoint if its multiplication map on finite tensors over L is injective. Equivalently, this injectivity holds after restricting each factor to any set-sized subfield containing the finitely many elements under consideration and the base when the base is a set.

All tensor arguments below can instead be read as finite matrix arguments. For a class-sized base, linear independence still quantifies only over a finite tuple of coefficients at a time. “Algebraically independent” also has its usual finite-polynomial meaning. We make no assertion about a proper-class-dimensional tensor object in the category of sets.

3 Independent ordered-group embeddings

The core construction is carried out on the *additive exponent group*, not initially on the field. Its flexibility comes from the ability to realize a set-sized cut while avoiding all previously used values.

Lemma 3.1 (A cut avoiding a set). *Let $L, R, E \subseteq \mathbf{No}$ be sets and suppose $L < R$. There is a surreal $y \notin E$ with $L < y < R$.*

Proof. Let $T = \{e \in E : L < e < R\}$. This is a set and $L \cup T < R$. Choose

$$y = \{L \cup T \mid R\}.$$

Then $L < y < R$. Were $y \in E$, it would belong to T , contradicting $T < y$. This also covers empty L , R , or T . $\square$

Lemma 3.2 (Ordered one-point extension with avoidance). *Let D be a set-sized $\mathbb{Q}$ -vector subspace of $\mathbf{No}$, let $f : D \rightarrow \mathbf{No}$ be an ordered additive embedding, and let E be a set-sized $\mathbb{Q}$ -vector subspace containing $f(D)$. For $x \notin D$, there is an ordered additive embedding*

$$f^+ : D + \mathbb{Q}x \longrightarrow \mathbf{No}$$

extending f and such that $f^+(x) \notin E$.

Proof. Set

$$L = \{f(d) : d \in D, d < x\}, \quad R = \{f(d) : d \in D, x < d\}.$$

These are sets with $L < R$. By lemma 3.1, choose $y \notin E$ realizing this cut. Define

$$f^+(d + qx) = f(d) + qy \quad (d \in D, q \in \mathbb{Q}).$$

The representation $d + qx$ is unique because $x \notin D$, so the formula is well defined and additive. If $q > 0$, positivity of $d + qx$ is equivalent to $x > -d/q$; by the defining cut this is equivalent to $y > f(-d/q)$, hence to $f(d) + qy > 0$. For $q < 0$, the same argument reverses the comparison. The case $q = 0$ follows from the order preservation of f . Thus f^+ preserves strict positivity and is an ordered embedding. Its new value is $y \notin E$. $\square$

Definition 3.3 (Independence over an additive subgroup). Subgroups $(G_i)_{i \in I}$ containing H are *independent over H* if, whenever $J \subseteq I$ is finite and $g_i \in G_i$,

$$\sum_{i \in J} g_i \in H \implies g_i \in H \text{ for every } i \in J.$$

Equivalently the images G_i/H have internal direct sum in the quotient additive group. The quotient here is algebraic; H need not be convex.

Theorem 3.4 (Independent group copies over any set-sized base). *Let $H \subseteq \mathbf{No}$ be a set-sized divisible additive subgroup. There is an ordinal-indexed class family of ordered additive embeddings*

$$h_\alpha : \mathbf{No} \longrightarrow \mathbf{No}$$

such that $h_\alpha|_H = \text{id}_H$ and the ranges $G_\alpha = h_\alpha(\mathbf{No})$ are independent over H .

Proof. A divisible subgroup of the torsion-free additive group of $\mathbf{No}$ is a $\mathbb{Q}$ -vector subspace. Fix the set-like enumeration of $\mathbf{On} \times \mathbf{No}$ described in section 2. We process each requirement (α, x) in that order, meaning that x must eventually be in the domain of the α th partial embedding.

At a stage the state consists of partial ordered $\mathbb{Q}$ -linear embeddings $f_\alpha : D_\alpha \rightarrow \mathbf{No}$ for a set of active indices. Their domains and graphs are sets, they contain the identity on H , and their ranges are independent over H . An inactive index has the implicit partial map id_H ; these implicit copies of H need not be stored separately.

Suppose the next requirement is (α, x) . Activate α if needed. If $x \in D_\alpha$, do nothing. Otherwise let

$$E = \text{span}_{\mathbb{Q}} \left(H \cup \bigcup_{\beta \text{ active}} f_\beta(D_\beta) \right).$$

The state is a set, so E is a set-sized vector space. Apply lemma 3.2 to extend f_α to $D_\alpha + \mathbb{Q}x$, choosing its new value y outside E .

We verify that independence survives. A new finite relation modulo H can involve at most one changed component. Write that component as $f_\alpha(d) + qy$. If $q \neq 0$, the relation would express y as a rational linear combination of old range elements and elements of H , contradicting $y \notin E$. Therefore $q = 0$ and the old independence assertion applies. This proves the invariant at successor stages.

At a limit ordinal, take unions of the compatible graphs. There have been only a set of preceding stages, so the resulting state is still a set. Every finite relation already occurred at an earlier stage, so independence is retained. The set-valued recursion is legitimate in the stated class framework.

Finally define h_α to be the union of its partial graphs over all stages. Each requirement was processed, hence its domain is all of $\mathbf{No}$. Every ordered-additive identity is witnessed at some stage, so h_α is an ordered additive embedding fixing H . Any finite relation among final range elements is also witnessed at one stage. Consequently the final ranges are independent over H . $\square$

Corollary 3.5. *For distinct α, β , one has $G_\alpha \cap G_\beta = H$. More generally, for $\alpha \notin J$,*

$$G_\alpha \cap \left(H + \sum_{\beta \in J} G_\beta \right) = H,$$

where the sum means finite sums, even when J is infinite. Every G_α is a proper subgroup of $\mathbf{No}$.

Proof. The first two assertions are exactly the independence condition. To prove properness, choose $\beta \neq \alpha$ and $x \notin H$. Injectivity and the identity on H imply $h_\beta(x) \notin H$. Thus it cannot lie in G_α , although it lies in $\mathbf{No}$. $\square$

Remark 3.6 (The role of the base). The maps h_α fix H as a set of *exponent values*. The field maps constructed next fix $\mathcal{K}_{\mathbb{R}}(H)$. These are different assertions: for a general subgroup H , it need not be true that $H \subseteq \mathcal{K}_{\mathbb{R}}(H)$. No identification of those two roles is used.

4 Strong field lifts and omnific compatibility

Theorem 4.1 (Monomial lift). *Let $h : \mathbf{No} \rightarrow \mathbf{No}$ be an ordered additive embedding. Then*

$$\Phi_h \left(\sum_g r_g \omega^g \right) = \sum_g r_g \omega^{h(g)} \quad (4.1)$$

defines a real-fixing ordered field embedding of $\mathbf{No}$ with image exactly $\mathcal{K}_{\mathbb{R}}(h(\mathbf{No}))$. It preserves and reflects Hahn summability, and its complexification has image $\mathcal{K}_{\mathbb{C}}(h(\mathbf{No}))$.

Proof. An order embedding sends a reverse-well-ordered set to a reverse-well-ordered set, so the formula defines a surreal. It preserves the leading coefficient and therefore the sign of a nonzero real-coefficient series. Coefficientwise addition is preserved. For multiplication, the identity $h(g + g') = h(g) + h(g')$ identifies each convolution coefficient on the two sides; the support lemma ensures that only finitely many terms contribute to each such coefficient. Since $h(0) = 0$, real constants are fixed. The map is injective by uniqueness of normal form, hence is a field embedding.

For a set-indexed family, the union of image supports is the h -image of the union of original supports. Reverse well-ordering and pointwise finiteness are equivalent under this injection. Thus summability is both preserved and reflected, and the sums agree coefficientwise.

Finally, a series supported in $h(\mathbf{No})$ can be pulled back term by term along the inverse of h on its image. Its support remains a set and is reverse well ordered. This proves that the image is the *full* Hahn field on $h(\mathbf{No})$, not merely the field generated by its monomials. Taking real and imaginary coefficients gives the complex assertion. $\square$

Proposition 4.2 (What the lift remembers). *The map Φ_h preserves the three normal-form projections to positive, zero, and negative exponents. It preserves and reflects membership in $\mathbf{Oz}$, and*

$$\Phi_h(\mathbf{Oz}) = \Phi_h(\mathbf{No}) \cap \mathbf{Oz}. \quad (4.2)$$

It commutes with the omnific floor map and with standard part on finite surreals. Its complexification commutes with conjugation, preserves and reflects $\mathbf{Oz}[i]$, and preserves the real-valued modulus in the sense

$$|\Phi_h^{\mathbb{C}}(z)| = \Phi_h(|z|).$$

Proof. Because h is ordered and additive, $h(g) > 0$, $h(g) = 0$, and $h(g) < 0$ are equivalent to the respective conditions on g . The coefficient at zero is unchanged. An omnific integer is precisely a series supported in $\mathbf{No}^{>0} \cup \{0\}$ with integral constant coefficient, so membership is preserved and reflected. The image formula follows by pulling back a series in the intersection using [theorem 4.1](#).

If $a = \lfloor x \rfloor_{\mathbf{Oz}}$, then $a \in \mathbf{Oz}$ and $a \leq x < a + 1$. Applying Φ_h gives an omnific integer in the corresponding unit interval for $\Phi_h(x)$, hence the unique omnific floor. If $x = r + \varepsilon$ is finite with $r \in \mathbb{R}$ and ε infinitesimal, the image has the same decomposition with infinitesimal part $\Phi_h(\varepsilon)$. This proves the standard-part assertion.

Complex conjugation is coefficientwise. Gaussian omnific membership is omnific membership of both real and imaginary parts. Finally $|z|$ is the unique nonnegative square root of $z\bar{z}$. The image of this equation, order preservation on the real part, and uniqueness of that square root prove the modulus identity. $\square$

Remark 4.3 (A floor boundary that must not be dropped). The positive-exponent truncation plus the ordinary floor of the constant coefficient is not always the omnific floor. For example, if $0 < \varepsilon$ is infinitesimal and $n \in \mathbb{Z}$, then $\lfloor n - \varepsilon \rfloor_{\mathbf{Oz}} = n - 1$. The order-and-uniqueness proof above retains this endpoint case automatically.

Proposition 4.4 (Classification within the monomial-preserving class). *Every strongly additive, real-fixing field embedding $\Phi : \mathbf{No} \rightarrow \mathbf{No}$ that maps each Conway monomial to a Conway monomial is uniquely Φ_h for an ordered additive embedding $h : \mathbf{No} \rightarrow \mathbf{No}$.*

Proof. Define $h(g)$ uniquely by $\Phi(\omega^g) = \omega^{h(g)}$. Multiplication of monomials proves additivity, and injectivity proves injectivity of h . A field embedding between real closed ordered fields preserves positivity: a positive element is a nonzero square. Hence it preserves order. For $g > 0$, the monomial ω^g exceeds every positive real number. Its image has the same property because the reals are fixed. A Conway monomial has this property exactly when its exponent is positive, so $h(g) > 0$. Thus h is ordered. Applying strong additivity to each normal form gives (4.1). Uniqueness is immediate from the monomials. $\square$

[write] [Theorem 4.1](#) and [propositions 4.2](#) and [4.4](#) are the case $\rho = \text{id}$ of the bottom-gap classification `opa:as:thm:classification` of the omnific-preserving report, written after the pin. Without the real-fixing hypothesis the exact-monomial embeddings preserving $\mathbf{Oz}$ are the maps $J_{h,\rho}$, which may move reals ([section 1.6](#)).

Warning 4.5 (Preserving monomials is not preserving the omega-map). The equality $\Phi_h(\omega^g) = \omega^{h(g)}$ does not say $\Phi_h(\omega^g) = \omega^{\Phi_h(g)}$. In general $h \neq \Phi_h$ as maps on surreal numbers. Nor have we shown compatibility with the Gonshor exponential, the Berarducci–Mantova derivation, birthdays, or simplicity. These extra structures are outside the assertions of this article.

5 Coset slices and linear disjointness

The next argument upgrades an exponent-group intersection calculation to linear disjointness of *full* series fields. The subgroups need not be convex, and no quotient ordering is needed.

Definition 5.1 (Normalized coset slice). Let $H \subseteq G$ be additive subgroups and $g \in G$. For $f = \sum_e a_e \omega^e \in \mathcal{K}_k(G)$ set

$$\lambda_{g,H}(f) = \omega^{-g} \sum_{e \in (g+H) \cap \text{supp}(f)} a_e \omega^e \in \mathcal{K}_k(H). \quad (5.1)$$

The value depends on the chosen representative g by the expected monomial normalization. We do not choose a global transversal of G/H .

Lemma 5.2 (Slice linearity and separation). *Every $\lambda_{g,H}$ is $\mathcal{K}_k(H)$ -linear, and the family of slices separates points of $\mathcal{K}_k(G)$.*

Proof. A subset of a reverse-well-ordered support is reverse well ordered, and translation preserves that property. Thus the definition is valid even when H is not convex. If $a \in \mathcal{K}_k(H)$, a product term with exponent $e + h$ belongs to $g + H$ exactly when e belongs to $g + H$. Therefore slicing the product af gives a times the slice of f . The equality is verified coefficientwise in finite convolution sums. Additivity follows directly. If $f \neq 0$, choose any g in its support. Its slice contains the nonzero coefficient a_g at exponent zero, so the slice is nonzero. $\square$

Lemma 5.3 (A finite separating matrix). *If $f_1, \dots, f_m \in \mathcal{K}_k(G)$ are linearly independent over $L = \mathcal{K}_k(H)$, there are $g_1, \dots, g_m \in G$ such that*

$$A = (\lambda_{g_a, H}(f_b))_{1 \leq a, b \leq m} \text{ is invertible over } L.$$

Proof. On the m -dimensional space $V = \text{span}_L(f_1, \dots, f_m)$, the restrictions of the slices have common kernel zero by [lemma 5.2](#). Start with V . If the current common kernel is nonzero, choose a nonzero vector in it and a slice not vanishing on that vector. Intersecting with this new kernel strictly lowers dimension. After at most m steps the common kernel is zero. The resulting linear map from V to at most m copies of L is injective, so exactly m independent coordinate functionals can be used, and their matrix in the stated basis is invertible. This is a finite argument, also for class-sized L . $\square$

Theorem 5.4 (Full Hahn fields are disjoint over their full intersection). *Let G_1, G_2 be additive subgroups of one ordered abelian group and put $H = G_1 \cap G_2$. Inside $\mathcal{K}_k(G_1 + G_2)$,*

$$\mathcal{K}_k(G_1) \cap \mathcal{K}_k(G_2) = \mathcal{K}_k(H),$$

and the two fields are linearly disjoint over $\mathcal{K}_k(H)$. The assertion applies to set or class subgroups, with set-sized individual supports, and to any coefficient field k .

Proof. The intersection identity follows from uniqueness of coefficients and support. Put $L = \mathcal{K}_k(H)$ and let $f_1, \dots, f_m \in \mathcal{K}_k(G_1)$ be L -linearly independent. Choose the invertible slice matrix of lemma 5.3 with representatives $g_a \in G_1$.

On the larger field $\mathcal{K}_k(G_1 + G_2)$ take slices λ_{g_a, G_2} with values in $\mathcal{K}_k(G_2)$. They are $\mathcal{K}_k(G_2)$ -linear. For $f \in \mathcal{K}_k(G_1)$, the identity

$$G_1 \cap (g_a + G_2) = g_a + (G_1 \cap G_2) = g_a + H$$

shows that $\lambda_{g_a, G_2}(f) = \lambda_{g_a, H}(f)$. Suppose $\sum_b b_b f_b = 0$ with $b_b \in \mathcal{K}_k(G_2)$. Applying all of these enlarged slices gives $A(b_1, \dots, b_m)^\top = 0$. Since A is invertible already over L , all b_b vanish. This is precisely linear disjointness. $\square$

Corollary 5.5 (Independence of any finite number of full Hahn fields). *If $(G_i)_{i \in I}$ are independent over H , then the full fields $\mathcal{K}_k(G_i)$ are jointly linearly disjoint over $\mathcal{K}_k(H)$. For $i \notin J$, the field $\mathcal{K}_k(G_i)$ is linearly disjoint over $\mathcal{K}_k(H)$ even from the full field $\mathcal{K}_k(H + \sum_{j \in J} G_j)$.*

Proof. The second assertion follows from theorem 5.4 and the corresponding group intersection. For a finite family, induct on its size. The compositum of the preceding fields is contained in the full Hahn field on the sum of their groups, and the next field is linearly disjoint from that larger field. Extension of scalars over a field preserves an injection of vector spaces. Applying this to the already injective finite-tensor multiplication map proves injectivity for the enlarged family. Equivalently, choose a basis of the finite span of the preceding products and apply linear disjointness to that basis. $\square$

Corollary 5.6 (Polynomial-relation descent). *Let $P = \mathcal{K}_k(G_1)$, $Q = \mathcal{K}_k(G_2)$, and $L = \mathcal{K}_k(G_1 \cap G_2)$. For a finite tuple a from P , let $I_L(a) \subseteq L[X_1, \dots, X_n]$ be its polynomial relation ideal. Then*

$$I_Q(a) = I_L(a) Q[X_1, \dots, X_n].$$

Proof. Take a polynomial relation $F(a) = 0$ with coefficients in Q . A finite L -basis $b_1, \dots, b_r$ for the span of its coefficients gives $F = \sum_j b_j F_j$ with $F_j \in L[X_1, \dots, X_n]$. Linear disjointness is symmetric (as is the finite-tensor criterion), so the b_j remain independent over P . It follows that every $F_j(a) = 0$. Thus F belongs to the extended ideal. The other inclusion is immediate. $\square$

6 Prescribed intersections and Boolean meet diagrams

Proof of Theorem A. Apply theorem 3.4 to obtain (h_α) and their independent ranges G_α , and set $\Phi_\alpha = \Phi_{h_\alpha}$. By theorem 4.1, the image is $K_\alpha = \mathcal{K}_\mathbb{R}(G_\alpha)$. Since h_α is the identity on H , it fixes every monomial with exponent in H , and strong additivity makes Φ_α the identity on all of $L_\mathbb{R}$. Each G_α is proper by corollary 3.5; a monomial with exponent outside it witnesses properness of the corresponding field.

The intersection and independence assertions follow from theorem 5.4 and corollary 5.5. Proposition 4.2 gives the omnific assertions. The same groups and the same slice argument with coefficients in $\mathbb{C}$ give the entire complex package. $\square$

Corollary 6.1 (Fixing every element of a prescribed set). *For every set $A \subseteq \mathbf{No}$ there is a set-sized real closed full Hahn field $L \supseteq \mathbb{R} \cup A$ which is the exact pairwise intersection of an ordinal-indexed family of independent strong omnific-preserving copies of $\mathbf{No}$ fixing L pointwise. For a set $A \subseteq \mathbf{No}(i)$, the analogous statement holds with an algebraically closed $L \supseteq \mathbb{C} \cup A$.*

Proof. Take H to be the $\mathbb{Q}$ -span of the union of the normal-form supports of all elements of A . In the complex case take the real and imaginary supports, equivalently the complex support. Replacement and union make this a set, and rational span remains a set. Then $A \subseteq \mathcal{K}_k(H)$. Apply [Theorem A](#). $\square$

This corollary does *not* assert that the intersection can be prescribed to be the ordinary field generated by A , its real closure, or an arbitrary non-Hahn subfield. It prescribes the full support closure just constructed. The distinction is essential to both the proof and the classification below. [write] For two maps, and without intersection statements, proper strong exact-monomial copies fixing $A \cup \mathbb{R}$ and preserving and reflecting **Oz** are `opa:as:thm:parameters` of the omnific-preserving report; the corollary extends that result to an **On**-indexed family with an exact common field ([section 1.6](#)). [merge] The same support-closure step localizes a set of parameters in two later results. In the dilation report, `adr:sr:thm:tail` places a set of surcomplex parameters in the Hahn field of the real span of their supports and finds a tail of an explicit ordinal family whose positive real dilation orbits are algebraically independent over that whole field, and `adr:sr:thm:kappa` gives set-sized families of independent orbits over any set-sized base. In this report, [lemma 23.1](#) applies the step one normal-form level deeper, to the outer coefficients of a parameter set, and [theorem 23.3](#) keeps a tail of a fixed ordinal family differentially independent over the compositum with any set of parameters adjoined.

Theorem 6.2 (Exact classification of set-sized intersections in this class). *The set-sized fields that occur as intersections of two images of strong, real-fixing, monomial-preserving self-embeddings of **No** are exactly the fields $\mathcal{K}_{\mathbb{R}}(H)$ with $H \subseteq \mathbf{No}$ a set-sized divisible additive subgroup.*

Proof. For necessity, [proposition 4.4](#) writes the two images as full Hahn fields on divisible groups G_1, G_2 . Their intersection is $\mathcal{K}_{\mathbb{R}}(G_1 \cap G_2)$. In a torsion-free divisible ambient group, each divisible subgroup is closed under the uniquely defined rational division; therefore their intersection H is divisible. If the intersection field is a set, its monomials form a set and their exponents, recovered by the inverse of the omega-map on monomials, form a set. Thus H is set-sized. Sufficiency is [Theorem A](#). $\square$

[write] The real-fixing hypothesis is used through [proposition 4.4](#). Exact-monomial embeddings that preserve **Oz** but move reals exist whenever the bottom gap is nonzero (`opa:as:thm:taylor criterion`); their images meet $\mathbb{R}$ in a proper real closed subfield (`opa:as:prop:overlap`). The theorem says nothing about intersections in that larger class.

6.1 All intersections in an entire diagram

Fix a set of indices I from the family of [Theorem A](#). For $J \subseteq I$ put

$$G_J = H + \sum_{j \in J} G_j, \quad E_J = \mathcal{K}_k(G_J), \quad G_{\emptyset} = H. \quad (6.1)$$

Every sum in G_J is finite; no infinite additive closure is intended.

Theorem 6.3 (Boolean meet realization). *The assignment $J \mapsto E_J$ is injective and order preserving. For $J, T \subseteq I$,*

$$E_J \cap E_T = E_{J \cap T}.$$

The analogous identity holds for every nonempty set-indexed intersection of subsets of I . Moreover E_J and E_T are linearly disjoint over $E_{J \cap T}$. Their full Hahn join is $E_{J \cup T}$. All E_J are real closed for $k = \mathbb{R}$ and algebraically closed for $k = \mathbb{C}$.

Proof. Independence over H means that an element of G_I/H has a unique finite set of nonzero components in the direct sum of the groups G_i/H . It lies in G_J/H exactly when all of these components have indices in J . This proves all the group intersection identities and $G_J + G_T = G_{J \cup T}$. Uniqueness of normal form transfers the intersection identities to full Hahn fields, and [theorem 5.4](#) proves the asserted linear disjointness.

If $j \in J \setminus T$, choose $g \in G_j \setminus H$; then $\omega^g \in E_J \setminus E_T$. This proves injectivity and reflection of containment. The full Hahn join is, by definition, the full Hahn field on the sum of the two exponent groups, hence is $E_{J \cup T}$. Every G_J is divisible, which gives the closedness assertions. $\square$

Remark 6.4 (The word “join”). The theorem realizes a Boolean meet diagram with *Hahn* joins. It does not identify ordinary field composita with those joins. [Theorem C](#) shows that the difference can already be transcendental for two factors.

Corollary 6.5 (Independent choices and absence of new relations). *For distinct indices $i_1, \dots, i_m$, choose $a_j \in K_{i_j} \setminus L_{\mathbb{R}}$. Then $(a_1, \dots, a_m)$ is algebraically independent over $L_{\mathbb{R}}$. The same assertion holds over $L_{\mathbb{C}}$ for the complexified copies. More generally all polynomial relations across any square of [theorem 6.3](#) descend to its intersection field.*

Proof. A real closed subfield has no proper algebraic extension inside an ordered field, so each a_j is transcendental over $L_{\mathbb{R}}$. In the complex case use algebraic closedness. The polynomial algebras $L[a_j]$ therefore have no relations individually, and their finite tensor product is a polynomial algebra on m variables. Joint linear disjointness makes its multiplication map injective. The last statement is [corollary 5.6](#). $\square$

6.2 A limited model-theoretic conclusion

For each fixed first-order formula in the language of ordered fields, the inclusion of any E_J into **No** is elementary, by quantifier elimination for real closed fields. The corresponding complex statement follows from quantifier elimination for algebraically closed fields of characteristic zero [\[4\]](#). This is a formula-by-formula use of ordinary field theory, not the construction of a class satisfaction predicate.

Nothing here asserts elementary embedding in the expanded language with a predicate for **Oz**, a symbol for the omega-map, or a derivation. The independence proved is algebraic independence and linear disjointness; it is not being silently identified with a model-theoretic independence notion in any richer theory.

7 Explicit copies with intersection exactly the constants

The general construction can fix an arbitrary set-sized Hahn core, but uses class bookkeeping. When $H = 0$ there is a direct formula. It also furnishes the separated scales needed for [Theorem C](#).

Lemma 7.1 (Compression of the surreal order). *The function*

$$\psi(x) = \frac{1}{2} + \frac{x}{2(1 + |x|)}$$

*is an order isomorphism from **No** onto the open interval $(0, 1)$.*

Proof. On the nonnegative half-line, $x/(1+x)$ is strictly increasing and belongs to $[0, 1)$. On the nonpositive half-line, $x/(1-x)$ is strictly increasing and belongs to $(-1, 0]$. The formulas agree at zero, proving strict monotonicity and the stated range containment. For $u \in (0, 1)$ put $s = 2u - 1$ and $x = s/(1 - |s|)$. Direct substitution on the two signs of s gives $\psi(x) = u$. All denominators are positive in the ordered field. $\square$

For every ordinal α let

$$I_\alpha = (3\alpha, 3\alpha + 1), \quad \sigma_\alpha(a) = 3\alpha + \psi(a),$$

where the arithmetic in 3α is surreal field arithmetic. These intervals are pairwise disjoint. Indeed $\alpha < \beta$ for ordinals implies $\alpha + 1 \leq \beta$ in the surreal order, hence $3\alpha + 1 < 3\beta$.

Define an *inner* map on normal forms by

$$h_\alpha \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^{\sigma_\alpha(a)}. \quad (7.1)$$

This map is an ordered, real-linear, strongly additive *group* embedding. Additivity only uses the injective relabeling of coefficients; no additivity of σ_α is needed. It is not a unital field map:

$$h_\alpha(1) = \omega^{3\alpha+1/2}. \quad (7.2)$$

Its image is precisely the additive class of all surreal normal forms supported in I_α . Pulling back the support along the order isomorphism σ_α proves fullness of that image.

Now apply the *outer* construction of [theorem 4.1](#):

$$\Phi_\alpha \left(\sum_g c_g \omega^g \right) = \sum_g c_g \omega^{h_\alpha(g)}. \quad (7.3)$$

The distinction between (7.1) and (7.3) is crucial: the outer exponents g are themselves surreal numbers with their own normal forms. [write] The inner map is the inner lift ι_{σ_α} of the omnific-preserving report `(opa:as:eq:innerlift, opa:as:lem:inner)`, and (7.3) is its monomial map $M_{1, \iota_{\sigma_\alpha}}$. That report's order isomorphisms $\mu_{11}, \mu_{12}, \mu_{13}$ have unbounded images; the bounded, pairwise disjoint images of the σ_α are what makes the ranges independent here.

Theorem 7.2 (An explicit independent ordinal family). *The maps (7.3) are strong, real-fixing, monomial-preserving proper field embeddings. Their image fields are jointly linearly disjoint over $\mathbb{R}$, have pairwise intersection $\mathbb{R}$, and their omnific image rings have pairwise intersection $\mathbb{Z}$. Their complexifications have the corresponding intersections $\mathbb{C}$ and $\mathbb{Z}[i]$.*

Proof. The inner ranges G_α have disjoint normal-form support intervals. In a finite sum of elements from different ranges, no nonzero coefficient from one interval can be canceled by another interval. Therefore these ranges are independent over zero. Apply [theorem 4.1](#), [proposition 4.2](#), and [corollary 5.5](#). $\square$

For example,

$$\Phi_0(\omega) = \omega^{\omega^{1/2}}, \quad \Phi_1(\omega) = \omega^{\omega^{7/2}}.$$

Both maps fix every real number. Since $\Phi_0(1) = 1$, the first identity also explicitly shows

$$\Phi_0(\omega^1) \neq \omega^{\Phi_0(1)}.$$

Thus preservation of the set of Conway monomials is much weaker than commutation with the omega-map as a unary operation on all of **No**.

Remark 7.3 (A tempting formula that does not give a proper copy). Replacing every exponent g by cg with a fixed $c > 0$ gives an automorphism, not a proper self-embedding, because $c\mathbf{No} = \mathbf{No}$. The compression above is applied one normal-form level below the field lift; that is what makes the exponent-group image genuinely proper.

8 A transcendental gap above an ordinary compositum

Take G_0, G_1 from [section 7](#). These groups are not merely independent. Their scales are separated:

$$|a| < b/n \quad (a \in G_0, 0 < b \in G_1, n \in \mathbb{N}_{>0}). \quad (8.1)$$

Indeed the inner leading exponent of a nonzero a lies in $(0, 1)$, whereas that of b lies in $(3, 4)$. Leading-monomial comparison proves (8.1). Thus G_0 is convex in the ordered group $G_0 + G_1$, and the sign of $a + b$ for $b \neq 0$ is the sign of b .

8.1 Iterated Hahn coordinates

Lemma 8.1 (Re-expansion at separated scales). *Suppose A, B are additive subgroups with $A \cap B = 0$ and every element of A is infinitesimal relative to every positive element of B , in the sense of (8.1). Then regrouping by B -coordinate gives a field isomorphism*

$$\mathcal{K}_k(A + B) \simeq \mathcal{K}_{\mathcal{K}_k(A)}(B), \quad \sum_{a+b} c_{a,b} \omega^{a+b} \mapsto \sum_b \left(\sum_a c_{a,b} \omega^a \right) \omega^b. \quad (8.2)$$

On the right the notation means iterated Hahn series; the coefficient field may be a class, but each individual series uses only a set of coefficients. The isomorphism is order preserving for ordered k .

Proof. The decomposition $a + b$ is unique, and the order on $A + B$ is lexicographic with B the dominant coordinate. The image in B of a reverse-well-ordered support is reverse well ordered: the inverse image of a nonempty subset of that image has a greatest element, whose projection is greatest in the subset. Each fiber is reverse well ordered in A .

Conversely, a reverse-well-ordered set of B -coordinates, with a reverse-well-ordered support in each fiber, yields a reverse-well-ordered lexicographic union. Its size is a set by replacement and union. The two regroupings are inverse. Addition and multiplication agree coefficientwise: there are finitely many pairs of B -coordinates contributing to each outer product coefficient, and within each resulting coefficient Hahn multiplication in A applies. The dominant B -coordinate and then the leading coefficient in $\mathcal{K}_k(A)$ also give exactly the original order. $\square$

For $k = \mathbb{R}$ or $\mathbb{C}$, put

$$K_0 = \mathcal{K}_k(G_0), \quad K_1 = \mathcal{K}_k(G_1), \quad E = \mathcal{K}_k(G_0 + G_1), \quad M = K_0 K_1.$$

In the iterated coordinates, K_0 is the coefficient field and K_1 has coefficients in the original k .

Lemma 8.2 (Finite coefficient-field control). *Every finite subset of the ordinary compositum M is contained, in the iterated coordinates, in a field $\mathcal{K}_D(G_1)$ where $D \subseteq K_0$ is a field finitely generated over k .*

Proof. Each of the finitely many chosen elements is obtained by finitely many field operations from finitely many elements of K_0 and K_1 . Let $a_1, \dots, a_r \in K_0$ include all the first-kind elements used, and put $D = k(a_1, \dots, a_r)$. The elements of K_1 have coefficients in $k \subseteq D$, while each a_j is a constant in the outer G_1 -series coordinate. The field $\mathcal{K}_D(G_1)$ is closed under the required field operations, proving the assertion. $\square$

This lemma does not claim that every series with coefficients in such a D is in M . It gives a necessary coefficient-field constraint, which is all that is needed. [merge] The constraint is far from sufficient: a series whose coefficients all lie in a field $\mathbb{R}(x)$ generated by one element can be differentially transcendental over M (corollary 19.5). The exact description of M in these coordinates is the finite coefficient-rank denominator test of corollary 22.1.

Theorem 8.3 (A coefficient-transcendence obstruction). *Let $s_1, s_2, \dots \in K_0$ be algebraically independent over k , where $k = \mathbb{R}$ or $\mathbb{C}$, and choose $0 < g \in G_1$. The well-defined series*

$$S = \sum_{n=1}^{\infty} s_n \omega^{-ng} \in E$$

is transcendental over M .

Proof. The outer support $\{-ng : n \geq 1\}$ is strictly decreasing, so the series is valid by lemma 8.1. Suppose a nonzero polynomial over M annihilates S . By lemma 8.2, all its coefficients lie in $\mathcal{K}_D(G_1)$ for one finitely generated field $D \subseteq K_0$ over k .

For $k = \mathbb{R}$, let D' be the relative algebraic closure of D in the real closed field K_0 . It is a set-sized real closed field. To see closedness, a square root of a positive element of D' and a root of an odd-degree polynomial over D' can be taken in K_0 ; each is algebraic over D , so belongs to D' . It is a set because a set of polynomials has only a set of finite root sets. For $k = \mathbb{C}$, the same argument with arbitrary polynomial roots gives a set-sized algebraically closed relative closure $D' \subseteq K_0$.

Since G_1 is divisible, the iterated Hahn field $\mathcal{K}_{D'}(G_1)$ is real closed in the real case, and algebraically closed in the complex case. For a proper-class G_1 this follows, as before, by working with the set-sized divisible subgroup generated by any finite collection of coefficient supports. Thus an element of E algebraic over $\mathcal{K}_D(G_1)$ must belong to $\mathcal{K}_{D'}(G_1)$: use relative algebraic closedness in an ordered field in the real case, or algebraic closedness in the complex case.

Applied to S , uniqueness of its outer expansion now gives $s_n \in D'$ for every n . But

$$\text{trdeg}_k D' = \text{trdeg}_k D < \infty,$$

whereas the s_n are an infinite algebraically independent family over k . This contradiction proves the theorem. $\square$

Explicit witness and proof of Theorem C. Take

$$g_n = h_0(\omega^n) = \omega^{\psi(n)}, \quad g = h_1(1) = \omega^{7/2}, \quad s_n = \omega^{g_n} \quad (n \geq 1).$$

The g_n are $\mathbb{Q}$ -linearly independent: they are distinct inner Conway monomials, so uniqueness of normal form precludes a finite rational linear relation. Consequently the s_n are algebraically independent over $\mathbb{R}$, and also over $\mathbb{C}$ in the complexified field. Indeed the distinct monomials of a polynomial in the s_n acquire distinct exponents $\sum_n m_n g_n$, and no nonzero coefficient can cancel.

The promised actual normal form is

$$S = \sum_{n=1}^{\infty} \omega^{\psi(n) - n\omega^{7/2}}. \quad (8.3)$$

Its exponents strictly decrease: the difference between consecutive exponents is $g + g_n - g_{n+1} > 0$ by scale separation. Thus (8.3) is a legitimate countably supported surreal, not a formal expression awaiting an analytic interpretation. It belongs to E and is transcendental over the appropriate ordinary compositum by [theorem 8.3](#). Transcendence also excludes membership in its relative algebraic closure. $\square$

[merge] The second manuscript extends this witness. With $b = \omega^{15/4} \in G_1$, the element $\omega^b S$ is the first member Y_0 of an ordinal family Y_α of purely infinite omnific integers of order type ω ([theorem 22.2](#)); all their Euler jets are algebraically independent over M , and a tail of the family stays independent after any set of parameters is adjoined ([theorem 23.3](#)). In the real case S itself is then differentially transcendental over M for that Euler derivation. [merge] A third manuscript gives a second proof that S is transcendental over M , real and complex, by a mixed-support annihilator that needs no iterated coordinates ([remark 36.2](#)).

Remark 8.4 (Why the infinite list of coefficients matters). A rational expression in the two fields can use arbitrary elements of each field, but only finitely many of them. In the separated-scale expansion this bounds the transcendence degree of its coefficient field over the constants. Algebraic extension does not remove that finite bound. The series (8.3) defeats it by displaying infinitely many independent coefficients. This argument is different from simply observing that a series is an infinite sum; many infinite sums, such as a geometric series, already belong to an ordinary compositum.

9 Omnific intersections and their fraction fields

For $k = \mathbb{R}, \mathbb{C}$ write $D_{\mathbb{R}} = \mathbb{Z}$ and $D_{\mathbb{C}} = \mathbb{Z}[i]$, and define

$$\Pi_k(H) = \{x \in \mathcal{K}_k(H) : \text{supp}(x) \subseteq H^{>0}\}, \quad \mathcal{A}_k(H) = D_k \oplus \Pi_k(H).$$

The direct sum is additive; multiplication makes $\Pi_k(H)$ a nonunital ring and an ideal in $\mathcal{A}_k(H)$. In particular,

$$\mathcal{A}_{\mathbb{R}}(H) = \mathcal{K}_{\mathbb{R}}(H) \cap \mathbf{Oz}, \quad \mathcal{A}_{\mathbb{C}}(H) = \mathcal{K}_{\mathbb{C}}(H) \cap \mathbf{Oz}[i].$$

We use the ordinary class fraction field of a class domain to mean the subfield of the ambient field consisting of ratios of two of its elements, with nonzero denominator. No quotient set of a proper class is invoked.

Lemma 9.1 (Common denominators for a set of surreals). *Every set $X \subseteq \mathbf{No}$ has a positive monomial $d \in \mathbf{Oz}$ such that $dX \subseteq \mathbf{Oz}$. The same holds for a set of surcomplex numbers with $\mathbf{Oz}[i]$ in place of $\mathbf{Oz}$, and a positive real monomial denominator. In particular*

$$\text{Frac}(\mathbf{Oz}) = \mathbf{No}, \quad \text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}(i).$$

Proof. The union S of the supports of elements of X is a set. Choose a surreal $a > 0$ with $a > -s$ for every $s \in S$, using the cut property. Then $d = \omega^a$ and every product dx have exclusively positive exponents, except that dx may be zero. Thus they lie in the required integer ring. In the complex case use the union of the real and imaginary supports. Applying the assertion to a singleton gives the fraction-field identities. These identities are classical; compare [3, Proposition 2.4.5]. $\square$

[write] In the collection this is the global monomial clearing `odg:thm:fractions` of the Diophantine report, with the same proof.

Transporting the lemma through an embedding gives more than merely one denominator for each individual element.

Proposition 9.2 (Separate uniform denominators, no shared uniform one). *Let $K_\alpha, R_\alpha, L_\mathbb{R}$ be as in Theorem A. Then $\text{Frac}(R_\alpha) = K_\alpha$, and there is a nonzero $d_\alpha \in R_\alpha$ with*

$$d_\alpha L_\mathbb{R} \subseteq R_\alpha.$$

For two distinct indices, no nonzero element of $R_\alpha \cap R_\beta$ can satisfy both such containments. The same conclusions hold in the complex and Gaussian setting.

Proof. The image under Φ_α of a ratio of omnific integers is a ratio of elements of R_α , and every element of K_α arises this way by lemma 9.1. Since $L_\mathbb{R}$ is a set and is fixed pointwise, apply that lemma to $L_\mathbb{R}$ and transport its denominator by Φ_α . This yields the uniform denominator d_α .

A hypothetical common nonzero denominator d belongs to $R_\alpha \cap R_\beta \subseteq L_\mathbb{R}$. If both containments held, then $dL_\mathbb{R} \subseteq \mathcal{A}_\mathbb{R}(H)$. But $d \neq 0$ is in the field $L_\mathbb{R}$, so $dL_\mathbb{R} = L_\mathbb{R}$, which is not contained in its omnific part (it contains $1/2$). This is impossible. The same proof uses $1/2 \notin \mathbb{Z}[i]$ in the complex case. $\square$

[write] The proof shows slightly more than the statement: since $dL_\mathbb{R} = L_\mathbb{R} \not\subseteq \mathbf{Oz}$, a nonzero element of $R_\alpha \cap R_\beta$ satisfies *neither* containment, not merely not both. The separate denominators d_α lie outside $L_\mathbb{R}$.

This proposition already separates two notions of a denominator. To prove Theorem B, however, we need an element with *no shared denominator at all*, not merely the absence of a shared denominator uniform on the entire common field. That stronger assertion is proved next.

9.1 The bounded-support field is exactly the shared fraction field

For this subsection and section 10, use the increasing-support Hahn convention

$$t^h = \omega^{-h}.$$

As $H = -H$, this changes the presentation of $\mathcal{K}_k(H)$, not its underlying field. The support of a nonzero t -series is well ordered and has a least element, its valuation v . Set $v(0) = +\infty$. Let

$$\mathcal{B}_k(H) = \{f \in k((t^H)) : \text{supp}_t(f) \text{ is bounded above in } H\}.$$

It is a subring: finite unions of upper bounds bound sums, and the sum of two upper bounds bounds a product. Its elements can have infinite support inside a bounded interval. In this convention $\mathcal{A}_k(H)$ consists of series with negative exponents and a constant coefficient in D_k .

Proposition 9.3 (Identification of the fraction fields). *If $H \neq 0$, then*

$$\text{Frac } \mathcal{A}_k(H) = \text{Frac } \mathcal{B}_k(H). \tag{9.1}$$

In particular this fraction field contains k and every single monomial t^h with $h \in H$.

Proof. One has $\mathcal{A}_k(H) \subseteq \mathcal{B}_k(H)$. Conversely, take $f \in \mathcal{B}_k(H)$ and choose an upper bound $u \in H$ of its support. Choose $a \in H$ with $a > u$ and $a > 0$, possible in any nonzero ordered abelian group. Both t^{-a} and $t^{-a}f$ have strictly negative support, so belong to $\mathcal{A}_k(H)$. Their ratio is f , giving the reverse inclusion of fraction fields. Constants and monomials have bounded support. $\square$

The identity explains why a mere missing constant is not the obstruction for nonzero H . For example, $r = (r\omega^h)/\omega^h$ for any $r \in \mathbb{R}$ and any positive $h \in H$. The unresolved issue is the unboundedness of supports *within the fixed set group* H . [write] [Proposition 9.3](#) is the localization of `bst:prop:localdensity` of the bounded-support report, with the constant coefficient restricted to D_k . Since batch 32 that report also prints the general form, for every field K and every unital subring $D \subseteq K$, as `bst:gr:prop:fractions`.

10 A self-contained cofinal gap argument

The mechanism in this section belongs to bounded-support Hahn arithmetic, not to the new embedding construction. It is included in full so that the transcendental conclusion in [Theorem B](#) does not depend on an unreviewed repository result. The repository's bounded-support report gives stronger independent-family assertions [8], and the non-fraction obstruction is already in [3]. [write] In that report, [theorem 10.2](#) is `bst:cor:onegap` at countable cofinality and, in characteristic zero, the one-series case of `bst:thm:independent`; the single-series form at uncountable cofinality over an arbitrary coefficient field is not stated there, but its proof is that report's tail estimate and root count ([section 1.6](#)). No priority is claimed for it.

Lemma 10.1 (Cofinal sequences with polynomial gaps). *Let $H \neq 0$ be a set-sized ordered abelian group and put $\kappa = \text{cf}(H)$. There is a strictly increasing cofinal family $(a_\alpha)_{\alpha < \kappa}$ in $H^{>0}$ such that for every positive integer d and every $c \in H$,*

$$a_{\alpha+1} - da_\alpha > c \quad \text{for all sufficiently large } \alpha < \kappa. \quad (10.1)$$

Proof. The cofinality of a nonzero ordered group is an infinite regular cardinal: a finite subset is bounded, and a cofinal union of fewer than κ subsets of size less than κ would contradict minimality of κ by taking bounds for those subsets. Fix a cofinal list $(u_\alpha)_{\alpha < \kappa}$.

When $\kappa = \aleph_0$, choose recursively $a_0 > 0, u_0$ and

$$a_{n+1} > u_{n+1}, \quad a_{n+1} > (n+2)a_n.$$

The list is cofinal. For fixed d , once $n+2 \geq d+1$ the difference in (10.1) exceeds a_n , and this eventually exceeds c .

When $\kappa > \aleph_0$, at stage α the set

$$\{0, u_\alpha\} \cup \{na_\beta : \beta < \alpha, n \in \mathbb{N}_{>0}\}$$

has cardinality less than κ and hence is bounded above in H . Choose a_α strictly above it. The resulting family is increasing and cofinal. At a successor stage, $a_{\alpha+1} > (d+1)a_\alpha$ for every positive integer d . Thus again the difference in (10.1) exceeds a_α and is eventually larger than c . Since an infinite cardinal has no last ordinal below it, all the successor indices used are still below κ . $\square$

Theorem 10.2 (One transcendental series over all bounded-support fractions). *For any field k and nonzero set-sized ordered abelian group H , choose a sequence from lemma 10.1. Then*

$$F = 1 + \sum_{\alpha < \kappa} t^{a_\alpha} \quad (10.2)$$

is transcendental over $\text{Frac } \mathcal{B}_k(H)$.

Proof. The support is a well-ordered set, so F is a Hahn series and $v(F) = 0$. Suppose a nonzero polynomial over $\text{Frac } \mathcal{B}_k(H)$ annihilates F . Clearing its finitely many denominators gives

$$Q(Y) = \sum_{j=0}^d b_j Y^j \in \mathcal{B}_k(H)[Y] \setminus \{0\}, \quad Q(F) = 0,$$

where $d \geq 1$. Each nonzero coefficient has support bounded both below and above: below by its least exponent and above by the definition of $\mathcal{B}_k(H)$. Choose common bounds $b, c \in H$ for the supports of all nonzero coefficients, with $b \leq c$.

For $\alpha < \kappa$ let

$$P_\alpha = 1 + \sum_{\beta \leq \alpha} t^{a_\beta}.$$

These are distinct elements of the same Hahn field, including at limit indices; they need not be polynomials in t . Their supports lie in $[0, a_\alpha]$, and

$$v(P_\alpha) = 0, \quad v(F - P_\alpha) = a_{\alpha+1}.$$

Finite Hahn multiplication gives

$$\text{supp}_t Q(P_\alpha) \subseteq [b, c + da_\alpha]. \quad (10.3)$$

The elementary factorization

$$F^j - P_\alpha^j = (F - P_\alpha) \sum_{\ell=0}^{j-1} F^{j-1-\ell} P_\alpha^\ell$$

and the nonnegative valuations of the factors in the finite sum give

$$v(Q(F) - Q(P_\alpha)) \geq b + a_{\alpha+1}. \quad (10.4)$$

By lemma 10.1, eventually

$$b + a_{\alpha+1} > c + da_\alpha.$$

Since $Q(F) = 0$, (10.4) says that every nonzero exponent of $Q(P_\alpha)$ is above the upper bound in (10.3). Therefore $Q(P_\alpha) = 0$ for every sufficiently large α . This gives infinitely many distinct roots of the nonzero finite-degree polynomial Q , a contradiction. The argument uses no characteristic-zero assumption. $\square$

Corollary 10.3 (The factorial witness). *In $\mathbb{R}((t^{\mathbb{Q}}))$ or $\mathbb{C}((t^{\mathbb{Q}}))$, the series*

$$F = 1 + \sum_{n=1}^{\infty} t^{n!}$$

is transcendental over the fraction field of all bounded-above-support series, hence over $\text{Frac } \mathcal{A}_k(\mathbb{Q})$.

Proof. The positive integers $n!$ are cofinal in $\mathbb{Q}$ and satisfy $(n+1)! - dn! = (n+1-d)n! \rightarrow +\infty$ for each fixed d . Thus the proof of [theorem 10.2](#) applies to this explicit sequence. For clarity, its partial sums start with $1+t$, $1+t+t^2$, $1+t+t^2+t^6$; there is no duplicate exponent from a $0!$ term. $\square$

[write] The same exponents carry the continuum of independent actual infinitesimals of `bst:thm:actualreal`. The omnific-preserving report's `opa:as:ex:smallfrac` uses $\sum t^{n!}$ at $\Gamma = \mathbb{Z}$ for the weaker statement that it is not a fraction of $\mathcal{R}_{\mathbb{Z}}$.

Proof of Theorem B. By [Theorem A](#), the common field is $L_k = \mathcal{K}_k(H)$ and the common integer ring is $\mathcal{A}_k(H)$. Each full integer copy has fraction field equal to its field copy by [proposition 9.2](#). If $H \neq 0$, combine [proposition 9.3](#) and [theorem 10.2](#) to obtain an element of L_k transcendental over $\text{Frac } \mathcal{A}_k(H)$. If $H = 0$, the common ring is $\mathbb{Z}$ or $\mathbb{Z}[i]$ and its fraction field is $\mathbb{Q}$ or $\mathbb{Q}(i)$. The ambient constant field $\mathbb{R}$ or $\mathbb{C}$ contains a transcendental element over that countable field: its algebraic elements are a countable union of finite root sets, whereas the constant field is uncountable. Finally [corollary 10.3](#) supplies the specified witness for $H = \mathbb{Q}$. This proves every assertion. $\square$

Corollary 10.4 (Failure witnessed by the denominator equation). *For $H = \mathbb{Q}$ and F from [corollary 10.3](#), there are nonzero $d_\alpha \in R_\alpha$ and $d_\beta \in R_\beta$ with $d_\alpha F \in R_\alpha$ and $d_\beta F \in R_\beta$, but there are no $a, d \in R_\alpha \cap R_\beta$ with $d \neq 0$ and $dF = a$.*

Proof. The separate denominators follow even uniformly from [proposition 9.2](#). A shared solution would express $F = a/d$ in $\text{Frac } \mathcal{A}_{\mathbb{R}}(\mathbb{Q})$, contradicting transcendence. The same proof works for the Gaussian rings. $\square$

10.1 Where the two transcendence arguments differ

The gap above an ordinary compositum in [section 8](#) is detected by an infinite algebraically independent *coefficient* family in a separated scale decomposition. The gap above the shared fraction field in this section is detected by a cofinal sequence of *support exponents*, even with all nonzero coefficients equal to one. Neither proof is a cardinality-only argument, and neither concludes merely that an element lacks a particular chosen representation: each excludes every finite-degree algebraic relation over the stated base.

11 A set-sized version of the construction

The class construction has a direct conditional set-sized analogue. This both isolates the exact saturation used and gives a more conventional target for formalization.

Definition 11.1. For an uncountable regular cardinal κ , a linear order is an η_κ -order if every cut $L < R$ with $|L \cup R| < \kappa$ has a strict intermediate point. Empty sides are allowed.

Theorem 11.2 (Independent embeddings in a saturated ordered group). *Let κ be uncountable regular, and let V be a divisible ordered abelian group of cardinality κ whose order is an η_κ -order. If $H \subseteq V$ is a divisible subgroup of cardinality less than κ and $2 \leq |I| \leq \kappa$, then there are ordered additive proper embeddings $h_i : V \rightarrow V$ fixing H whose ranges are independent over H .*

Proof. Enumerate the requirements $I \times V$ in κ stages. Before each stage, fewer than κ elements and partial graph entries have been introduced. Their rational span still has size less than κ ; uncountability and regularity ensure this also at limit stages below κ .

The one-point proof works verbatim. To avoid the current span E , form the set T of its elements realizing the current cut and realize the enlarged cut $L \cup T < R$. All these sets have size less than κ , so the η_κ hypothesis provides the new value. Independence is preserved by exactly the same finite-relation argument. At the end all domains are V , and another independent range witnesses properness of each one. $\square$

This theorem is conditional on the existence of the stated group; it does not assume that every cardinal admits such a structure. The uncountability hypothesis is purposeful: in a countable group, adjoining one rational vector already gives a countable domain, so mere realization of finite order cuts would not justify the same proof.

Lifting these embeddings to full set-sized Hahn fields gives the same intersection and disjointness statements. Those Hahn fields need not have cardinality κ ; the theorem places no unwarranted cardinality bound on all their allowable supports.

12 Further research questions

The questions below are proposed by the present theorems. They are not presented as a list of previously published open problems, nor as claims that their answers must be difficult. The first four most directly extend the proved results.

Research question 1 (Intersections outside the full Hahn category). Which set-sized real closed subfields $L \subseteq \mathbf{No}$ containing $\mathbb{R}$ can occur as the exact intersection of two strong, omnific-preserving self-embedding images if the monomial-preserving hypothesis is removed?

Why it is the next step. [Theorem 6.2](#) answers the monomial-preserving case exactly. A non-Hahn example, or an obstruction forcing Hahn closure for a larger class of embeddings, would identify the precise information carried by the chosen monomial cross-section. Strongness alone should not be assumed to settle it. [write] *Status in the collection:* *open and new*. It is related to `opa:as:q:units`, which asks to classify embeddings without exact monomial preservation, but that question does not concern intersections.

Research question 2 (Relative classification of pairs of copies). Suppose two independent pairs have the same core $\mathcal{K}_{\mathbb{R}}(H)$. Which invariants determine whether an omnific-preserving automorphism of $\mathbf{No}$ fixing that core carries one pair to the other?

Concrete first case. Compare pairs whose exponent groups form different ordered amalgams over H , even when their abstract additive quotients agree. The ambient ordering of mixed sums is an evident invariant for monomial lifts, but it is not proved here to be complete. [write] *Status in the collection:* *open*. It is the pair version, at a fixed Hahn core, of `opa:as:q:conjugacy`, whose partial answer `opa:as:thm:continuum` concerns single copies that move reals.

Research question 3 (An exact description of ordinary composita). For independent groups G_1, G_2 over H , characterize the subfield $\mathcal{K}_k(G_1)\mathcal{K}_k(G_2)$ inside $\mathcal{K}_k(G_1 + G_2)$. Determine the transcendence degree of the full Hahn join over the ordinary compositum in natural set-sized and class-sized examples.

Available starting point. In the separated case, [lemma 8.2](#) gives a necessary finite coefficient-field condition, and [theorem 8.3](#) supplies a transcendental obstruction even after algebraic closure. A necessary-and-sufficient condition would be much stronger. For a proper-class extension, formulate

dimension statements as independent families of specified set cardinalities rather than casually assigning a set cardinal to the whole class. [merge] *Status (batch 34): answered in the separated case, open in general.* For the separated explicit copies of [Theorem C](#) (trivial core, G_0 infinitesimal relative to G_1), the compositum is characterized by a finite coefficient-rank denominator test ([theorem 17.2](#) and [corollary 22.1](#)), and the necessary condition of [lemma 8.2](#) is shown not to be sufficient ([corollary 19.5](#)). In the natural set-sized Laurent examples the algebraic and Euler-differential transcendence degrees of $F((t))$ over $Fk((t))$ are both $|F|^{\aleph_0}$ when $|k| \leq \mathfrak{c}$ and $\text{trdeg}(F/k) > 0$ ([theorem 21.1](#)). In the class-sized explicit join, as this question asks, the answer is given by independent families of every set cardinal of countably supported omnific integers ([theorem 22.2](#)), a tail independent over any set of adjoined parameters ([theorem 23.3](#)), and the absence of set-sized generation ([corollary 23.4](#)). Open: groups over a common core H that are not separated, prescribed nonzero common cores, infinite algebraic coefficient extensions, and $|k| > \mathfrak{c}$ with small positive $\text{trdeg}(F/k)$ (Research questions [11](#), [12](#), [13](#) and [14](#)). [merge] *Status (batch 35): the second part is also answered for explicit non-separated examples.* For interleaved real closed factors $\mathcal{K}_k(A_\kappa), \mathcal{K}_k(B_\kappa)$ with intersection k , the join has transcendence degree exactly 2^κ over the compositum at every infinite cardinal κ , with omnific witnesses ([Theorem E](#)); for dense Archimedean groups it is $\mathfrak{c}$, with witnesses prime in **Oz** and **Oz**[i] ([Theorem F](#)); a fixed pair of explicit class support subfields has an ordinal-indexed independent family and no set-sized transcendence basis ([Theorem G](#)). All rest on the mixed-support criterion [Theorem D](#), which needs no convex separation. The first part (an exact description of the compositum) remains open outside the separated case (Research questions [11](#) and [24](#)), as do prescribed nonzero cores (Research questions [14](#) and [26](#)).

Research question 4 (Descent for the arithmetic intersection). How do integral closure, complete integral closure, and finite-type algebraic geometry over $\mathcal{A}_k(H)$ compare with those over $\text{Frac } \mathcal{A}_k(H)$ and $\mathcal{K}_k(H)$? Which descent failures besides the single denominator equation are forced by the transcendental gap?

Important boundary. The elementary fraction-field failure proved here does not by itself calculate a normalization, a Picard group, or a nontrivial projective module. Those require additional arguments and should be compared with the repository's dedicated arithmetic reports.

Research question 5 (Large fixed bases). For which proper-class divisible subgroups $H \subsetneq \mathbf{No}$ can independent ordered-group copies fixing H be constructed? What cut-size or cofinality condition replaces the set-sized hypothesis of [theorem 3.4](#)?

Obstacle to the present proof. Its first partial domain and its forbidden range span must be sets. A proper-class H invalidates that step. This is a limitation of the argument, not a proof that every such extension is impossible.

Research question 6 (Compatibility with additional surreal operations). Can versions of the prescribed-intersection theorem be proved for embeddings commuting with the Gonshor exponential or with a fixed surreal derivation? What cores would necessarily be closed under those operations?

First necessary checks. If an operation is preserved by both embeddings, their images and intersection must be closed under that operation. The explicit maps of [section 7](#) already fail to commute with the omega-map. Therefore they cannot simply be relabeled as solutions in richer languages. Published work on exponential surreal embeddings provides a relevant comparison [\[5\]](#). [write] *Status in the collection: open.* It lies inside `opa:as:q:exp` of the omnific-preserving report, restricted to prescribed intersections.

Research question 7 (Simplicity and initiality). Which prescribed intersections remain possible when each image is required to be initial for the simplicity relation, or when a specified bound on birthdays is imposed?

Reason for separating this question. The saturation proof uses the simplest realizer of an enlarged cut to choose one point, but it does not preserve the global simplicity relation. Local use of a simplest number is not an initial-embedding theorem. [write] *Status in the collection:* open. Simplicity relations are also named in `opa:as:q:exp`.

Research question 8 (Uniform maximal transcendence at a fixed core). For a fixed nonzero set-sized H , obtain an independently checked sharp calculation of $\text{trdeg}(\mathcal{K}_k(H)/\text{Frac } \mathcal{A}_k(H))$ and relate it to $\text{cf}(H)$ and support cardinality. How much of this calculation can be made uniform across all realized intersection diagrams?

Prior work to retain. This is not an untouched direction: the repository’s bounded-support report already proposes a $2^{\text{cf}(H)}$ -sized independent family and optimal support thresholds [8]. A useful next contribution is verification, extension, or a genuinely different uniformity statement, not rediscovery of its one-witness special case. [write] *Status in the collection:* the programme of the bounded-support report (`bst:thm:independent`, `bst:cor:exactcard`, `bst:cor:realgroup`); its uniformity over realized diagrams is open.

Research question 9 (Elementary embeddings of the field–integer pair). Can the embeddings be chosen elementary in $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$ while retaining a prescribed set-sized intersection and the stated algebraic independence? What extra restrictions does the integer-ring predicate impose?

What has actually been proved. The field embeddings preserve and reflect membership in $\mathbf{Oz}$ and commute with floor, but these facts do not establish preservation of all formulas with alternating quantifiers. Ordinary real-closed-field elementarity is a weaker assertion. [write] *Status in the collection:* open. For the maps here it is `opa:as:q:elementary`, with a prescribed intersection added; the nonelementarity criterion of `opa:as:thm:elementary` needs a moved real and does not apply.

Research question 10 (A verified set-level kernel). Can the coset-slice lemma, finite separating-matrix argument, and saturated ordered-group extension theorem be formalized as reusable library results, then connected to a universe-aware model of the surreal construction?

Concrete target. The next section gives declarations and obligations at the level of mathematical specifications. A verified finite support computation is not a substitute for these support and class-size proofs. [write] *Status in the collection:* open. Its Hahn-field kernel overlaps `opa:as:q:formal`; none of the proposed identifiers exists in the collection’s Lean library.

13 Proof dependencies, verification boundary, and conclusion

13.1 A dependency map

The proof of the main copy theorem follows the chain

$$\begin{aligned} & \text{set-cut property} \implies \text{one-point avoidance} \\ & \implies \text{independent group ranges} \implies \text{strong monomial lifts.} \end{aligned}$$

The upgrade from support intersections to algebraic independence is a separate chain:

$$\begin{aligned} & \text{coset slices} \implies \text{finite separating matrix} \\ & \implies \text{linear disjointness} \implies \text{polynomial-relation descent.} \end{aligned}$$

The arithmetic application adds the common-denominator lemma and the cofinal gap argument. The compositum application instead adds the explicit separated-scale construction, iterated Hahn coordinates, and finite coefficient-field control. Neither application is used to justify the other, and neither is needed to justify the group construction.

13.2 A practical formalization decomposition

A set-level implementation could first express full Hahn series in the usual increasing-support convention and prove the following specifications. The names below are proposed identifiers, not claims that declarations with these names already exist.

- (1) `cosetSlice_linear`: restriction of a Hahn series to one coset of a subgroup, followed by monomial normalization, is linear over the Hahn field on that subgroup. No convexity assumption is permitted.
- (2) `cosetSlice_separates` and `finite_slice_matrix`: the family separates points and yields an invertible finite matrix on a linearly independent tuple.
- (3) `hahnSubfields_linearlyDisjoint`: use the same slice matrix after enlarging the scalar subgroup from the intersection to the second subgroup. This is the main reusable field-theoretic lemma.
- (4) `orderedEmbedding_extend_avoid`: realize a cut of the current rational span while avoiding a prescribed small subspace, with explicit cardinality hypotheses for [theorem 11.2](#).
- (5) `boundedSupport_gap_transcendental`: formalize the two support bounds (10.3) and (10.4), followed by the finite root bound. Start with the factorial sequence over $\mathbb{Q}$ before the transfinite cofinality version.
- (6) `iteratedHahn_coefficientField`: formalize separated ordered sums, the regrouping isomorphism, and finite coefficient-field control of a compositum. This is the kernel of the second transcendence proof.

The final proper-class theorem additionally needs a class or universe framework, a set-like enumeration, and proof that every recursion stage and every support pullback is a set. A set-level Hahn formalization by itself does not discharge that final obligation. The repository contains relevant Hahn and normal-form infrastructure [6], but this article adds no Lean code and makes no machine-verification claim.

13.3 Claims deliberately not made

The following boundaries prevent several plausible but invalid upgrades. The common field is a *full Hahn field* on a *set-sized* subgroup, not an arbitrary prescribed real closed field. Group sums are finite sums, not hidden summable closures. Boolean joins are Hahn joins, not ordinary composita. Strongness concerns Hahn-summable set families, not convergence of all real series. The embeddings preserve omnific membership, not an unproved complete first-order theory of the pair. They are

not asserted to preserve the omega-map, exponentiation, derivation, birthdays, or simplicity. The proofs establish the stated theorems but do not establish historical priority or settle a named general factorization conjecture for **Oz**. [write] [Section 14](#) collects these boundaries with every other limitation stated in the source, and adds those that come from the collection.

13.4 Conclusion

Full surreal copies can be arranged with far more control than a bare universality statement provides. An arbitrary set-sized full Hahn core can be fixed exactly, an ordinal-indexed family of copies can be made independent over it, and their arithmetic parts remain genuine copies of the entire omnific ring. The same construction simultaneously controls the surcomplex field and Gaussian omnific integers.

The two descent failures are complementary. Passing from intersecting omnific rings to their fraction field can lose transcendental elements that belong to both full field copies, even though each copy provides one uniform denominator for the whole common set field. Passing from two full field copies to their ordinary compositum can likewise miss transcendental normal forms allowed by their combined exponent group. Thus support closure, ordinary field generation, and arithmetic localization are three distinct operations. The prescribed-intersection construction makes their difference visible within the actual surreal and surcomplex fields, rather than only in abstract auxiliary examples.

14 Consolidated ledger of non-claims

[write] The source states its limitations where they arise, and each is kept in place. This ledger collects them with their locations. “README” refers to the delivered README, which is not shipped; its content is in the collection README. Items S1–S24 are the source’s. Items W1–W8 were added when the report joined the collection. [merge] The limitations of the second manuscript (Sections 15–27) are listed separately, as items C1–C23 and M1–M5 of [section 25.5](#); those of the third manuscript (Sections 28–41) as items J1–J25 and JM1–JM6 of [section 38.4](#).

- S1. The common field is a *full* Hahn field on a *set-sized* divisible subgroup. It is not an arbitrary prescribed real closed or set-sized subfield, not the ordinary field generated by a parameter set, not its real closure, and not a non-Hahn subfield (after [corollary 6.1](#); [section 13.3](#); README).
- S2. Group sums G_J are finite sums, not hidden summable closures (after (6.1); [section 13.3](#)).
- S3. The joins of [theorem 6.3](#) are Hahn joins, not ordinary composita; [Theorem C](#) shows that the difference can be transcendental ([remark 6.4](#); [section 13.3](#)).
- S4. Strongness concerns Hahn-summable set-indexed families, not convergence of real series or continuity; ordinary analytic convergence is not used in any proof ([section 2](#); [section 13.3](#)).
- S5. Elementarity is claimed only formula by formula in the languages of ordered fields and of fields, by quantifier elimination for real closed and algebraically closed fields. Nothing is asserted in the language with a predicate for **Oz**, the omega-map or a derivation, and algebraic independence is not identified with a model-theoretic independence notion of a richer theory. Preservation and reflection of **Oz** and commutation with the floor do not give elementarity of the pair ([section 6.2](#); research question 9; [section 13.3](#)).

- S6. The embeddings are not asserted to preserve the omega-map, the Gonshor exponential, the Berarducci–Mantova derivation, birthdays or simplicity. Monomial preservation is not omega-map commutation: $h \neq \Phi_h$ in general, and $\Phi_0(\omega^1) \neq \omega^{\Phi_0(1)}$ for the explicit maps, so they cannot be relabelled as solutions in richer languages ([warning 4.5](#); [section 7](#); research question 6; [section 13.3](#)).
- S7. No proper-class base H is treated. The obstacle is a limitation of the argument, whose first partial domain and forbidden span must be sets, not a proof that such extensions are impossible (research question 5).
- S8. [Theorem 11.2](#) is conditional on the existence of the stated saturated group; it does not assume that every cardinal admits one. Uncountability is needed for the proof. The lifted Hahn fields need not have cardinality κ ([section 11](#)).
- S9. The fraction-field failure does not by itself compute a normalization, a Picard group or a nontrivial projective module (research question 4).
- S10. The bounded-support transcendence mechanism of [section 10](#) is not new. It belongs to bounded-support Hahn arithmetic; the bounded-support report gives stronger independent families, and the non-fraction obstruction is Proposition 2.4.5 of [3]. It is reproved so that [Theorem B](#) does not depend on an unreviewed repository result, not claimed as a discovery ([section 1.4](#); [section 10](#); research question 8; README).
- S11. The ingredients are not claimed as new: the set-cut property, normal forms, Hahn multiplication and inversion, and the closedness criteria. The group extension is a concrete saturation construction, not a new saturation principle; the monomial lift is standard Hahn functoriality; the coset-slice lemma is an elementary reusable lemma, not a certified new theorem of Hahn field theory ([section 1.4](#); title page).
- S12. Priority is not certified. Targeted literature searches do not certify that the combined formulation is absent from published or unpublished work, and no claim of having audited every report or every later repository revision is made ([section 1.4](#); [section 13.3](#); README).
- S13. The article is unrefereed and not formalized. It adds no Lean code and makes no machine-verification claim. The identifiers of [section 13.2](#) are proposals, not existing declarations; a set-level Hahn formalization does not discharge the class-size obligations, and a verified finite support computation is no substitute for them (title page; [section 13.2](#); research question 10; README).
- S14. No named longstanding or published conjecture is claimed solved, and no general factorization conjecture for $\mathbf{Oz}$ is settled (title page; [section 13.3](#); README).
- S15. [Theorem B](#) is the failure of one descent operation, not of $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$, which the argument uses (after [Theorem B](#)).
- S16. [Lemma 8.2](#) gives a necessary coefficient-field condition, not membership in M of every series with coefficients in D . The compositum argument is not the invalid inference that every infinite sum lies outside a compositum; a geometric series already lies in one (after [lemma 8.2](#); [remark 8.4](#); README).
- S17. The class framework uses no tensor product indexed by a proper class and no proper-class-dimensional tensor object, no class truth or satisfaction predicate, and no recursion with a proper-class state at a stage. The cut property is never applied to proper-class sides ([section 2](#); [section 6.2](#)).

- S18. [Theorem 6.2](#) is a classification inside the strong, real-fixing, monomial-preserving class only. Strongness alone should not be assumed to settle research question 1.
- S19. The maps h_α fix H as a set of exponent values and the Φ_α fix $\mathcal{K}_\mathbb{R}(H)$; the two roles are not identified ([remark 3.6](#)).
- S20. The positive-exponent truncation plus the ordinary floor of the constant coefficient is not always the omnific floor ([remark 4.3](#)). Scaling all exponents by a constant gives an automorphism, not a proper copy ([remark 7.3](#)).
- S21. The saturation proof uses a simplest realizer locally but does not preserve the simplicity relation; this is not an initial-embedding theorem (research question 7).
- S22. The research questions are proposed by these theorems; they are not presented as previously published open problems or as claims of difficulty. Research question 8 is not an untouched direction ([section 12](#)).
- S23. For coefficient fields other than $\mathbb{R}$ and $\mathbb{C}$ the Hahn-field notation is formal, not a subfield of **No** or **No**(i); fraction fields of class domains are ratios inside the ambient field, not quotients of a proper class ([section 2](#); [section 9](#)).
- S24. The encyclopedia page is orientation only; Ehrlich–Kaplan is a comparison, not a proof input; the repository reports were used for scope and nonduplication, not as dependencies. The delivered production checks (compilation, rendering and visual inspection of the PDF) are not formal verification of the mathematics (bibliography; README).
- W1. No `isc:` label has a Lean mapping in `FORMALIZATION.md`. Independent proof review and formalization are pending. The package shipped no verification code, data or audit, so the report has no suite to rerun ([section 1.6](#); [Appendix A.6](#)). [merge] This holds for source 01; the second manuscript shipped a finite-check suite, rerun in [Appendix A.8](#), which tests none of source 01’s theorems; so did the third ([Appendix A.10](#)), which tests none of them either.
- W2. [Theorem 4.1](#) and [propositions 4.2](#) and [4.4](#) and the inner lift of [section 7](#) are the real-fixing case of the omnific-preserving report’s classification and its inner lift; no priority is claimed for them relative to that report. [Corollary 6.1](#) extends `opa:as:thm:parameters` but does not claim its topological assertions ([section 1.6](#)).
- W3. [Lemma 9.1](#) is `odg:thm:fractions`, [proposition 9.3](#) is the localization of `bst:prop:localdensity`, the coset slice is `bst:eq:projection`, and [theorem 10.2](#) and [corollary 10.3](#) duplicate `bst:cor:onegap` and the one-series case of `bst:thm:independent` ([section 1.6](#)).
- W4. Real fixing is a choice. For the explicit copies the bottom gap is nonzero, so exact-monomial embeddings with the same exponent maps that preserve **Oz** and move reals exist; nothing is claimed about them or about intersections in that larger class ([section 1.6](#)).
- W5. The large-cardinal report’s question “Joint algebraic dependence of the images” is not answered: its image $J(\mathbf{No})$ is not a full Hahn field on a subgroup ([section 1.6](#)).
- W6. None of the omnific-preserving report’s questions is answered; in particular `opa:as:q:composition`, `opa:as:q:conjugacy`, `opa:as:q:elementary` and `opa:as:q:exp` stay open. None of the bounded-support report’s questions is answered ([section 1.6](#)).
- W7. The source’s statements about the collection are those of its pin and are kept as provenance ([section 1.4](#); [Appendix A.2](#)).

W8. The proofs were read when the report was written (Appendix A.4). That reading is not an independent proof review.

15 Beyond composita of surreal copies: a later manuscript

[merge] Sections 15 to 27 print a second manuscript, written against this report. In the report's local numbering (Appendix A.7) the original manuscript is source 01 and the addition is source 02: manuscript 05 of batch 34 of the collection, archive *Surreal_Composita_Research*, *Beyond Composita of Surreal Copies: Exact rank tests, differential transcendence, and omnific witnesses*, a 29-page PDF dated 23 September 2026 and pinned to repository commit 7b256e0792df7718995b816ad12a77bf56731f8f. At that pin this report's `article.tex` was the blob 7d5cfe644e7c902f2b594179c337a6c633350fc3. Since then only three [write] sentences about the bounded-support report have changed (sections 1.4, 1.6 and 9.1), so the source's description of sections 7, 8 and 12 is accurate. The source takes Research question 3 of section 12 as its target.

Section n of the source is Section $n + 14$ here for $n = 1, \dots, 12$, and every numbered statement keeps its position: the source's Theorem $n.m$ is Theorem $(n + 14).m$ here, and likewise for lemmas, corollaries, remarks, definitions and examples. Its Questions 1–12 are Research questions 11–22. Its Appendices A and B form section 27; its notation index (Appendix C) is folded into the table of section 15.5. Where the source cites this report by section number, the citation is replaced by a cross-reference to the same section; the numbering of Sections 1–12 has not changed since the pin. Where the source reproduces material already printed here (the explicit copies of section 7 and the re-expansion of section 8), it is printed once and the source is credited (section 22.1). Text added in the merge is marked [merge]; symbols renamed from the source are listed in section 15.5. Every result, proof, example, question and limitation of the source is printed.

15.1 Field generation is not Hahn summation

A Hahn field permits well-ordered, potentially infinite supports. An ordinary field compositum permits only finitely many field operations on finitely many elements at a time. These are different closure operations. Even when two Hahn subfields have a transparent intersection, their ordinary compositum need not contain all the series supported on the sum of their exponent groups.

This report constructs explicit strong surreal self-embeddings whose images K_0, K_1 intersect in the real constants (theorem 7.2), and section 8 proves that the full Hahn join E contains an element transcendental over $M = K_0 K_1$ (Theorem C). Research question 3 asks for an exact description of ordinary composita and the transcendence degree of the join over the compositum. Source 02 takes that question as its target. The explicit copies and the one-series coefficient-transcendence obstruction of theorem 8.3 are this report's; neither is claimed by source 02 as a new construction.

There are two genuinely different ways for a series to escape the compositum. Its coefficients can use infinitely much algebraically independent information. More subtly, its coefficients can all lie in one field $k(x)$ and nevertheless have complexity in x that defeats every finite algebraic or differential relation over the compositum. The second phenomenon shows that the necessary finite coefficient-field condition of lemma 8.2 is far from sufficient.

15.2 Main conclusions

The first conclusion is an exact test. If the *coefficient rank* of a Hahn series means the dimension over k of the span of its coefficients, then

$$x \in Fk((t^\Gamma)) \iff \exists q \neq 0 (\text{crk}_k(q) < \infty \text{ and } \text{crk}_k(qx) < \infty). \quad (15.1)$$

This is an intrinsic test once the coefficient field and Hahn coordinates have been fixed. It does not assert decidability for arbitrary infinite input. The proof is elementary tensor algebra and is presented as such.

The second conclusion is a maximal-cardinality theorem. Write $\mathfrak{c} = 2^{\aleph_0}$. For $|k| \leq \mathfrak{c}$ and any coefficient extension F/k of positive transcendence degree,

$$\text{dtrdeg}(F((t))/Fk((t))) = \text{trdeg}(F((t))/Fk((t))) = |F|^{\aleph_0}. \quad (15.2)$$

Thus even $F = k(x)$, a single transcendental coefficient extension, produces the full continuum of differential freedom when k is $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$. Here the derivation acts only on the outer series variable.

The third conclusion concerns actual omnific integers. For the explicit separated copies of [section 7](#), with the order isomorphism $\psi(u) = \frac{1}{2} + \frac{u}{2(1+|u|)}$ of [lemma 7.1](#) and ordinal arithmetic used only in the label $\omega \cdot \alpha + n$, define

$$Y_\alpha = \sum_{n \geq 1} \omega^{\omega^{15/4} - n\omega^{7/2} + \omega^{\psi(\omega \cdot \alpha + n)}} \quad (\alpha \in \mathbf{On}). \quad (15.3)$$

Every exponent in this normal form is positive, and the exponents strictly decrease with n . Hence Y_α is a purely infinite omnific integer with countable support. All its Euler jets, together with the jets of all the other Y_α , are algebraically independent over the compositum in the precise finite-relation sense. Moreover, given *any set* $P \subseteq E$, there is an ordinal α_0 such that the tail $\{Y_\alpha : \alpha \geq \alpha_0\}$ remains differentially algebraically independent over $M\langle P \rangle$.

This last assertion is stronger than producing a fresh witness separately for each parameter set: a tail of one predefined family works. It also avoids assigning a set cardinal to a proper-class transcendence degree. [merge] For $\alpha = 0$ the label is $\beta_{0,n} = n$, and $Y_0 = \omega^{\omega^{15/4}} S$, where S is the witness (8.3) of [Theorem C](#). Source 02 thus extends this report's witness to an ordinal family inside $\mathbf{Oz}$, as it says.

15.3 Scope, sources, and the meaning of new

Hahn arithmetic, linear disjointness, algebraic extension of derivations in characteristic zero, and the surreal normal-form construction are established tools. Standard Hahn derivation frameworks appear in Kuhlmann–Matusinski [11]; recurrence descriptions of generalized series subfields are studied by Krapp–Kuhlmann–Serra [10]. The finite-rank characterization below is a coefficient-space counterpart, not a replacement for their recurrence theory. Omnific arithmetic has a substantial factorization literature, including L’Innocente–Mantova [3]; no factorization conjecture is addressed here.

The candidate contributions of source 02 are the rapid coefficient-degree differential-independence theorem, its combination with independent coefficient trees to give (15.2), and the explicit omnific tail-resilience theorem and arithmetic consequences. The literature comparison is targeted, not exhaustive. These claims are proved in the source but are not certified as previously unpublished.

The differential statements use a specified diagonal Euler derivation, *not* the exponential-compatible surreal derivation of Berarducci–Mantova [12].

Source 02’s own source audit (shipped as 02-composita-SOURCE_AUDIT.md) lists six candidate contributions: the rapid coefficient-degree theorem over $k((t))(x)$ with arbitrary Laurent coefficients in the relation; private coefficient-block independence over the whole finitely-generated-coefficient envelope and its tree amplification; their combination into the exact degrees $|F|^{\aleph_0}$; the explicit purely infinite omnific family with support of order type ω and the sharpness of its support cardinality; tail resilience of that predefined family under arbitrary set-sized parameter adjunction, including differential parameters; and persistent nonintegrality of the arithmetic join after set adjunction, with the Gaussian versions. It calls them proposed research contributions with written proofs; a targeted source search did not find them elsewhere, and that absence does not establish novelty.

15.4 What the addition settles in this report

[merge]

1. *Research question 3* (an exact description of ordinary composita) is answered in the *separated* case with trivial common core, which is the case of [Theorem C](#), and in the stated cardinal cases. Its first part, the characterization of the compositum, is the finite coefficient-rank denominator test ([theorem 17.2](#)) and, for the explicit copies, [corollary 22.1](#). Its second part, the transcendence degree of the join over the compositum, is answered in the natural set-sized Laurent examples by [theorem 21.1](#): both the algebraic and the Euler-differential transcendence degree of $F((t))$ over $Fk((t))$ are $|F|^{\aleph_0}$ when $|k| \leq \mathfrak{c}$ and $\text{trdeg}(F/k) > 0$. In the class-sized explicit join it is answered in the form the question asks for, by independent families of every set cardinal ([theorem 22.2](#)), a tail that survives every set of parameters ([theorem 23.3](#)), and the absence of set-sized generation ([corollary 23.4](#)); no set cardinal is assigned to the class extension. Open: independent groups that are not separated (neither infinitesimal relative to the other), copies with a prescribed nonzero common core H , infinite algebraic coefficient extensions, and $|k| > \mathfrak{c}$ with small positive $\text{trdeg}(F/k)$. These are Research questions 11, 14, 13 and 12. [merge] Since batch 35 the transcendence degree is also known for explicit non-separated examples ([section 28.4](#)); the description of the compositum there is still open.
2. [Lemma 8.2](#) is shown to be far from sufficient: one transcendental coefficient x already gives a series with all coefficients in $k(x)$ that is differentially transcendental over the compositum ([corollary 19.5](#)).
3. [Theorem 8.3](#) is strengthened for the explicit witness: $S = \omega^{-b}Y_0$, with $\omega^{-b} \in K_1$ a constant of the Euler derivation Δ of [section 22.3](#), so S is differentially transcendental over $\mathcal{C}_{\text{fg}}(K_0/\mathbb{R}; G_1) \supseteq M$ by [theorem 22.2](#) (see the [merge] note after the proof of [Theorem C](#)).
4. No other question of this report is answered. Research question 4 (descent for the arithmetic intersection $\mathcal{A}_k(H)$) is untouched: the nonintegrality result, [theorem 24.2](#), concerns the join of two copies, not their intersection, and computes no normalization.

15.5 Conventions and renamed symbols

[merge] Source 02 uses both exponent conventions of this report: the decreasing large- ω convention $\sum c_g \omega^g$ with reverse-well-ordered support, written $\mathcal{K}_k(G)$ as here, and the increasing small- t convention $F((t^\Gamma))$ with well-ordered support, identified by $t^\gamma = \omega^{-\gamma}$ as in [section 9.1](#). In [sections 16](#)

to 21 the fields are abstract t -series fields over arbitrary coefficient fields, not subfields of **No**. From section 22 on, the fields are the explicit surreal fields of sections 7 and 8, with $k = \mathbb{R}$ (and $\mathbb{C}$ in theorem 24.4).

The following symbols of the source are renamed; everything else is as in the source.

Here	Source 02	Reason
$\Gamma, \Gamma_0, \Gamma'$: the outer ordered group, a subgroup, a set subgroup	H, H_0, H'	H is the prescribed common core of Theorem A, and the source itself uses H in that sense in its Question 4 (Research question 14)
$g > 0$ in Γ with $\lambda(g) = 1$; $g = \omega^{7/2}$ in sections 22 to 24	h	h, h_α are ordered additive embeddings here; this report already writes $g = h_1(1) = \omega^{7/2}$ (proof of Theorem C) and uses g in this role in theorem 8.3
ω^b , the shift multiplier	s	$s_n = \omega^{g_n}$ are the coefficients of (8.3)
$F[L]$, the ring generated by F and L	T	T is also the source's inner support set
$\Theta, G_\Theta, C_\Theta$: inner support set, its support group and localized coefficient field	T, H_T, C_T	as above; G_Θ is a subgroup of G_0
$R_0[R_1]$, the ring generated by $R_0 \cup R_1$; $R_0[R_1][S]$; $R_0[R_1][i]$	$B; B[S]; B_{\mathbb{C}}$	B is a subgroup in lemma 8.1, and $\mathcal{B}_k(H)$ is the bounded-support ring
$j = 0, 1$, index of the two explicit copies	i	$i = \sqrt{-1}$ in section 24; Theorem C writes K_j
$\mathfrak{A}$, a family of infinite subsets of $\mathbb{N}_{>0}$	$\mathcal{A}$	$\mathcal{A}_k(H)$ is the integer ring of section 9
$p_{ij} = P_{Y_{ij}}(\mathbf{f})$, evaluated partials	A_{ij}	A_i are members of $\mathfrak{A}$ in the same proof
$W_i(X, x)$, the leading polynomials in (19.4)	$H_i(T, x)$	H and T as above
$w_i, \mathbf{w}$: tails in the proof of theorem 19.2	$R_i, \mathbf{R}$	R_j are the omnific copies; R is a cut side
ν , the degree in x in the same proof	D	D is a finitely generated coefficient field, as in lemma 8.2
$\gamma_0 + \gamma_1$ with $\gamma_j \in G_j$, and e : generic exponents from section 22 on	$g_0 + g_1, g$	g is $\omega^{7/2}$ above
crk_k , coefficient rank	crk_k , macro <code>\rk</code>	same symbol; only the macro name differs

No normalization is changed by these renamings. In particular $\omega^{7/2} = g$ and $\omega^{15/4} = b$ are the same surreals as in the source, and (15.3) is the source's formula verbatim.

Agreements. $\mathcal{K}_k(G)$, $E = \mathcal{K}_{\mathbb{R}}(G_0 + G_1)$, $M = K_0 K_1$, $K_j = \mathcal{K}_{\mathbb{R}}(G_j)$, $G_j = h_j(\mathbf{No})$, $I_j = (3j, 3j+1)$, ψ, h_j, Φ_j , the convention $t^\gamma = \omega^{-\gamma}$, the finitely generated field D of lemma 8.2, and a series $S_i = \sum a_{b_i(n)} t^{ng}$ of the shape of theorem 8.3 mean the same here as in sections 7 and 8. The source's $g_{\alpha,n}$ and $a_{\alpha,n}$ extend this report's $g_n = \omega^{\psi(n)}$ and $s_n = \omega^{g_n}$: $g_{0,n} = g_n$ and $a_{0,n} = s_n$.

Local letters and tempting false readings.

- In sections 16 to 21, $E = F((t^\Gamma))$, $L = k((t^\Gamma))$ and $M = FL$. This L is *not* the common field $L_k = \mathcal{K}_k(H)$ of Theorem A and not a cut side; this F is a coefficient field, not the witness F of Theorem B. In the surreal instance K_0 plays the role of F , $K_1 \simeq \mathbb{R}((t^{G_1}))$ that of L , and $\mathbb{R}$ that of k .
- $\lambda : \Gamma \rightarrow k$ is an additive *Euler weight*. It is not a coset slice $\lambda_{g,H}$ of (5.1), and not a class functional.
- ∂ and Δ are Euler derivations, d and d_b coefficient derivations. The omnific-preserving report reserves d and D for derivations; here D is always a field, and d is also a total degree, d_n a degree sequence, and $d \in G_j$ a denominator exponent in lemma 24.1; none of these is an Euler derivation. $D_{\mathbb{R}} = \mathbb{Z}$ and $D_{\mathbb{C}} = \mathbb{Z}[i]$ keep their meaning from section 9.
- $\mathcal{J}(y_i : i \in I)$ is a set of jets, not an index set J . I is an index set and I_j an interval, as in section 1.5. The index set B of a coefficient family $(a_b)_{b \in B}$ and the label maps b_i are local to section 20; $b = \omega^{15/4}$ is a positive element of G_1 , as the letter b is in (8.1).
- κ is an infinite cardinal in corollary 20.4 and a set cardinal in theorem 22.2; it is not $\text{cf}(H)$ of section 10. $\mu = |F|$ in theorem 21.1. $\mathfrak{c}$ is $2^{\aleph_0}$, with no assumption on its value as an aleph.
- P is a polynomial relation in sections 19 and 20 and a set of parameters in sections 23 and 24; $M\langle P \rangle$ is the differential field generated by M and all jets of P . $O = E \cap \mathbf{Oz}$. Y_α and Z_α are the omnific witnesses (the letter Y is also a jet variable Y_{ij}). $\tau = \omega^{-g}$ is a monomial, not an exponent map.
- *Tempting false reading*: “ $\text{crk}_k(x) < \infty$ is necessary for $x \in M$.” It is necessary only for $x \in F[L]$; the compositum M is the fraction field of $F[L]$ and contains series of infinite coefficient rank (example 17.4).
- *Tempting false reading*: “ Δ is a surreal derivation.” It is a diagonal Euler derivation of the explicit join, defined through the iterated coordinates, killing K_0 ; it is not the Berarducci–Mantova derivation, not Hardy type, and not compatible with the exponential.
- *Strong* means preserving Hahn summability, as in section 1.5, not convergence in an order topology.

Notation index of the source (its Appendix C), with the names used here. $F((t^\Gamma))$: small- t Hahn field, well-ordered support in Γ . $\mathcal{K}_k(G)$: large- ω Hahn field, reverse-well-ordered support in G . E, L, M in the abstract sections: $F((t^\Gamma))$, $k((t^\Gamma))$ and the ordinary field compositum FL . $F[L]$: the finite coefficient-rank ring, the image of $F \otimes_k L$. $\text{crk}_k(z)$: dimension over k of the span of all coefficients of z . $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$: union of $D((t^\Gamma))$ over finitely generated coefficient fields D/k inside F . ∂ : abstract outer Euler derivation, with weight map $\lambda : \Gamma \rightarrow k$. h_j, Φ_j : inner additive embedding and outer surreal field embedding. G_j, K_j, R_j : exponent-group image, full field image, and omnific image of copy j . E, M from section 22 on: the explicit Hahn join $\mathcal{K}_{\mathbb{R}}(G_0 + G_1)$ and its ordinary compositum $K_0 K_1$. g, b, ω^b : $\omega^{7/2}$, $\omega^{15/4}$, and the monomial that shifts all witness exponents into the positive cone. Δ : Euler derivation extracting the inner $\omega^{7/2}$ coefficient from the dominant exponent coordinate, with the sign in (22.5). $\beta_{\alpha,n}$: the ordinal label $\omega \cdot \alpha + n$, the only place where ordinal arithmetic is used. $g_{\alpha,n}, a_{\alpha,n}$: independent inner exponents and the corresponding outer monomial coefficients. Y_α, Z_α : real omnific witness and its Gaussian multiple $(1+i)Y_\alpha$. $\Theta, G_\Theta, C_\Theta$: set of inner support labels, its full additive support group, and the localized coefficient field. $R_0[R_1], O$: the ring generated by the two omnific copies, and the full

omnific ring inside their Hahn join. $M\langle P \rangle$: differential field generated by M and all jets of the set P .

15.6 Place in the collection

[merge] Source 02 cites only this report from the collection. Checked against the collection at the placement commit [a7a435f](#), the following reports bear on it; none is a dependency of its proofs.

1. *This report.* [Lemma 18.3](#) is a variant of the coset-slice result, [theorem 5.4](#): there two full Hahn fields with the same coefficient field are disjoint over the field of the intersection group; here $F((t^{\Gamma_0}))$ and $k((t^{\Gamma}))$, with different coefficient fields and $\Gamma_0 \subseteq \Gamma$, are disjoint over $k((t^{\Gamma_0}))$, by the same slice argument. The source credits the coset slice as part of the elementary Hahn toolkit. The first clause of [lemma 24.1](#), $\text{Frac}(R_j) = K_j$, is the first clause of [proposition 9.2](#) for the explicit copies, by the monomial clearing of [lemma 9.1](#), as the source says. [Lemma 16.5](#) is [lemma 8.2](#) in the abstract setting. The localization of a parameter set in [lemma 23.1](#) is the support-closure step of [corollary 6.1](#), applied one normal-form level deeper.
2. *Three duals of Hahn vector spaces* (`surcomplex/three-duals-of-hahn-vector-spaces`, prefix `duals:`). Parts (i)–(ii) of [theorem 17.2](#) are the injectivity paragraph before `duals:prop:algebraic` and that proposition, for the k -vector space $V = F$: the scalar extension $K \otimes_k V$, $K = k((t^{\Gamma}))$, embeds in $V_{\Gamma} = F((t^{\Gamma}))$ with image the vectors of finite total coefficient rank. Source 02 does not cite that report and presents the tensor fact as standard. Part (iii), the fraction-field test, is new but immediate from (ii).
3. *Tail spans and differential transcendence* (`surreal/tail-spans-and-differential-transcendence`, prefix `tail:`). `tail:thm:surreal` uses the same binary-prefix almost disjoint family as [corollary 19.4](#), private indices and a Vandermonde argument to give $\mathfrak{c}$ differentially independent series. The base $(\mathbb{Q}((t^{\mathbb{Q}})))$, the derivation (Berarducci–Mantova, via Euler jets) and the mechanism (the tail-span theorem `tail:thm:tail` for multiquadratic coefficients) differ from those of [theorems 19.2](#) and [20.2](#), whose base is a compositum or a finite-generation envelope and whose tool is coefficient degree or a coefficient derivation. Source 02 does not cite it; the two are related, not duplicates.
4. *Bounded-support transcendence* (`surreal/transcendence-over-bounded-support`, prefix `bst:`). `bst:thm:independent` obtains $2^{\text{cf}(G)}$ independent series over the bounded-support fraction field by coefficient codes realized at cofinally many indices, a private-index device over a different base, with a support-gap estimate instead of a derivation. Source 02 does not cite it.
5. *Dilation rigidity* (`surcomplex/autonomous-dilation-relations`, prefix `adr:`). `adr:sr:thm:tail` localizes a set of surcomplex parameters in the Hahn field of the real span of their supports and finds a tail of an explicit ordinal family whose dilation orbits are independent over that whole field; `adr:sr:thm:kappa` gives set-sized families over a prescribed set base. [Theorem 23.3](#) has the same shape, with Euler jets instead of dilation orbits and localization at the inner normal-form level.
6. *Omnific-preserving automorphisms* (`opa:`). The explicit copies are the maps $M_{1,\iota_{\sigma_j}}$ of that report’s Part III ([section 1.6](#)). Research questions [17](#) and [18](#) lie inside `opa:as:q:exp`.

No `isc:cp:` label has a Lean mapping in `FORMALIZATION.md`.

16 Conventions and foundational lemmas of the addition

16.1 Two support conventions

For a field F and an ordered abelian group Γ , the Hahn field $F((t^\Gamma))$ consists of formal sums

$$x = \sum_{\gamma \in \Gamma} x_\gamma t^\gamma$$

whose nonzero coefficient support is a well-ordered set. All supports are sets. Addition is coefficient-wise. In a product, the coefficient at γ is the finite sum of $x_\alpha y_\beta$ over pairs with $\alpha + \beta = \gamma$.

Surreal normal forms instead use the large- ω convention $x = \sum_g c_g \omega^g$, with reverse-well-ordered support: every nonempty subset has a greatest element. As in [section 2](#), $\mathcal{K}_k(G)$ is the full field of these normal forms with coefficients in k and exponents in an additive subgroup $G \subseteq \mathbf{No}$. Whenever the conventions change, the identification is

$$t^\gamma = \omega^{-\gamma}, \tag{16.1}$$

as in [section 9.1](#). The reversal in (16.1) is essential in the final omnific construction.

For set-sized groups, ordinary Hahn field algebra suffices. For proper-class exponent groups, each finite calculation occurs in the subgroup generated by a set of supports. The inverse of a nonzero series can likewise be constructed in a set-sized Hahn field containing its support. Thus the formulas define class fields without allowing class-sized sums. The class assertions are formulated in NBG with global choice, with ZFC for sets, as in [section 2](#). The displayed class families and derivations are given by formulas; no class-state recursion, class transcendence basis, or class satisfaction predicate is used.

16.2 Euler derivations and differential independence

Definition 16.1. Let $k \subseteq F$ have characteristic zero and let $\lambda : \Gamma \rightarrow (k, +)$ be additive. The associated outer Euler derivation is

$$\partial \left(\sum_{\gamma} a_\gamma t^\gamma \right) = \sum_{\gamma} \lambda(\gamma) a_\gamma t^\gamma. \tag{16.2}$$

It kills F . When $\Gamma = \mathbb{Z}$ and $\lambda(n) = n$, it is $t d/dt$.

Lemma 16.2. *Formula (16.2) is a derivation of $F((t^\Gamma))$. It preserves Hahn-summable families and their sums. If $d : F \rightarrow F$ is a k -derivation, its coefficientwise extension to $F((t^\Gamma))$ is also a derivation and commutes with ∂ .*

Proof. Both maps only delete terms or change coefficients, so the support condition is preserved. For a product coefficient, finiteness of the convolution and $\lambda(\alpha + \beta) = \lambda(\alpha) + \lambda(\beta)$ give the Leibniz identity. The same finite calculation works for d . A Hahn-summable family has a well-ordered union of supports and only finitely many nonzero contributions at each exponent; neither property is destroyed. Finally $d(\lambda(\gamma)) = 0$ because $\lambda(\gamma) \in k$, proving commutation coefficient by coefficient. $\square$

Definition 16.3. A family $(y_i)_{i \in I}$ in a differential field extension E/M is *differentially algebraically independent over M* if the family

$$\mathcal{J}(y_i : i \in I) = \{\partial^j y_i : i \in I, j \geq 0\}$$

is algebraically independent over M . Every relation and every collection of variables in a polynomial are finite. We write $M\langle P \rangle$ for the field generated by M and all $\partial^j p$, $p \in P$, $j \geq 0$.

For set fields, differential transcendence degree has its usual meaning. All lower bounds below are proved by exhibiting independent families of the stated cardinality. For class fields, only the finite-relation definition and set-indexed subfamilies are used; no class transcendence basis is postulated, and no cardinal equal to the size of **No**.

16.3 The coefficient-field envelope

Throughout the set-sized theory, put

$$E = F((t^\Gamma)), \quad L = k((t^\Gamma)), \quad M = FL. \quad (16.3)$$

The compositum is taken in E and is a field. In contrast, the ring $F[L]$ generated by F and L consists of finite sums of products.

Definition 16.4. Define the finite-generation coefficient envelope by

$$\mathcal{C}_{\text{fg}}(F/k; \Gamma) = \bigcup_{\substack{k \subseteq D \subseteq F \\ D/k \text{ finitely generated}}} D((t^\Gamma)). \quad (16.4)$$

Here “finitely generated” means as a field, not as a ring.

Lemma 16.5. *The union in (16.4) is a field containing M and is stable under every Euler derivation (16.2). Every finite subset of M is contained in one $D((t^\Gamma))$ with D/k finitely generated.*

Proof. The compositum of two finitely generated coefficient fields is again finitely generated, so the union is directed. Each constituent is a Hahn field. A finite field expression in elements of F and L uses only finitely many elements $a_1, \dots, a_r$ of F , and therefore belongs to $k(a_1, \dots, a_r)((t^\Gamma))$. The same argument handles finitely many expressions simultaneously. Formula (16.2) introduces only scalars from k into coefficients. $\square$

The last lemma is the finite coefficient-field control of lemma 8.2, in the abstract setting. The later constructions show both strictness of its necessary condition and independence over the entire envelope.

17 An exact coefficient-rank test for the compositum

Definition 17.1. For $x = \sum x_\gamma t^\gamma \in E$, define

$$\text{crk}_k(x) = \dim_k \text{span}_k \{x_\gamma : \gamma \in \Gamma\}.$$

In particular, $\text{crk}_k(0) = 0$. Let $F[L] \subseteq E$ be the ring generated by F and L .

Theorem 17.2 (Finite rank and finite-rank denominators). *For arbitrary fields $k \subseteq F$ and any ordered abelian group Γ :*

- (i) *The natural multiplication map $F \otimes_k L \rightarrow E$ is injective, with image $F[L]$.*
- (ii) $F[L] = \{x \in E : \text{crk}_k(x) < \infty\}$.

- (iii) $M = \text{Frac}(F[L])$, and an element $x \in E$ belongs to M exactly when there is $q \in E \setminus \{0\}$ for which both $\text{crk}_k(q)$ and $\text{crk}_k(qx)$ are finite.

Proof. A finite tensor can be written as $\sum_{i=1}^r b_i \otimes \ell_i$ with the $b_i \in F$ linearly independent over k . If its image vanishes, then comparison at each exponent γ gives

$$\sum_{i=1}^r b_i (\ell_i)_\gamma = 0.$$

All $(\ell_i)_\gamma$ belong to k , so each is zero. Thus every $\ell_i = 0$, proving injectivity. The image is the finite sums of products of elements of F and L , namely $F[L]$.

For $x = \sum_{i=1}^r b_i \ell_i$, every coefficient of x lies in the finite-dimensional space spanned by the b_i . Conversely, suppose the coefficient span has basis $b_1, \dots, b_r$. Write uniquely

$$x_\gamma = \sum_{i=1}^r b_i c_{i,\gamma}, \quad c_{i,\gamma} \in k.$$

For every i , the support of $\ell_i = \sum_\gamma c_{i,\gamma} t^\gamma$ is contained in $\text{supp}(x)$, hence is well ordered. Therefore $\ell_i \in L$ and $x = \sum b_i \ell_i \in F[L]$.

Since $F[L]$ is a subring of a field, its fraction field inside E is precisely the smallest field containing F and L , which is M . Finally $x = p/q$ with $p, q \in F[L]$, $q \neq 0$, is equivalent to $q, qx \in F[L]$ for such a q . Part (ii) gives the asserted test. $\square$

[merge] Parts (i) and (ii) are, for the k -vector space $V = F$, the injectivity paragraph before `duals:prop:algebraic` of the three-duals report and that proposition ([section 15.6](#)); the source does not cite it and presents the tensor fact as standard.

Remark 17.3 (What the criterion does and does not do). The theorem describes the subfield exactly in terms of its coefficients, rather than merely as an unspecified compositum. Nevertheless, the existential search over q is an infinite-input problem. No algorithm deciding arbitrary Hahn-series membership follows. The tensor argument is a standard linear-disjointness mechanism; neither a new tensor-product theorem nor a new general theory of rational series is claimed. The comparison with recurrence criteria in [10] is conceptual, not a claim that the two formulations have identical inputs.

Example 17.4 (Infinite rank inside the compositum). Let $u \in F$ be transcendental over k and let $g \in \Gamma$ be positive. The geometric series

$$z = \frac{1}{1 - ut^g} = \sum_{n \geq 0} u^n t^{ng}$$

has infinite coefficient rank, because the powers of u are linearly independent over k . Yet $z \in M$: the multiplier $q = 1 - ut^g$ has rank two and $qz = 1$ has rank one. The support $\{ng : n \geq 0\}$ is well ordered in every ordered abelian group. No cofinality assumption on g is needed.

Corollary 17.5 (Finite coefficient extensions). *If $[F : k] < \infty$, then $E = F[L] = M$.*

Proof. Every coefficient span is a subspace of the finite-dimensional k -space F . Apply [theorem 17.2](#). $\square$

For finite ranks one also has

$$\text{crk}_k(x + y) \leq \text{crk}_k(x) + \text{crk}_k(y), \quad \text{crk}_k(xy) \leq \text{crk}_k(x) \text{crk}_k(y).$$

Indeed a coefficient of the product lies in the span of pairwise products of basis elements for the two coefficient spaces. Infinite rank therefore obstructs membership in $F[L]$, but not in its fraction field.

18 Linear disjointness under two enlargements

The next lemmas prevent an independence proof over a small coefficient field or a cyclic exponent group from being lost when the ambient Hahn field is enlarged. The arguments are included instead of silently treating base extension as harmless. Linear disjointness has its usual definition via injectivity of the tensor multiplication map [13], as in section 2.

Lemma 18.1 (Coefficient extension). *If $F_0 \subseteq F$, then F and $F_0((t^\Gamma))$ are linearly disjoint over F_0 inside $F((t^\Gamma))$.*

Proof. A finite relation with coefficients in $F_0((t^\Gamma))$ among elements of F linearly independent over F_0 vanishes coefficientwise only when every coefficient series is zero. This is the first argument of theorem 17.2 with $k = F_0$. $\square$

Lemma 18.2 (Intermediate base change). *Suppose fields A and B are linearly disjoint over K in an ambient field, and $K \subseteq A_0 \subseteq A$. Then A and A_0B are linearly disjoint over A_0 .*

Proof. Let $a_1, \dots, a_m \in A$ be linearly independent over A_0 and suppose $\sum_i a_i c_i = 0$ with $c_i \in A_0B$. Clear a common nonzero denominator from the ring generated by A_0 and B . Each resulting coefficient is a finite sum of products. Choose a K -linearly independent list $b_1, \dots, b_r \in B$ spanning all the finitely many B -entries used. The relation becomes

$$\sum_{j=1}^r \left(\sum_{i=1}^m a_i a_{ij} \right) b_j = 0, \quad a_{ij} \in A_0.$$

Linear disjointness over K makes the b_j linearly independent over A . Hence every inner sum vanishes. Independence of the a_i over A_0 gives all $a_{ij} = 0$, so all the original c_i are zero. $\square$

Lemma 18.3 (Coset slices). *Let $\Gamma_0 \subseteq \Gamma$ be an additive subgroup. Inside $F((t^\Gamma))$, the fields*

$$E_0 = F((t^{\Gamma_0})), \quad L = k((t^\Gamma))$$

are linearly disjoint over $L_0 = k((t^{\Gamma_0}))$. No convexity assumption on Γ_0 is required.

Proof. Suppose $u_1, \dots, u_m \in E_0$ are linearly independent over L_0 and $\sum_i u_i v_i = 0$, $v_i \in L$. For a coset C of Γ_0 meeting one of the supports of the v_i , choose a representative $c \in C$. Let $v_{i,C}$ be the sum of the terms of v_i supported on C , multiplied by t^{-c} . Its support is a translate of a subset of a well-ordered set, and is contained in Γ_0 , so $v_{i,C} \in L_0$.

Multiplication by a series supported in Γ_0 does not change the coset of an exponent. Extracting the C -slice of the original relation and translating by $-c$ therefore gives

$$\sum_i u_i v_{i,C} = 0.$$

Independence over L_0 yields $v_{i,C} = 0$ for all i . Every support exponent belongs to a coset that was considered; hence every $v_i = 0$. Only a set of cosets is involved, even in the class-supported setting. $\square$

[merge] Up to the change of convention (16.1), $v_{i,C}$ is a normalized coset slice as in (5.1), and the lemma is a variant of theorem 5.4 with different coefficient fields on the two sides (section 15.6).

Corollary 18.4 (Independence survives both enlargements). *Let $x \in F$ be transcendental over k , let $g > 0$ in Γ , and let $\lambda : \Gamma \rightarrow k$ satisfy $\lambda(g) = 1$. If a family in $k(x)((t))$ is differentially algebraically independent over $k((t))(x)$ for $t d/dt$, then replacing t^n by t^{ng} gives a family differentially algebraically independent over $Fk((t^\Gamma))$ in $F((t^\Gamma))$.*

Proof. First apply lemma 18.1 to $k(x) \subseteq F$ and the cyclic exponent group. Then apply lemma 18.2 with $A = k(x)((t))$ and $A_0 = k((t))(x)$: the field A is linearly disjoint from $Fk((t))$ over A_0 . Distinct monomials in any finite collection of jets, which were linearly independent over A_0 , remain so over $Fk((t))$.

Next identify this cyclic Hahn field with $F((t^{\mathbb{Z}g}))$. Apply lemma 18.3, followed by lemma 18.2 with the intermediate field $Fk((t^{\mathbb{Z}g}))$. This preserves independence after adjoining $k((t^\Gamma))$. Finally $\partial(t^{ng}) = nt^{ng}$, so the transported elements have exactly the transported jets. $\square$

19 Differential independence from one coefficient variable

The next theorem is the key to the small-transcendence-degree case. The exponents in the outer variable need not be sparse. Instead, the degrees in a single coefficient variable grow so fast that a late coefficient cannot be canceled by any fixed differential polynomial.

Definition 19.1. A family $\mathfrak{A}$ of infinite subsets of $\mathbb{N}_{>0}$ has the *finite private-index property* if, for every finite subfamily $A_1, \dots, A_r$ of distinct members and every i , the set

$$A_i \setminus \bigcup_{j \neq i} A_j$$

is infinite. In particular, every pairwise almost disjoint family of infinite subsets has this property.

Theorem 19.2 (Rapid coefficient-degree independence). *Let k have characteristic zero, let x be transcendental over k , and let $(d_n)_{n \geq 1}$ be a strictly increasing sequence of positive integers such that*

$$\frac{d_n}{d_{n-1}} \longrightarrow +\infty. \quad (19.1)$$

For a family $\mathfrak{A}$ with the finite private-index property, put

$$f_A = \sum_{n \in A} x^{d_n} t^n \in k(x)((t)). \quad (19.2)$$

Then $(f_A)_{A \in \mathfrak{A}}$ is differentially algebraically independent over $k((t))(x)$ for $\partial = t d/dt$ and $\partial x = 0$.

Proof. Suppose there is a relation in finitely many jets of distinct $f_{A_1}, \dots, f_{A_r}$. Among all nonzero polynomial relations over $k((t))(x)$ in finite lists of these jets, choose one of least total jet degree d . Write it as

$$P((\partial^j f_{A_i})_{1 \leq i \leq r, 0 \leq j \leq m}) = 0,$$

allowing unused variables. A degree-zero nonzero relation is impossible, so $d \geq 1$.

Clear the denominators in x of the finitely many coefficients of P , then multiply by a power of t large enough to remove their negative outer exponents. We may therefore assume

$$P \in k[[t]][x][Y_{ij} : 1 \leq i \leq r, 0 \leq j \leq m]. \quad (19.3)$$

Let ν be its degree in x . These operations do not change total jet degree or minimality. We use $\mathbf{f}$ to denote the tuple of all the indicated jets.

For each variable set

$$p_{ij} = P_{Y_{ij}}(\mathbf{f}) \in k[x][[t]].$$

If the formal polynomial $P_{Y_{ij}}$ is nonzero, its total jet degree is smaller than d , so minimality forces $p_{ij} \neq 0$. At least one formal partial is nonzero, because the characteristic is zero and $d \geq 1$. Consequently

$$\ell = \min_{p_{ij} \neq 0} \text{ord}_t(p_{ij})$$

is a nonnegative integer. Define polynomials in a new indeterminate X by

$$W_i(X, x) = \sum_{j=0}^m [t^\ell] p_{ij} X^j \in k[x][X]. \quad (19.4)$$

At least one W_i is nonzero: its coefficients occupy distinct powers of X , and at least one p_{ij} attains the valuation ℓ . Fix such an index i_* .

For an integer n , truncate every original series strictly before t^n :

$$v_i = \sum_{\substack{a \in A_i \\ a < n}} x^{d_a} t^a, \quad w_i = f_{A_i} - v_i.$$

Every jet of w_i has t -valuation at least n . Let $\mathbf{v}$ and $\mathbf{w}$ be the corresponding tuples of jets. Evaluating any polynomial from (19.3) at $\mathbf{f}$ or $\mathbf{v}$ changes it only by a multiple of t^n . Therefore, when $n > \ell$, the partial evaluations $P_{Y_{ij}}(\mathbf{v})$ have zero coefficients below ℓ and the same coefficient at ℓ as p_{ij} .

Choose n in the infinite private-index set

$$A_{i_*} \setminus \bigcup_{i \neq i_*} A_i.$$

We may require simultaneously

$$n > \max\{\ell, 1\}, \quad W_{i_*}(n, x) \neq 0, \quad d_n > \nu + d d_{n-1}. \quad (19.5)$$

The middle condition excludes only finitely many integers, since a nonzero polynomial in X over the field $k(x)$ has finitely many roots and characteristic zero gives distinct integer elements. The last condition holds eventually by (19.1).

The polynomial Taylor expansion about $\mathbf{v}$ is

$$P(\mathbf{v} + \mathbf{w}) = P(\mathbf{v}) + \sum_{i,j} P_{Y_{ij}}(\mathbf{v}) \partial^j w_i + \text{terms of tail degree at least two}. \quad (19.6)$$

Every omitted term is divisible by t^{2n} . Since $n + \ell < 2n$, these terms do not contribute at exponent $n + \ell$. In the linear sum, a tail coefficient after exponent n could contribute only by multiplying a partial coefficient before ℓ , and those coefficients vanish. At exponent n , only w_{i_*} is nonzero, and

$$[t^n] \partial^j w_{i_*} = n^j x^{d_n}.$$

Taking $[t^{n+\ell}]$ in (19.6), using $P(\mathbf{f}) = 0$, gives the exact polynomial identity

$$0 = [t^{n+\ell}]P(\mathbf{v}) + x^{d_n}W_{i_*}(n, x). \quad (19.7)$$

Every coefficient of every truncated jet has degree in x at most d_{n-1} . Thus the first summand in (19.7) has degree at most $\nu + d d_{n-1}$. The second is nonzero and has degree at least d_n . The final inequality in (19.5) makes cancellation impossible. This contradiction proves the theorem. $\square$

Remark 19.3 (Why arbitrary Laurent coefficients are allowed). The coefficients of P can be arbitrary elements of $k((t))(x)$, not just rational functions of t and x over k . Clearing denominators makes their degree in x uniformly finite, while their t -series can remain infinite. The proof uses only this finite x -degree and a common lower bound on outer exponents. It does not impose a recurrence on those coefficient series.

Corollary 19.4 (A continuum family in one coefficient field). *For every characteristic-zero field k and transcendental x , the field $k(x)((t))$ contains $\mathfrak{c}$ elements differentially algebraically independent over $k((t))(x)$, all with coefficients in $k(x)$.*

Proof. Take $d_n = n!$. Encode the nonempty finite binary strings injectively by positive integers. For each infinite binary sequence η , let A_η consist of the labels of its finite nonempty prefixes. Distinct branches have only finitely many common prefixes, so their sets are pairwise almost disjoint. There are $\mathfrak{c}$ branches. Apply [theorem 19.2](#). $\square$

[merge] The binary-prefix family is the one of `tail:eq:Aalpha` in the tail-spans report, where it serves `tail:thm:surreal` over a different base and with a different derivation ([section 15.6](#)).

Corollary 19.5 (Failure of the finite coefficient-field criterion). *Suppose F/k has positive transcendence degree and there are $g > 0$ in Γ and additive $\lambda : \Gamma \rightarrow k$ with $\lambda(g) = 1$. Then $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$ contains an element differentially transcendental over $M = Fk((t^\Gamma))$. In particular, containment of all coefficients in a single finitely generated field does not imply membership in M , or even algebraicity over M .*

Proof. Choose a transcendental $x \in F$ and use

$$f = \sum_{n \geq 1} x^{n!} t^{ng}.$$

The singleton family $\{\mathbb{N}_{>0}\}$ satisfies the private-index condition. Apply [theorem 19.2](#) and [corollary 18.4](#). The series belongs to $k(x)((t^\Gamma)) \subseteq \mathcal{C}_{\text{fg}}(F/k; \Gamma)$. $\square$

Example 19.6 (The first terms and the mechanism). The last witness starts as

$$xt^g + x^2 t^{2g} + x^6 t^{3g} + x^{24} t^{4g} + x^{120} t^{5g} + \dots$$

The outer support here is the entire positive cyclic progression, not a factorially sparse support. A polynomial differential relation of total jet degree d and coefficient degree ν sees only degree at most $\nu + d(n-1)!$ in the truncation before t^{ng} . A surviving linear response to the next coefficient has degree at least $n!$. This is the obstruction. Merely pointing to gaps in the outer support would not prove this example.

[merge] The source's table in section 25 applies corollary 19.5 to the explicit join of section 8, with $F = K_0$, $k = \mathbb{R}$, $\Gamma = G_1$ and λ as in section 22.3; the lemmas of section 18 use only finite relations, so the class-sized F causes no difficulty. The example is then a series with every coefficient in the one-generator field $\mathbb{R}(x)$, $x \in K_0$ transcendental over $\mathbb{R}$. Lemma 8.2 places every finite subset of M in some $\mathcal{K}_D(G_1)$ with D finitely generated; the example shows that the converse fails for a single generator.

20 Independent coefficient blocks and tree amplification

The preceding theorem uses one coefficient variable. The next results exploit many algebraically independent coefficients and obtain independence over a larger base than the compositum itself.

20.1 Coefficient derivations with finite avoidance

Lemma 20.1 (Avoiding a finitely generated coefficient field). *Let F/k have characteristic zero, let $(a_b)_{b \in B} \subseteq F$ be algebraically independent over k , and let $D \subseteq F$ be finitely generated over k . There is a finite set $B_0 \subseteq B$ such that, for every $b \in B \setminus B_0$, a k -derivation $d_b : F \rightarrow F$ exists with*

$$d_b(a_b) = 1, \quad d_b(a_c) = 0 \ (c \neq b), \quad d_b(D) = 0.$$

Proof. Extend the given independent set to a transcendence basis $\mathcal{T}$ of the set field F/k . Each member of a finite generating list for D/k is algebraic over $k(\mathcal{T}_0)$ for some finite $\mathcal{T}_0 \subseteq \mathcal{T}$; take one such finite set for the whole list. Let B_0 consist of the b with $a_b \in \mathcal{T}_0$.

On the rational function field $k(\mathcal{T})$ define the coordinate derivation that sends a_b to 1 and all other basis elements to 0. It extends uniquely over the algebraic extension $F/k(\mathcal{T})$, because this extension is separable. Explicitly, for an algebraic element z with minimal polynomial $p(X)$, differentiation of $p(z) = 0$ forces

$$d_b(z) = -\frac{(d_b p)(z)}{p'(z)}. \quad (20.1)$$

The denominator is nonzero. This formula gives the usual compatible extension through finite algebraic subextensions and hence through their union. If $b \notin B_0$, the derivation kills $k(\mathcal{T}_0)$, so (20.1) makes it kill each of the chosen generators of D . It therefore kills D . $\square$

20.2 The block theorem

Theorem 20.2 (Private coefficient blocks). *Let F/k have characteristic zero. Let $(a_b)_{b \in B} \subseteq F$ be algebraically independent over k , and choose $g > 0$ in Γ and additive $\lambda : \Gamma \rightarrow k$ with $\lambda(g) = 1$. For each $i \in I$, let*

$$b_i : \mathbb{N}_{>0} \longrightarrow B$$

be injective. Assume that for every finite set of distinct indices $i_1, \dots, i_r$ and every $i = i_q$, there are infinitely many n such that the label $b_i(n)$ occurs in no other sequence b_{i_p} , at any position. Put

$$S_i = \sum_{n \geq 1} a_{b_i(n)} t^{ng}. \quad (20.2)$$

Then $(S_i)_{i \in I}$ is differentially algebraically independent over $C_{\text{fg}}(F/k; \Gamma)$ for the Euler derivation associated to λ .

Proof. Every displayed series has well-ordered support contained in $\{ng : n \geq 1\}$. Suppose a nonzero polynomial relation exists among finitely many jets of finitely many distinct S_i , with coefficients in $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$. Choose a relation P of least total jet degree. Its coefficients all lie in one $D((t^\Gamma))$ with D/k finitely generated. The total degree is positive.

Fix one index i involved in the relation. By the private-label hypothesis, there are infinitely many positions n whose coefficient label occurs only in this series among those under consideration. After excluding the finite exceptional set of lemma 20.1, infinitely many remain. For such a position, extend $d_{b_i(n)}$ coefficientwise to the Hahn field. By lemma 16.2, it kills every coefficient of P , commutes with ∂ , and satisfies

$$d_{b_i(n)}(\partial^j S_i) = n^j t^{ng}, \quad d_{b_i(n)}(\partial^j S_{i'}) = 0 \quad (i' \neq i).$$

Differentiating $P = 0$ by this coefficient derivation and dividing by the nonzero monomial t^{ng} gives

$$\sum_{j=0}^m n^j P_{Y_{ij}}((\partial^a S_{i'})_{i',a}) = 0. \quad (20.3)$$

Here m bounds all jet orders used. The coefficients in (20.3) do not depend on n . Since it holds for infinitely many distinct integers in a characteristic-zero field, the polynomial in n is identically zero. Thus every evaluated partial $P_{Y_{ij}}$ is zero. Repeat for each index i in the relation.

A nonconstant polynomial in characteristic zero has some nonzero formal partial derivative. That partial has smaller total jet degree than P , yet its evaluation vanishes. This contradicts minimality. Therefore no relation exists. $\square$

Remark 20.3 (A finite Vandermonde version). For a relation involving jets only through order m , it is enough to use $m + 1$ distinct private positions outside the finite exceptional set. The matrix $(n_a^j)_{0 \leq a, j \leq m}$ has determinant $\prod_{a < b} (n_b - n_a) \neq 0$. The polynomial-root argument in the proof is simply a convenient way to avoid choosing a particular finite list.

Corollary 20.4 (Tree amplification). *If $\text{trdeg}(F/k) \geq \kappa$ for an infinite cardinal κ , and Γ, g, λ satisfy the preceding hypotheses, then $F((t^\Gamma))$ contains a family of cardinality $\kappa^{\aleph_0}$ differentially algebraically independent over $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$.*

Proof. The set of nonempty finite sequences in κ has cardinality κ . Choose an algebraically independent coefficient a_s for every such finite sequence s . For each branch $\eta \in \kappa^{\mathbb{N}_{>0}}$, use

$$S_\eta = \sum_{n \geq 1} a_{\eta|n} t^{ng}.$$

Two distinct branches share only finitely many prefix labels. Relative to any finite collection of other branches, every sufficiently long prefix is private. Apply theorem 20.2. $\square$

20.3 Local use for class coefficient fields

Proposition 20.5 (Set-local form). *The private coefficient-block argument remains valid when F or Γ is a class, provided that k is a set field, every series has set support, and algebraic independence of the coefficient family means independence of every finite subfamily. The conclusion concerns every finite polynomial relation. No class transcendence basis is required.*

Proof. Fix a proposed relation. It involves finitely many S_i , so the union of their coefficient labels is a set. Its finitely many coefficient series belong to $D((t^\Gamma))$ for a single set field D finitely generated over k . Form the set field F' generated by D , k , and all the coefficients in the finitely many chosen S_i . The subgroup Γ' generated by g and all supports of the relation coefficients is a set. Restrict λ to Γ' .

Every expression in the relation lies in $F'((t^{\Gamma'}))$. The algebraically independent coefficient family used there is a set, so [lemma 20.1](#) applies to F'/k in ordinary set theory. All differentiations, finite convolution identities, and the contradiction now occur in this set-sized field. This proves the class statement one finite relation at a time. $\square$

Corollary 20.6 (Three distinct closure levels). *Assume $\text{trdeg}(F/k)$ is infinite and that Γ admits $g > 0$ and λ with $\lambda(g) = 1$. Then*

$$F[L] \subsetneq M \subsetneq \mathcal{C}_{\text{fg}}(F/k; \Gamma) \subsetneq E. \quad (20.4)$$

Moreover, the second strict inclusion has a differential-transcendental witness over M , and the third has one over $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$.

Proof. The first strictness is [example 17.4](#), the second is [corollary 19.5](#), and the third follows from a single sequence of algebraically independent coefficients in [theorem 20.2](#). $\square$

21 Exact transcendence degrees for Laurent-series composita

Theorem 21.1 (Maximal Laurent transcendence). *Let $k \subseteq F$ be set fields of characteristic zero. Suppose*

$$|k| \leq \mathfrak{c} = 2^{\aleph_0}, \quad \text{trdeg}(F/k) > 0.$$

Inside $E = F((t))$, put $M = Fk((t))$ and use $\partial = t d/dt$, acting trivially on F . Then

$$\boxed{\text{dtrdeg}(E/M) = \text{trdeg}(E/M) = |F|^{\aleph_0}.} \quad (21.1)$$

No continuum hypothesis or other cardinal-arithmetic hypothesis is imposed.

Proof. Write $\mu = |F|$, an infinite cardinal. The set of Laurent series has cardinality $\mu^{\aleph_0}$: a series is specified by an integer lower bound and a sequence of coefficients, and arbitrary coefficient sequences already occur in $F[[t]]$. Thus

$$\text{dtrdeg}(E/M) \leq \text{trdeg}(E/M) \leq \mu^{\aleph_0}.$$

The first inequality also follows directly from the definition: the zeroth jets of a differentially independent family are algebraically independent.

If $\mu \leq \mathfrak{c}$, then

$$\mathfrak{c} \leq \mu^{\aleph_0} \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c},$$

so $\mu^{\aleph_0} = \mathfrak{c}$. Choose a transcendental $x \in F$. The continuum family from [corollary 19.4](#) remains differentially independent over M by [corollary 18.4](#), with $\Gamma = \mathbb{Z}$ and $g = 1$. This reaches the upper bound.

If $\mu > \mathfrak{c}$, let $\mathcal{T}$ be a transcendence basis for F/k . A field algebraic over $k(\mathcal{T})$ has cardinality

$$\max\{|k|, |\mathcal{T}|, \aleph_0\}.$$

This follows by counting polynomials and their finitely many roots. Since $|k| \leq \mathfrak{c} < \mu$, it follows that $|\mathcal{T}| = \mu$. The tree family from [corollary 20.4](#), with $\kappa = \mu$, has size $\mu^{\aleph_0}$ and is differentially independent even over $\mathcal{C}_{\text{fg}}(F/k; \mathbb{Z})$, hence over M . Again it reaches the upper bound. The two cases prove both equalities. $\square$

Corollary 21.2. *For $k = \mathbb{Q}, \mathbb{R}$, or $\mathbb{C}$ and x transcendental over k ,*

$$\text{dtrdeg}(k(x)((t))/k((t))(x)) = \text{trdeg}(k(x)((t))/k((t))(x)) = \mathfrak{c}.$$

More generally, [theorem 21.1](#) applies to every positive-transcendence-degree extension of any of these three fields, irrespective of the cardinality of F .

21.1 Boundary cases that must not be suppressed

If F/k is finite, [corollary 17.5](#) gives $E = M$, so both transcendence degrees are zero. An infinite algebraic coefficient extension is not covered by either the finite-extension conclusion or [theorem 21.1](#); the formula is not silently extended to that case.

If $\Gamma = 0$, there is no outer series variable and $E = M = F$. If Γ is nonzero but no specified map $\lambda : \Gamma \rightarrow k$ takes a chosen positive element to 1, the particular differential statements here have not been formulated for it. The purely algebraic finite-rank test still applies. For divisible groups and characteristic-zero coefficient fields, suitable maps can often be chosen by linear algebra over $\mathbb{Q}$, but the theorems keep the needed map explicit.

Characteristic zero is substantive. In characteristic $p > 0$, the standard Euler derivation on $F((t))$ satisfies

$$\partial^p = \partial,$$

because $n^p = n$ in the prime field. Every element then satisfies the differential polynomial $Y^{(p)} - Y^{(1)} = 0$. The differential-independence conclusion is therefore impossible for that derivation in positive characteristic. This does not settle the corresponding ordinary algebraic-transcendence problem.

The hypothesis $|k| \leq \mathfrak{c}$ enters only in the cardinal upper-bound matching when the coefficient transcendence degree is small. Both independence constructions themselves work over every characteristic-zero base field. For larger k , they still give their explicit lower bounds, but it is not claimed that the bounds always equal $|F|^{\aleph_0}$.

22 Separated surreal copies and explicit omnific witnesses

The set-local results now apply to actual surreal normal forms. The source reproduces the explicit two-copy construction of [section 7](#) to make its later proofs independent of any unreviewed general class-embedding theorem, in particular of [theorem 3.4](#). Only normal-form algebra and the elementary order comparison are needed.

22.1 The inner and outer levels, printed once

[merge] The source's Subsections 8.1 and 8.2 re-derive, with the same proofs, material printed in [sections 7](#) and [8](#); it is printed once, there, and the source is credited as an independent re-derivation. The correspondence is: the order isomorphism ψ and its surjectivity proof are [lemma 7.1](#); the

intervals $I_j = (3j, 3j + 1)$, the inner maps h_j of (7.1), their non-unitality (7.2) and the description of $G_j = h_j(\mathbf{No})$ as all normal forms supported in I_j are in section 7 (with $\alpha = j \in \{0, 1\}$); the outer lifts Φ_j of (7.3) with images $K_j = \mathcal{K}_{\mathbb{R}}(G_j)$ are theorems 4.1 and 7.2; the scale separation is (8.1); and the re-expansion is lemma 8.1, whose proof the source repeats. The source adds the following, each immediate from those statements.

Conway's omnific integers are

$$\mathbf{Oz} = \left\{ \sum_{e>0} r_e \omega^e + n : r_e \in \mathbb{R}, n \in \mathbb{Z}, \text{ supp reverse well ordered} \right\}. \quad (22.1)$$

A *purely infinite* omnific integer has constant term zero; the term does not require every coefficient to be positive. Since h_j preserves the sign and zero of an exponent, Φ_j preserves and reflects the condition in (22.1) (proposition 4.2), so

$$R_j := \Phi_j(\mathbf{Oz}) = K_j \cap \mathbf{Oz}, \quad R_0 \cap R_1 = \mathbb{Z}, \quad (22.2)$$

as in theorem 7.2, and Φ_j preserves and reflects purely infinite elements. None of this says that Φ_j commutes with the unary omega-map: $\Phi_0(\omega) = \omega^{\omega^{1/2}} \neq \omega^{\Phi_0(1)} = \omega$ (warning 4.5). By (8.1), $G_0 + G_1$ is a direct sum as a group, ordered lexicographically with G_1 dominant. With

$$E = \mathcal{K}_{\mathbb{R}}(G_0 + G_1), \quad M = K_0 K_1,$$

as in Theorem C, regrouping by the G_1 -coordinate (lemma 8.1) and using $t^\gamma = \omega^{-\gamma}$ identifies

$$E \simeq K_0((t^{G_1})), \quad K_1 \simeq \mathbb{R}((t^{G_1})). \quad (22.3)$$

Here the coefficient field K_0 is a class, but each individual outer series uses only a set of coefficients.

22.2 The exact test for the explicit copies

Corollary 22.1 (The exact test for these copies). *Expand $z \in E$ as a Hahn series in G_1 with coefficients in K_0 . Then $z \in M$ if and only if some nonzero $q \in E$ has both its coefficient space and the coefficient space of qz finite-dimensional over $\mathbb{R}$.*

Proof. Apply the coefficientwise proof of theorem 17.2 in the coordinates (22.3). Every tensor, finite-dimensional span, and fraction used in that proof is set-sized even though the ambient fields are classes. $\square$

22.3 A concrete Euler derivation

Define

$$g = \omega^{7/2} \in G_1, \quad b = \omega^{15/4} \in G_1, \quad \tau = \omega^{-g}. \quad (22.4)$$

Here τ is the single cyclic monomial indexed by g in the full small- t notation $t^\gamma = \omega^{-\gamma}$, which is retained only for the full iterated Hahn field. The element g is $h_1(1)$, the same g as in the proof of Theorem C.

Let $\lambda : G_1 \rightarrow \mathbb{R}$ extract the coefficient of the inner monomial $\omega^{7/2}$ from an element of G_1 . Coefficient extraction is additive, with

$$\lambda(g) = 1, \quad \lambda(b) = 0.$$

The associated derivation Δ on E kills K_0 and, in Conway notation, satisfies

$$\Delta(\omega^{\gamma_0+\gamma_1}) = -\lambda(\gamma_1)\omega^{\gamma_0+\gamma_1} \quad (\gamma_j \in G_j). \quad (22.5)$$

In particular $\Delta\tau = \tau$ and $\Delta\omega^b = 0$. It preserves both K_1 and M . By lemma 16.2 it is a strong derivation for Hahn summation. It does not alter support exponents, and therefore maps $E \cap \mathbf{Oz}$ into its purely infinite ideal: any constant term is killed.

This is an explicitly specified Euler derivation adapted to the separated coordinates. It is not asserted to be a Hardy-type derivation, the Berarducci–Mantova derivation, or compatible with surreal exponentiation [11, 12].

22.4 The countably supported family

For every ordinal α and integer $n \geq 1$, let

$$\beta_{\alpha,n} = \omega \cdot \alpha + n, \quad g_{\alpha,n} = \omega^{\psi(\beta_{\alpha,n})}, \quad a_{\alpha,n} = \omega^{g_{\alpha,n}}. \quad (22.6)$$

The product and sum in the label $\beta_{\alpha,n}$ are ordinary *ordinal* operations. All arithmetic in $g_{\alpha,n}$, $a_{\alpha,n}$ and subsequent exponent formulas is surreal-field arithmetic. Distinct pairs (α, n) give distinct ordinal labels: the finite tail after $\omega \cdot \alpha$ lies strictly before $\omega \cdot (\alpha + 1)$.

Each $g_{\alpha,n}$ lies in G_0 . The family of all these $g_{\alpha,n}$ is linearly independent over $\mathbb{Q}$, because their inner normal forms are distinct monomials. Hence the family of all $a_{\alpha,n}$ is algebraically independent over $\mathbb{R}$: distinct monomials in any finite polynomial produce distinct sums of the $g_{\alpha,n}$ as outer exponents, and cannot cancel. [merge] For $\alpha = 0$ these are the $g_n = g_{0,n}$ and $s_n = a_{0,n}$ of the proof of Theorem C, with the same independence argument.

Theorem 22.2 (Explicit omnific differential freedom). *Define*

$$Y_\alpha = \omega^b \sum_{n \geq 1} a_{\alpha,n} \omega^{-ng} \quad (22.7)$$

$$= \sum_{n \geq 1} \omega^{b-ng+g_{\alpha,n}}. \quad (22.8)$$

Then:

- (i) Y_α is a purely infinite omnific integer in E whose support, read in decreasing order, has order type ω .
- (ii) Its jets are
$$\Delta^j Y_\alpha = \sum_{n \geq 1} n^j \omega^{b-ng+g_{\alpha,n}} \quad (j \geq 0). \quad (22.9)$$
- (iii) The family $(Y_\alpha)_{\alpha \in \mathbf{On}}$ is differentially algebraically independent over $\mathcal{C}_{\text{fg}}(K_0/\mathbb{R}; G_1)$, hence over M , in the finite-relation sense.
- (iv) For every set cardinal κ , the subfamily indexed by $\alpha < \kappa$ has cardinality κ and has the same independence property.

Proof. The inner leading monomial of b has exponent $15/4$, that of g has exponent $7/2$, and all the $g_{\alpha,n}$ have inner leading exponent below 1. Consequently

$$b - ng + g_{\alpha,n} > 0$$

for every finite n . Moreover

$$(b - ng + g_{\alpha,n}) - (b - (n+1)g + g_{\alpha,n+1}) = g + g_{\alpha,n} - g_{\alpha,n+1} > 0$$

by (8.1). Thus (22.8) is a reverse-well-ordered normal form with distinct positive exponents indexed in decreasing order by the positive integers. Its coefficients are all 1, and it has no constant term. This proves (i).

Formula (22.5) gives eigenvalue n on the n th summand, since $\lambda(b) = 0$, $\lambda(g) = 1$, and Δ kills the G_0 -part. Strongness gives (22.9).

The coefficient sequences $n \mapsto a_{\alpha,n}$ are disjoint blocks in an algebraically independent family. Apply theorem 20.2 and proposition 20.5 to the unscaled sums in (22.7), using $k = \mathbb{R}$, $F = K_0$, and $\Gamma = G_1$. The multiplier ω^b is a nonzero member of M with $\Delta\omega^b = 0$. Multiplying every jet by ω^b is an invertible substitution over the base, so the scaled family remains independent. This proves (iii). Part (iv) follows by restriction; independence in particular implies distinctness. $\square$

[merge] Since $Y_0 = \omega^b S$ with S the witness (8.3) and $\omega^{-b} \in K_1$ is a Δ -constant, part (iii) contains the statement that S is differentially transcendental over $\mathcal{C}_{\text{fg}}(K_0/\mathbb{R}; G_1)$, a strengthening of theorem 8.3 for this S in the real case. Theorem 8.3 itself needs neither characteristic-zero derivations nor the explicit coefficients: it holds for any algebraically independent s_n and also over $\mathbb{C}$.

Corollary 22.3 (Countable support is the sharp cardinal threshold). *Every element of E with finite Conway normal-form support belongs to M . Nevertheless, $E \cap \mathbf{Oz}$ contains countably supported purely infinite elements differentially transcendental over M . Thus $\aleph_0$ is the least possible support cardinality for this phenomenon in the explicit join.*

Proof. A finite-support element is a finite sum of terms

$$c\omega^{\gamma_0+\gamma_1} = (c\omega^{\gamma_0})\omega^{\gamma_1} \in K_0K_1 \quad (\gamma_j \in G_j).$$

Apply theorem 22.2 for the countable examples. The threshold is about support cardinality, not birthday or cofinality in the value group. $\square$

Remark 22.4 (Algebraic enlargements of the base). Every finite family of jets in theorem 22.2 remains algebraically independent after any algebraic base extension within a common ambient extension field. Indeed, for a finite tuple $\mathbf{y}$, if N/M is algebraic, then $N(\mathbf{y})/M(\mathbf{y})$ is algebraic; transcendence degree in the two towers gives $\text{trdeg}(N(\mathbf{y})/N) = \text{trdeg}(M(\mathbf{y})/M)$. Only finitely many algebraic coefficients are involved in a proposed relation. In particular, the witnesses remain differentially independent over the relative algebraic closure of M in E , with the induced derivation: differentiating a minimal polynomial shows that this relative algebraic closure is stable under the derivation. This observation is not a computation of that closure.

23 A fixed tail survives every set of parameters

The class-sized strength of the preceding construction is not just that its indices run through the ordinals. Set-supported normal forms allow every set of parameters to be localized to a bounded collection of inner supports. All sufficiently late coefficient labels escape that collection.

Lemma 23.1 (Coefficient localization for a parameter set). *For every set $P \subseteq E$, there is a set $\Theta \subseteq (0, 1)$ such that, if G_Θ is the additive group of all inner surreal normal forms supported in Θ and*

$$C_\Theta = \mathcal{K}_\mathbb{R}(G_\Theta) \subseteq K_0,$$

then G_Θ and C_Θ are sets and

$$P \subseteq C_\Theta((t^{G_1})). \quad (23.1)$$

Proof. Expand each element of P in the iterated coordinates (22.3). Collect all its outer coefficients in K_0 . This is a set, since P and all its supports are sets. Expand each of these coefficients in its Conway normal form, whose exponents lie in G_0 , and collect the resulting set of exponents. Expand each such exponent once more in its inner normal form. Its support is a subset of $(0, 1)$. The union Θ of all these supports is again a set.

Every collected exponent belongs to G_Θ , so every original outer coefficient belongs to C_Θ . This gives (23.1). The class of inner normal forms supported in a fixed set Θ injects into the set of functions $\Theta \rightarrow \mathbb{R}$, and is closed under addition and negatives; it is therefore a set-sized additive group. Likewise the full Hahn field on the set group G_Θ injects into the set of functions $G_\Theta \rightarrow \mathbb{R}$ and is a set. All these statements use ordinary replacement and union, not a class-sized support operation. $\square$

[merge] This is the support-closure step of [corollary 6.1](#), applied to the outer coefficients and then to their exponents: there the parameters are placed in $\mathcal{K}_\mathbb{R}(H)$ with H the rational span of their supports; here their outer coefficients are placed in $\mathcal{K}_\mathbb{R}(G_\Theta)$.

Lemma 23.2 (A tail is independent over the localized coefficient field). *For Θ and C_Θ as above, there is an ordinal α_0 such that the family*

$$\{a_{\alpha,n} : \alpha \geq \alpha_0, n \geq 1\}$$

is algebraically independent over C_Θ .

Proof. Because ψ is injective, the ordinals β satisfying $\psi(\beta) \in \Theta$ form a set: each member of Θ has at most one inverse image, and replacement collects the ordinal inverse images. Choose α_0 greater than every such ordinal. If $\alpha \geq \alpha_0$, then $\beta_{\alpha,n} = \omega \cdot \alpha + n \geq \alpha$ and hence $\psi(\beta_{\alpha,n}) \notin \Theta$.

The corresponding $g_{\alpha,n}$ are distinct inner monomials supported outside Θ . Every nonzero rational linear combination of finitely many of them still has a nonzero coefficient outside Θ , and thus does not belong to G_Θ . They are therefore linearly independent over $\mathbb{Q}$ modulo G_Θ .

In a nonzero polynomial in finitely many $a_{\alpha,n}$ with coefficients in C_Θ , the support of each coefficient is contained in G_Θ . Multiplication by a monomial in the $a_{\alpha,n}$ translates that support by the corresponding integer linear combination of the $g_{\alpha,n}$. Distinct polynomial monomials produce distinct cosets modulo G_Θ . Their supports are consequently disjoint, so their nonzero terms cannot cancel. This proves algebraic independence over C_Θ . $\square$

Theorem 23.3 (Set-parameter tail resilience). *For every set $P \subseteq E$, there is an ordinal α_0 such that the predefined tail*

$$(Y_\alpha)_{\alpha \geq \alpha_0}$$

is differentially algebraically independent over $M\langle P \rangle$. In fact it is independent over the larger field

$$C_{\text{fg}}(K_0/C_\Theta; G_1) \quad (23.2)$$

for a set coefficient field C_Θ supplied by [lemma 23.1](#).

Proof. Choose Θ , C_Θ , and α_0 from the preceding lemmas. Apply the private coefficient-block theorem with base field C_Θ , coefficient field K_0 , outer group G_1 , and the tail coefficients $a_{\alpha,n}$. Its hypotheses hold by lemma 23.2. The Euler weight map takes its values in $\mathbb{R} \subseteq C_\Theta$, so it is valid over this enlarged base. The set-local formulation proposition 20.5 applies because C_Θ is a set. This gives differential independence of the unscaled tail over (23.2), and then of the Y_α after multiplication by ω^b .

It remains to verify inclusion of the desired parameter field. The envelope (23.2) contains every element of K_0 as an outer constant, since an individual element generates a finitely generated extension of C_Θ . It contains K_1 because its coefficients lie in $\mathbb{R} \subseteq C_\Theta$. It contains P by (23.1). Finally it is stable under Δ , by the proof of lemma 16.5. Therefore it contains $M\langle P \rangle$, and independence descends to that smaller base. $\square$

Corollary 23.4 (No set-sized algebraic or differential generation). *For no set $P \subseteq E$ is E algebraic over $M(P)$. For no such P is E differential-algebraic over $M(P)$. In particular, no set of ordinary or differential generators, even followed by the respective algebraic closure operation within E , generates E over M .*

Proof. A member of the tail supplied by theorem 23.3 is differentially transcendental over $M\langle P \rangle$ and therefore algebraically transcendental over its subfield $M(P)$. This contradicts either asserted algebraicity. $\square$

Remark 23.5 (Why the set qualification is sharp for the method). A proper class of parameters could use every inner support label $\psi(\beta)$. The localization set Θ would then no longer be a set, and the ordinal bound in lemma 23.2 would not be available. The theorem is a genuine set-parameter result, not a statement about arbitrary proper-class parameter collections. It also does not assert independence over a differential-algebraic closure of $M\langle P \rangle$; that stronger base-change statement is unnecessary for corollary 23.4.

24 Arithmetic joins and the surcomplex extension

24.1 Fraction fields of the three integer rings

Let $R_j = K_j \cap \mathbf{Oz}$ as in (22.2). Let

$$R_0[R_1] = \text{the subring generated by } R_0 \cup R_1, \quad O = E \cap \mathbf{Oz}. \quad (24.1)$$

Thus $R_0[R_1]$ is formed by finite ring operations, whereas O permits every set-supported omnific normal form whose exponents lie in $G_0 + G_1$. Both are class domains. Fractions and integrality are interpreted by their ordinary finite formulas, as in section 9.

Lemma 24.1 (Monomial denominator clearing). *One has*

$$\text{Frac}(R_j) = K_j, \quad \text{Frac}(R_0[R_1]) = M, \quad \text{Frac}(O) = E. \quad (24.2)$$

Proof. For $x \in K_j$, its support is a set in G_j . Pull it back through the ordered-group isomorphism $h_j : \mathbf{No} \rightarrow G_j$. Every set of surreal numbers has an upper bound, so choose $d \in G_j$, $d > 0$, with $d + e > 0$ for every e in the support of x . Both ω^d and $\omega^d x$ have only positive exponents, and hence belong to R_j . Thus $x = (\omega^d x)/\omega^d$ is a fraction of elements of R_j . The reverse inclusion is immediate.

The fraction field of $R_0[R_1]$ contains both K_j and is contained in their compositum, giving $\text{Frac}(R_0[R_1]) = M$.

Finally take $x \in E$. Project its set of support exponents to a set in G_1 . Choose $d \in G_1$, $d > 0$, strictly greater than the negatives of all these projections. If $\gamma_0 + \gamma_1$ occurs in x , then $d + \gamma_1$ is positive in G_1 and dominates γ_0 by (8.1). Hence $d + \gamma_0 + \gamma_1 > 0$. The same denominator ω^d clears x into O , proving $\text{Frac}(O) = E$. $\square$

The first clearing mechanism is already part of this report's omnific fraction theory (lemma 9.1, and $\text{Frac}(R_\alpha) = K_\alpha$ in proposition 9.2). It is included here because the exact fraction field of the *two-copy generated ring* is needed in the next result.

Theorem 24.2 (Arithmetic failure cannot be repaired by a set). *With $R_0[R_1] \subseteq O$ as in (24.1):*

- (i) *For every set $J \subseteq \mathbf{On}$, substitution $X_\alpha \mapsto Y_\alpha$ embeds the polynomial $R_0[R_1]$ -algebra in the variables $(X_\alpha)_{\alpha \in J}$ into O .*
- (ii) *For every set $S \subseteq O$, the extension $O/R_0[R_1][S]$ is not integral. In fact E is not algebraic over $\text{Frac}(R_0[R_1][S]) = M(S)$.*
- (iii) *The obstructing element in (ii) can be chosen from a tail of the same family (22.8); it has purely positive, countable Conway support.*

Proof. Since $R_0[R_1] \subseteq M$, algebraic independence over M implies the injectivity in (i); the image lies in O by theorem 22.2. For (ii), apply theorem 23.3 with $P = S$. Any tail member is transcendental over $M(S)$, while lemma 24.1 gives $\text{Frac}(R_0[R_1][S]) = M(S)$. If it were integral over $R_0[R_1][S]$, a monic polynomial over that ring would make it algebraic over $M(S)$, a contradiction. The same element proves (iii). $\square$

Remark 24.3 (Not a normalization calculation). Nonintegrality of $O/R_0[R_1][S]$ does not compute the integral closure of $R_0[R_1]$ in M or in E . Nor does it compute a Picard group, a class group, or a projective module. The theorem isolates a much larger transcendental obstruction that no set of arithmetic parameters removes. Questions about normalization inside the fixed fraction field remain distinct.

24.2 Surcomplex and Gaussian omnific versions

Put

$$E_{\mathbb{C}} = E[i], \quad K_{j,\mathbb{C}} = K_j[i], \quad M_{\mathbb{C}} = M[i].$$

Because E is ordered, adjoining i gives a genuine quadratic field extension. Separating real and imaginary coefficients identifies these fields with the corresponding full complex-coefficient Hahn fields on the same groups. The union of two set supports is still a set, so this identification preserves the Hahn conventions.

Extend Δ by $\Delta i = 0$. Define

$$O_{\mathbb{C}} = E_{\mathbb{C}} \cap \mathbf{Oz}[i] = O[i].$$

The Gaussian integer images are $R_j[i]$, and the ring they generate is $R_0[R_1][i]$.

Theorem 24.4 (Surcomplex independence and Gaussian witnesses). *The following statements hold.*

- (i) *The real family (Y_α) remains differentially algebraically independent over $M_{\mathbb{C}}$ in $E_{\mathbb{C}}$.*

(ii) *The genuinely nonreal elements*

$$Z_\alpha = (1 + i)Y_\alpha \quad (24.3)$$

belong to $\mathbf{Oz}[i]$, have countable support, and are differentially algebraically independent over $M_{\mathbb{C}}$.

(iii) *For every set $P \subseteq E_{\mathbb{C}}$, a tail of each of these families is differentially algebraically independent over $M_{\mathbb{C}}\langle P \rangle$.*

(iv) *$\text{Frac}(R_0[R_1][i]) = M_{\mathbb{C}}$ and $\text{Frac}(O_{\mathbb{C}}) = E_{\mathbb{C}}$. For every set $S \subseteq O_{\mathbb{C}}$, the ring $O_{\mathbb{C}}$ is not integral over $R_0[R_1][i][S]$.*

Proof. For a polynomial relation over $M[i]$ among real jets, write each coefficient uniquely as $a + ib$, $a, b \in M$. Its real and imaginary parts give polynomial relations over M . At least one is a nonzero polynomial if the original was nonzero. This contradicts [theorem 22.2](#), proving (i). Multiplication by the nonzero constant $1 + i$ is an invertible substitution on all jets, proving (ii).

For (iii), take the set $P_{\mathbb{R}} \subseteq E$ consisting of the real and imaginary parts of all members of P . Apply [theorem 23.3](#) to $P_{\mathbb{R}}$. Splitting coefficients as above gives independence over $M\langle P_{\mathbb{R}} \rangle[i]$, which contains $M_{\mathbb{C}}\langle P \rangle$. Again rescale to obtain the Z_α version.

For (iv), the displayed fraction fields follow from [lemma 24.1](#) after adjoining i , or by clearing real and imaginary parts with a common monomial. A tail element supplied by (iii) is transcendental over $\text{Frac}(R_0[R_1][i][S]) = M_{\mathbb{C}}(S)$, so cannot be integral over $R_0[R_1][i][S]$. $\square$

No statement here depends on extending analytic functions to all surcomplex arguments, choosing a complex logarithm, or assuming an exponential-compatible derivation. The passage from real to complex is finite algebraic extension and coefficient separation.

25 What the addition resolves, and what it does not

25.1 Relation to Research question 3

Research question 3 has two parts: an exact description of the ordinary compositum and a dimension calculation for the Hahn join over it. The results of the addition give the following answers in specified domains.

Problem component	Conclusion proved by source 02
Membership in separated coordinates	The finite coefficient-rank denominator criterion, theorem 17.2 and corollary 22.1 .
Finite coefficient-field condition	Not sufficient even for algebraicity; a witness uses only $k(x)$, corollary 19.5 .
Natural set-sized dimensions	Exact algebraic and differential dimensions for $F((t))/Fk((t))$ under the hypotheses of theorem 21.1 .
Explicit class-sized join	Independent families of every set cardinal, with countable purely infinite supports, theorem 22.2 .
Dependence on set parameters	A fixed tail remains independent after any set adjunction, theorem 23.3 .
Arithmetic and complex forms	Nonintegrality persists after set adjunction; the same phenomena occur in $\mathbf{Oz}[i]$, theorems 24.2 and 24.4 .

These answers do not solve the complete problem for arbitrary non-separated ordered amalgams of exponent groups, arbitrary coefficient-field cardinalities, and arbitrary prescribed common Hahn cores. The rank criterion applies whenever an iterated coefficient interpretation has been established. The cardinal theorem does not assign a set cardinal to the class extension. The question addressed is a question posed in this report, not a claimed resolution of a separately established, long-standing published conjecture.

25.2 Provenance and proof dependencies

The source compared itself with this report at commit `7b256e0792df7718995b816ad12a77bf56731f8f`, where the report's source had blob identifier `7d5cfe644e7c902f2b594179c337a6c633350fc3`. The key portions consulted were the explicit-copy and re-expansion material of [sections 7 and 8](#) and the question list of [section 12](#). The report's README and the collection's main catalogue were also consulted. The source's audit (shipped as `02-composita-SOURCE_AUDIT.md`) records the paths and distinguishes sources read for substance from sources consulted only for orientation.

The explicit maps (7.1) and (7.3), the scale separation, and the finite coefficient-field observation are credited to this report. The coset-slice argument belongs to the same elementary Hahn toolkit and is not presented as a priority claim. The proofs of the new class conclusions do not import this report's general transfinite independent-copy recursion ([theorem 3.4](#)), nor, in the source's words, the repository's proposed automatic-strongness results, curve-rigidity results, or unreviewed quotient theorems.

The main dependency chains are

$$\begin{aligned}
 &\text{rapid degrees} \implies \text{continuum family} \implies \text{small-cardinality case,} \\
 &\text{coefficient derivations} \implies \text{private blocks} \implies \text{tree amplification,} \\
 &\text{tree amplification} \implies \text{large-cardinality case,} \\
 &\text{separated copies and private blocks} \implies \text{omnific witnesses,} \\
 &\text{inner-support localization} \implies \text{tail resilience,} \\
 &\text{tail resilience} \implies \text{arithmetic and complex consequences.}
 \end{aligned}$$

No numerical experiment replaces any arrow in these chains.

25.3 Critical proof checks

The rapid-degree argument depends on more than comparing leading terms informally. The polynomial relation is made integral in the outer variable before valuations are compared. Minimal total jet degree guarantees a nonzero evaluated partial derivative. The selected coefficient lies at $n + \ell < 2n$, so all quadratic tail terms vanish there. Partial coefficients below ℓ vanish, so later tail coefficients cannot interfere. Finally the coefficient degree ν is fixed even though the outer coefficient series are arbitrary Laurent series.

The private-block argument depends on a different finiteness principle. A finitely generated field D is algebraic over a field generated by finitely many elements of a transcendence basis. Coordinate derivations outside that finite set kill D after separable extension. This is why the argument controls whole coefficient series in $D((t^\Gamma))$, not merely their leading coefficients. Characteristic zero is used both for separability and for distinct integer eigenvalues.

The class argument uses two levels of support localization. An outer coefficient is a surreal series in exponents from G_0 ; those exponents themselves have inner supports in $(0, 1)$. Taking the union

of the inner supports for a *set* of parameters produces a set Θ . The predefined ordinal coefficient labels outside Θ remain independent modulo G_Θ . Replacing this argument with an unproved claim that “a class is larger than a set” would not establish the required independence over the parameter field.

The source’s proof audit (shipped as `02-composita-PROOF_AUDIT.md`) adds three points. The auxiliary coordinate derivations of [theorem 20.2](#) need not preserve any *inner* surreal summation; they are extended coefficientwise only in the explicitly used *outer* Hahn representation. The rapid-degree argument does not require sparse outer support, but it does require a uniform finite degree in x after denominators are cleared. In [theorem 22.2](#) the $g_{\alpha,n}$ are inner monomials in $(0, 1)$ and the $a_{\alpha,n}$ are independent outer coefficient monomials; the Euler derivation has weight $-\lambda(\gamma_1)$ in Conway notation, so the n th witness term has eigenvalue n , and the shift multiplier has derivative zero.

25.4 Limitations and nonclaims, as stated by the source

The proofs are mathematical arguments, not Lean-checked results. The finite program supplied with the source tests identities and finite models only. Independent review should prioritize [theorems 19.2](#), [20.2](#) and [23.3](#). The source search did not certify novelty or exhaustive priority. The adjective “strong” refers to Hahn summability, not convergence in an order topology.

No claim is made about Gonshor-exponential compatibility, preservation of simplicity, initiality of the two images, birthday optimality, or the Berarducci–Mantova derivation. The full algebraic closure of the compositum, its normalization, and its projective modules are not calculated. No general algorithm deciding compositum membership is proved. These distinctions are mathematically relevant, not merely editorial qualifications.

25.5 What the addition does not claim

[merge] This list collects every limitation stated by source 02, in its article, delivered README, source and proof audits, recorded run and build audit, with its location here, followed by those added in the merge.

- C1. The rank test is not a decision procedure: the existential search over q is an infinite-input problem, and no algorithm deciding arbitrary Hahn-series membership follows ([remark 17.3](#); Research question [16](#)).
- C2. The tensor fact is not new: it is a standard linear-disjointness mechanism. No new tensor-product theorem and no general theory of rational series is claimed ([remark 17.3](#); source audit).
- C3. The comparison with Krapp–Kuhlmann–Serra [\[10\]](#) is conceptual, not a claim that the formulations have identical inputs; that paper was used for the literature comparison, not as a proof of the compositum theorems ([remark 17.3](#); bibliography).
- C4. Infinite coefficient rank obstructs membership in $F[L]$, not in M ; a geometric series of infinite rank lies in M ([example 17.4](#) and [corollary 17.5](#)).
- C5. Base extension is not treated as harmless; coefficient extension uses coefficientwise independence of constants, intermediate base change clears denominators, and exponent-group extension uses coset slices, not an unjustified claim that arbitrary base extension preserves algebraic independence. No convexity is required for the coset slice, but the surreal iterated presentation requires the explicit scale separation ([section 18](#); proof audit).

- C6. The rapid-degree theorem does not rest on gaps in the outer support; merely pointing to such gaps would not prove [example 19.6](#). It requires characteristic zero, a transcendental coefficient variable, unbounded ratios d_n/d_{n-1} , infinitely many private indices relative to each finite subfamily, and a uniform finite x -degree after clearing denominators ([theorem 19.2](#) and [remark 19.3](#); proof audit).
- C7. The formula $|F|^{\aleph_0}$ is not extended to infinite algebraic coefficient extensions, to $\Gamma = 0$, to groups without a specified λ with $\lambda(g) = 1$ (only the algebraic rank test applies there), or to $|k| > \mathfrak{c}$ with small positive transcendence degree, where only the explicit lower bounds are claimed; the rapid-degree family need not reach the cardinality of the Laurent field for large k ([section 21.1](#); Research questions [12](#) and [13](#)).
- C8. Characteristic zero is essential for the differential statements: in characteristic p the Euler derivation satisfies $\partial^p = \partial$. This does not settle the ordinary algebraic-transcendence problem in characteristic p ([section 21.1](#)).
- C9. The derivation Δ is an explicitly specified diagonal Euler derivation. It is not asserted to be Hardy type, the Berarducci–Mantova derivation, or compatible with exponentiation. It is proved to be a derivation directly and is not identified with a Hardy-type derivation of Kuhlmann–Matusinski [\[11\]](#), whose Hardy-type theory the source did not review in full ([sections 15.3](#) and [22.3](#); source audit; Research question [17](#)).
- C10. The explicit copies do not commute with the omega-map, and strong Hahn summability alone does not provide compatibility with it or with the exponential ([section 22.1](#); Research question [18](#)).
- C11. [Remark 22.4](#) is not a computation of the relative algebraic closure of M in E .
- C12. The class results use only finite relations, set-indexed families and set-parameter resilience. No class transcendence basis is chosen, no set cardinal is assigned to the proper-class transcendence degree, and no cardinal equal to the size of $\mathbf{No}$ is postulated. No class-state recursion or class satisfaction predicate is used ([section 16](#); [proposition 20.5](#); delivered README; proof audit).
- C13. Tail resilience is for *set* parameters only; proper-class parameter collections are not covered. Independence is not asserted over the differential-algebraic closure of $M\langle P \rangle$ ([remark 23.5](#)).
- C14. Nonintegrality of $O/R_0[R_1][S]$ does not compute the integral closure of $R_0[R_1]$ in M or E , a normalization, a Picard group, a class group or a projective module ([remark 24.3](#); Research question [19](#)).
- C15. The complex versions use only finite algebraic extension and coefficient separation: no analytic continuation to surcomplex arguments, no complex logarithm, no exponential-compatible derivation ([section 24](#)).
- C16. Research question [3](#) is not solved for non-separated ordered amalgams, arbitrary coefficient-field cardinalities, or arbitrary prescribed common Hahn cores; the cardinal theorem assigns no set cardinal to the class extension; the question is this report’s own, not a long-standing published conjecture ([section 25](#)).
- C17. The explicit maps, scale separation and finite coefficient-field observation are this report’s; the coset slice is elementary. The proofs do not import this report’s transfinite recursion, nor automatic-strongness, curve-rigidity or quotient theorems of the repository. No numerical experiment replaces a proof step ([section 25](#)).

- C18. Nothing is claimed on Gonshor-exponential compatibility, simplicity, initiality, birthday optimality, the full algebraic closure of the compositum, normalization or projective modules; no general membership algorithm is claimed (section 25; Research question 20).
- C19. Status: the proofs are not Lean-checked, not refereed and not independently reviewed; the source’s proof audit is author-side, not a referee report; review should prioritize theorems 19.2, 20.2 and 23.3. Priority is not certified: the literature comparison was targeted, a search that finds nothing does not establish novelty, and not all related literature was read (sections 15.3 and 25; title page of the source; source and proof audits; build audit).
- C20. The source’s comparison with this report was targeted: it read two line ranges of the pinned source for substance, not every proof of the report. The Stacks Project was used for terminology; the encyclopedia page on surreal numbers for orientation only; L’Innocente–Mantova for context, with no factorization result claimed (source audit; bibliography).
- C21. The research questions are proposed by the proved statements. Except for Research question 3 of this report, they are not represented as previously published open problems (section 26).
- C22. The finite checks are narrower than the theorems. They test polynomial identities, finite coefficient ranks, finite combinatorial examples and finite ordered-group models; they do not implement **No** or infinite Hahn summation, do not represent countably supported surreals, and prove no independence, cardinal or proper-class assertion. The lexicographic model is a model of the comparisons, not an approximation to a surreal number. Neither typesetting nor finite computation is proof verification (section 27; recorded run).
- C23. The formalization decomposition names implementation targets, not declarations that exist (section 27.2).
- M1. Parts (i)–(ii) of theorem 17.2 are `duals:prop:algebraic` of the three-duals report and its preceding injectivity paragraph; part (iii) is new but immediate. The source does not cite that report. Lemma 18.3 is a variant of theorem 5.4 (section 15.6).
- M2. The private-index and tree mechanisms are related to `tail:thm:surreal` and `bst:thm:independent`, which use different bases, derivations and tools, and the tail resilience to `adr:sr:thm:tail`; the source cites none of them (section 15.6).
- M3. Research question 3 is answered only in the separated case with trivial core and in the cardinal cases of theorem 21.1; its remaining part is Research questions 11 and 14. [merge] Source 03 later answers the degree part for explicit non-separated examples (section 28.4). No other question of this report is answered, and no question of another report (section 15.4).
- M4. No `isc:cp:` label has a Lean mapping in `FORMALIZATION.md`. The proofs were read when the addition was merged (Appendix A.7); that reading is not an independent review.
- M5. The recorded run was repeated in the merge on a copy, with identical counts; the shipped `code/02-composita-build.sh` does not run unchanged under the shipped file names (Appendix A.8).

26 Research questions of the addition

The following questions are proposed by the proved statements of source 02. Except for Research question 3, which the source takes from this report, they are not represented as previously published

open problems. Each asks for a specific extension, obstruction, or effective refinement. [merge] The source's Question n is Research question $10 + n$ here.

Research question 11 (Non-separated ordered amalgams). Let G_0, G_1 be independent ordered subgroups but assume neither is infinitesimal relative to the other. Determine the image of $\mathcal{K}_k(G_0)\mathcal{K}_k(G_1)$ inside $\mathcal{K}_k(G_0 + G_1)$ and its transcendence defect. Can a finite-rank denominator criterion be formulated without an iterated Hahn coefficient field?

The obstacle is not the finite tensor argument itself, but choosing slices with support properties strong enough to characterize the entire join. This is the main part of Research question 3 left untreated. [merge] *Status (batch 35): the transcendence defect is known for explicit examples; the image and a denominator criterion are open.* For the interleaved groups of [Theorem E](#) the defect is 2^κ , for the dense Archimedean groups of [Theorem F](#) it is $\mathfrak{c}$, and for the class groups of [Theorem G](#) there is no set-sized transcendence basis ([section 28.4](#)). These groups are explicit subgroups, not the exponent groups of copies of **No** (the question allows any independent groups); the criterion [Theorem D](#) applies to any pair of full Hahn fields on groups meeting its hypotheses. The smallest open dense case is Research question 23; the image is Research question 24.

Research question 12 (Very large base fields). For $|k| > \mathfrak{c}$ and $0 < \text{trdeg}(F/k) < |k|$, must

$$\text{trdeg}(F((t))/Fk((t))) = |F|^{\aleph_0}$$

still hold? Does the same equality hold for differential transcendence degree under the outer Euler derivation?

The rapid-degree theorem gives a continuum family over any k , but that family need not reach the cardinality of the entire Laurent field. A stronger combinatorial coefficient construction or a counterexample would settle the gap in the cardinal argument.

Research question 13 (Infinite algebraic coefficient extensions). For an infinite algebraic extension F/k of characteristic zero, characterize $Fk((t))$ inside $F((t))$ and determine the algebraic and differential transcendence degrees of the latter over the former.

Finite extensions give equality of fields, while the transcendental-coordinate derivations no longer apply. Growth in the algebraic degrees needed to contain successive coefficients is a natural replacement invariant, but no theorem about it is claimed.

Research question 14 (Prescribed nonzero common Hahn cores). Can the explicit omnific tail-resilience theorem be extended to independent surreal copies with a prescribed set-sized common field $\mathcal{K}_{\mathbb{R}}(H)$? Which separation conditions on the ordered amalgam over H suffice?

This report already constructs copies with prescribed cores ([Theorem A](#)). What is needed is a compatible local coefficient decomposition and an independent-label family modulo the shared support group, not just an abstract pair of embeddings. [merge] Here H is the common core of [Theorem A](#), as in the source's question. [merge] *Status (batch 35): continued, not answered.* The mixed-support criterion [Theorem D](#) allows a nonzero core algebraically, but no example with $H \neq 0$ is constructed and no resilience is proved; the ordered construction is Research question 26, and full copies over a core are Research question 27.

Research question 15 (Presentation-independent coefficient rank). Which part of the finite-rank denominator criterion can be recovered from the pair of embedded fields alone, without naming a monomial cross-section or a chosen iterated Hahn presentation?

The compositum is intrinsic, but coefficient rank is a presentation-dependent certificate. An invariant description of the smallest possible certificate spaces could distinguish coordinate artifacts from structure of the pair.

Research question 16 (Effective membership on controlled inputs). For rational, recurrence-defined, automatic, or otherwise effectively presented supports and coefficients, when is the finite-rank denominator test decidable? Can one bound the denominator rank in a useful subclass?

The generalized recurrence framework of [10] is a relevant starting point. A useful algorithm must specify its representation of infinite series and should not treat a formal coefficient oracle as an effective finite input.

Research question 17 (Natural surreal derivations). Do comparable compositum-independence and set-parameter resilience statements hold for a Hardy-type surreal derivation, particularly the Berarducci–Mantova derivation, on subfields stable under it?

The proofs exploit a diagonal derivation that kills the entire lower-scale coefficient field. A natural surreal derivation generally differentiates those coefficients and may not preserve the explicit copy images. Establishing an appropriate stable pair is part of the problem, not an automatic corollary of [theorem 22.2](#). [merge] Research question 6 asks the analogous question for the prescribed-intersection theorem.

Research question 18 (Exponential-compatible copies). Can one obtain a similarly large gap for a pair of proper surreal self-embeddings commuting with the Gonshor exponential, or with the omega-map? Which common cores must then be closed under the additional operation?

The displayed copies fail to commute with the omega-map. Strong Hahn summability alone does not provide the desired compatibility. This question connects the compositum programme to the richer embedding questions already distinguished in this report (Research question 6).

Research question 19 (Normalization within the fraction field). For $R_0[R_1]$ from (24.1), compute or constrain its integral closure inside M . How does that ring compare with $M \cap \mathbf{Oz}$, and what can be said about finitely generated projective modules over it?

The nonintegrality of $O/R_0[R_1][S]$ comes from transcendence outside M and therefore does not answer this within-fraction-field problem. A normalization result would complement rather than restate [theorem 24.2](#). [merge] Research question 4 asks the corresponding questions for the intersection ring $\mathcal{A}_k(H)$.

Research question 20 (Birthday complexity of minimal-support witnesses). Among countably supported omnific elements transcendental over the explicit compositum, what birthday bounds are possible? Can one lower the nested normal-form complexity while retaining set-parameter tail resilience?

The support cardinality $\aleph_0$ is already sharp by [corollary 22.3](#). Birthday optimality is a different invariant and requires genuine estimates on surreal construction lengths.

Research question 21 (Several scales and commuting derivations). For three or more separated copies, construct several commuting Euler derivations and determine algebraic independence of mixed jets over the corresponding ordinary compositum. What is the correct tree-label condition for joint differential independence?

Coefficient derivations still offer finite avoidance, but one must separate multivariable eigenvalue interpolation from the ordered-group support problem. A multidimensional Vandermonde argument alone does not settle the Hahn support conditions. [merge] The algebraic counterpart for r factor fields, without separation, is Research question 30.

Research question 22 (Formalization of the core mechanisms). Can the coefficient-rank theorem, the coefficient-extension and coset-slice lemmas, the rapid-degree Taylor argument, and the private-block theorem be formalized as reusable Hahn-field results before specializing to surreal numbers?

A promising order is to formalize the finite polynomial and coefficient-space identities first, then the set-sized Hahn statements, and only afterwards the class-local interface. This would test the most important new proofs without requiring a monolithic formalization of the entire report. [merge] It overlaps Research question 10 in the coset-slice kernel, and Research question 32 in the coset and Euler-derivation kernel.

27 Finite verification and a formalization decomposition of the addition

27.1 Finite verification and reproducibility

The source’s program, shipped as `code/02-composita-verify.py`, performs exact finite checks. Its recorded output, shipped as `data/02-composita-verification.json`, records the actual checks and their counts: 2,530 checks in 21 categories. These checks are deliberately narrower than the theorems. They do not represent countably supported surreal numbers and do not prove an independence or cardinality statement.

The checked mechanisms include finite geometric-series convolution and coefficient ranks, Euler and coefficient-derivation identities on polynomial examples, Vandermonde determinants at distinct integer eigenvalues, finite prefix-tree private-label behaviour, factorial growth inequalities used in (19.5), and a multivariable Taylor coefficient calculation with quadratic tails excluded at the relevant order. A finite lexicographic exponent model tests the sign and strict descent used in (22.8). It is a model of the comparisons, not a numerical approximation to an actual surreal number.

A characteristic- p check illustrates the explicit boundary $\partial^p = \partial$ on monomials. Finite complex substitution checks confirm that multiplying variables by $1 + i$ is invertible; the proof for arbitrary polynomials remains the elementary field argument above.

The program uses the standard library and SymPy. It writes a file only when given `-output`. The source’s build script also supports repeated `pdflatex` passes when `latexmk` is unavailable, and its build audit records the compiler result, page count, reference checks, and layout checks for the delivered PDF. Neither successful typesetting nor finite computation constitutes proof verification. [merge] The shipped file names, the rerun and its comparison with the recorded output are in Appendix A.8.

27.2 A proposed formalization decomposition

The following are implementation targets, not declarations claimed to exist in the collection.

Coefficient-space layer. Represent a finite-dimensional coefficient span and prove that its coordinate functions define Hahn series with supports contained in the original support. Prove

tensor injectivity by coefficient comparison. Derive the rank-denominator criterion using the ordinary fraction field universal property.

Support and base-change layer. Define restriction to an exponent coset, prove its compatibility with multiplication by series on the subgroup, and establish the two linear-disjointness transfer lemmas. Make the chosen ambient field explicit to avoid conflating abstract tensor products with a specified compositum.

Polynomial perturbation layer. For a polynomial in finitely many jet variables over $k[[t]][x]$, prove a first-order Taylor congruence modulo t^{2n} after a perturbation divisible by t^n . Prove the coefficient-degree bound $\nu + d d_{n-1}$. These finite statements are the delicate reusable kernel of [theorem 19.2](#).

Derivation and interpolation layer. Formalize coefficientwise Hahn derivations, their commutation with Euler derivations, extension of a derivation over separable algebraic fields, and the finite Vandermonde elimination of evaluated partials. This yields the set-sized private-block theorem.

Cardinal and surreal layer. Add the prefix-tree construction and cardinal counting. For the explicit surreals, prove the two-level normal-form relabeling, lexicographic comparison, and set-support localization. Each proposed relation must be transported into a set-sized field before invoking a transcendence basis. No unrestricted class basis should be introduced just to shorten the proof.

[merge] The coset-slice and iterated-Hahn items overlap items (1)–(3) and (6) of [section 13.2](#).

28 Maximal transcendence in Hahn joins: a third manuscript

[merge] [Sections 28 to 41](#) print a third manuscript, written against this report. In the report’s local numbering ([Appendix A.9](#)) it is *source 03*: manuscript 01 of batch 35 of the collection, archive `Surreal_Hahn_Joins_Research`, *Maximal Transcendence in Hahn Joins: Mixed-support independence and prime omnific integers*, a 27-page PDF dated 23 September 2026 and pinned to repository commit `f388f950f44664f21f50f04b8ac74dab56c8ff2f`. At that pin this report’s `article.tex` was the blob `7d5cfe644e7c902f2b594179c337a6c633350fc3`, the same snapshot against which source 02 was written. Source 03 therefore saw [section 1–section 14](#) (source 01) but not [section 15–section 27](#) (source 02), which were merged afterwards; the two later sources are independent of each other. The source takes Research question 3 of [section 12](#) as its target.

Section n of the source is Section $n + 27$ here for $n = 1, \dots, 13$, and every numbered statement keeps its position: the source’s Lemma $n.m$ is Lemma $(n + 27).m$ here, and likewise for theorems, propositions, corollaries, definitions and remarks, and for equations. The source’s Theorems A, B, C and D share the lettered counter of this report’s Theorems A–C and are therefore printed as [Theorem D](#), [Theorem E](#), [Theorem F](#) and [Theorem G](#): its Theorem A is Theorem D here, B is E, C is F and D is G. Its Questions 1–10 are Research questions 23–32. Its Appendices A–C form [section 41](#); its Appendix D (sources and provenance) is folded into [Appendix A.9](#). Where the source cites this report, the citation is replaced by a cross-reference to the same place. Text added in the merge is marked [merge]; symbols renamed from the source are listed in [section 28.5](#). Every result, proof, example, question and limitation of the source is printed.

28.1 Finite field operations versus Hahn summation

Suppose two additive subgroups $A, B \subseteq \mathbf{No}$ have been chosen. Write, as in [section 1](#),

$$\mathcal{K}_k(A) = \left\{ \sum_{a \in S} c_a \omega^a : c_a \in k, S \subseteq A \text{ reverse well ordered and set-sized} \right\}, \quad k = \mathbb{R}, \mathbb{C}.$$

For $k = \mathbb{C}$ this notation denotes the corresponding subfield of $\mathbf{No}(i)$. Every individual monomial ω^{a+b} is already a product of an A -monomial and a B -monomial. It does not follow that an arbitrary allowable sum of these products belongs to the field generated by the two Hahn fields. There are two distinct objects:

$$M = \mathcal{K}_k(A)\mathcal{K}_k(B) \subseteq E = \mathcal{K}_k(A+B). \quad (28.1)$$

The left-hand side is the *ordinary compositum*: each of its elements uses only finitely many elements of the two fields and finitely many field operations. The right-hand side is the *full Hahn join*: it allows every set-sized reverse-well-ordered support in $A+B$.

The distinction is not a simple “finite sum versus infinite sum” observation. Each factor field already contains unrestricted Hahn sums on its own exponent group. Moreover, a geometric series in a mixed monomial is the inverse of a binomial and therefore belongs to M . The problem is to distinguish genuinely new mixed sums from sums disguised as rational expressions in arbitrary elements of the factor fields.

This report, in Research question 3 (headed *An exact description of ordinary composita*), asks both for a characterization of M inside E and for the transcendence degree of E/M in natural examples. Its compositum-gap argument ([Theorem C](#) and [theorem 8.3](#)) uses separated scales and infinitely many independent coefficients. Here the main mechanism is different: a finite differential annihilator works directly on the exponent group and does not require an iterated-series presentation. The degree question is solved in explicit families, including one with Archimedean, interleaved exponent groups.

28.2 Four principal statements

Theorem D (Mixed-support independence). *Let k be a field of characteristic zero, Γ_0, Γ_1 divisible ordered abelian subgroups of a common ordered abelian group, and $H = \Gamma_0 \cap \Gamma_1$. Suppose a set-indexed family $(a_i, b_i)_{i \in I}$ satisfies:*

- (i) $a_i \in \Gamma_0$, and the classes $a_i + H$ are $\mathbb{Q}$ -linearly independent;
- (ii) $b_i \in \Gamma_1$, and $b_i + H$ are pairwise distinct;
- (iii) $\{a_i + b_i : i \in I\}$ is well ordered.

Let $\mathcal{U}$ be a family of subsets of I such that, for every finite list of distinct $U_1, \dots, U_s \in \mathcal{U}$, each $U_j \setminus \bigcup_{\ell \neq j} U_\ell$ is infinite. Then the series

$$F_U = \sum_{i \in U} t^{a_i + b_i} \quad (U \in \mathcal{U})$$

are algebraically independent over $k((t^{\Gamma_0}))k((t^{\Gamma_1}))$ inside $k((t^{\Gamma_0 + \Gamma_1}))$.

The change from ω to t in this theorem is intentional: $t^\gamma = \omega^{-\gamma}$, so well-ordered t -supports correspond to reverse-well-ordered Conway supports. The theorem is proved in [section 31](#). Its

hypotheses concern the exponents of the proposed witnesses, not the supports of the coefficients of a putative relation.

Theorem E (The largest possible set-sized gap). *For every infinite cardinal κ there are explicit divisible subgroups $A_\kappa, B_\kappa \subseteq \mathbf{No}$, each of cardinality κ , with $A_\kappa \cap B_\kappa = 0$, such that, for $k = \mathbb{R}, \mathbb{C}$,*

$$\text{trdeg}(\mathcal{K}_k(A_\kappa + B_\kappa)/\mathcal{K}_k(A_\kappa)\mathcal{K}_k(B_\kappa)) = 2^\kappa.$$

The two factor fields are linearly disjoint over k . For $k = \mathbb{R}$ they are real closed; for $k = \mathbb{C}$ they are algebraically closed. There is a witnessing independent family of 2^κ elements of $\mathbf{Oz}$, valid simultaneously for both coefficient fields.

All three Hahn fields in this statement have cardinality 2^κ . Consequently the lower bound is not a cardinality comparison between a small base and a larger ambient field. It asserts maximal transcendence above a base that already has the same cardinality.

Theorem F (An interleaved family of omnific primes). *Let $(p_n)_{n \geq 0}$ enumerate the ordinary positive primes increasingly, and put*

$$\lambda_n = p_{2n}^{-3/2}, \quad \mu_n = p_{2n+1}^{-3/2}, \quad \rho_n = \lambda_n + \mu_n, \quad A = \text{span}_{\mathbb{Q}}\{\lambda_n\}, \quad B = \text{span}_{\mathbb{Q}}\{\mu_n\}.$$

There is an explicit family $(\mathfrak{B}_\eta)_{\eta \in 2^{\mathbb{N}}}$ of infinite, pairwise almost-disjoint subsets of $\mathbb{N}$ for which

$$P_\eta = 1 + \sum_{n \in \mathfrak{B}_\eta} \omega^{\rho_n} \tag{28.2}$$

are algebraically independent over $\mathcal{K}_{\mathbb{C}}(A)\mathcal{K}_{\mathbb{C}}(B)$. Each P_η is prime in both $\mathbf{Oz}$ and $\mathbf{Oz}[i]$. Moreover,

$$\text{trdeg}(\mathcal{K}_k(A + B)/\mathcal{K}_k(A)\mathcal{K}_k(B)) = \mathfrak{c} \quad (k = \mathbb{R}, \mathbb{C}), \quad \mathfrak{c} = 2^{\aleph_0}.$$

Here $A + B \subseteq \mathbb{R}$ is Archimedean and $A \cap B = 0$.

The primality input is classical, from Pitteloud and the transfer work of Biljakovic–Kochetov–Kuhlmann; it is also available through the broader criteria of L’Innocente–Mantova [17, 18, 3]. The proposed contribution in [Theorem F](#) is the simultaneous maximal algebraic independence over two entire Hahn subfields, together with the explicit prime realization, not a new one-row primality theorem.

Theorem G (One fixed class-sized example). *There are explicit proper-class divisible subgroups $A_*, B_* \subseteq \mathbf{No}$ with $A_* \cap B_* = 0$ such that the full Hahn join $\mathcal{K}_k(A_* + B_*)$ contains an ordinal-indexed, algebraically independent family of omnific integers over $\mathcal{K}_k(A_*)\mathcal{K}_k(B_*)$, for $k = \mathbb{R}, \mathbb{C}$. The extension therefore has no set-sized transcendence basis. For every infinite cardinal κ it contains a set of 2^κ algebraically independent omnific integers over that same class compositum.*

This last statement concerns explicitly defined class-sized support subfields. It does *not* identify either factor with a full copy of $\mathbf{No}$, and does not use this report’s construction of such copies ([Theorem A](#)), which the source treats as unreviewed. Algebraic independence of a class always means that every finite subfamily is algebraically independent.

28.3 How to read the source

The essential proof is [sections 30](#) and [31](#). Readers mainly interested in concrete surreal numbers may then go directly to [sections 34](#) and [35](#). Cardinal amplification and the class example are in [sections 32, 33](#) and [36](#). The exact scope of the advance over this report, failure cases, and further questions are recorded in [sections 38](#) and [39](#). [Section 41](#) gives a proof-dependency audit, finite exact verification, and a formalization roadmap.

28.4 What the addition settles in this report

[merge]

1. *Research question 11* (non-separated ordered amalgams) asks for the image of $\mathcal{K}_k(G_0)\mathcal{K}_k(G_1)$ in $\mathcal{K}_k(G_0 + G_1)$, for its transcendence defect, and for a finite-rank denominator criterion without an iterated Hahn coefficient field. Its *transcendence-defect part* is answered for explicit non-separated examples: for the interleaved groups A_κ, B_κ of [Theorem E](#) (neither is infinitesimal relative to the other: $f_0 \in B_\kappa$ lies between e_0 and e_1 of A_κ , and so on) the defect is 2^κ at every infinite cardinal κ ; for the dense Archimedean groups of [Theorem F](#) it is $\mathfrak{c}$; for the class groups of [Theorem G](#) there is no set-sized transcendence basis. The image of the compositum and a denominator criterion are *not* obtained; they remain open, and the source's own Research question [24](#) asks for them.
2. *Research question 3*, second part (the transcendence degree of the join over the compositum in natural examples), was answered in batch 34 in the separated case and the Laurent cardinal cases. Source 03 adds the explicit non-separated examples just listed, with omnific and prime witnesses. The first part (an exact description of the compositum) remains answered only in the separated case ([theorem 17.2](#)).
3. *Research question 14* (prescribed nonzero common cores) is continued, not answered. [Theorem D](#) allows a nonzero intersection $H = \Gamma_0 \cap \Gamma_1$ algebraically (hypotheses (i) and (ii) are modulo H), but every example constructed has $H = 0$, and no tail resilience is proved; the ordered construction for $H \neq 0$ is the source's Research question [26](#).
4. [Theorem C](#) receives a second route: the witness S of [\(8.3\)](#) is an instance of [Theorem D](#) in the class-local form of [lemma 36.1](#) ([remark 36.2](#)).
5. No other question of this report is answered. Research questions [21](#), [22](#) and 10 are continued by Research questions [30](#) and [32](#); Research question 3 and [14](#) by Research questions [26](#) and [27](#).

28.5 Conventions and renamed symbols

[merge] Source 03 uses both exponent conventions of this report: the decreasing large- ω convention $\sum c_g \omega^g$, written $\mathcal{K}_k(G)$ as here, and the increasing small- t convention $k((t^\Gamma))$ with well-ordered support, identified by $t^\gamma = \omega^{-\gamma}$ as in [sections 9.1](#) and [16](#). In [sections 30](#) and [31](#) and in [Theorem D](#) the fields are abstract t -series fields over an arbitrary characteristic-zero field k ; from [section 33](#) on they are subfields of **No** or **No**(i), with $k = \mathbb{R}$ or $\mathbb{C}$.

The following symbols of the source are renamed; everything else is as in the source. No normalization is changed.

Here	Source 03	Reason
$M = K_0 K_1$, the ordinary compositum; M' , M^{alg} , M_*	L ; L' , L^{alg} , L_*	this report writes M for the ordinary compositum (Theorem C , source 02); L is the common core $L_k = \mathcal{K}_k(H)$ of Theorem A and, in section 16 , the field $k((t^\Gamma))$
Γ_0, Γ_1 ; $K_0 = k((t^{\Gamma_0}))$, $K_1 = k((t^{\Gamma_1}))$; $K_{0,*}$, $K_{1,*}$	Γ_1, Γ_2 ; K_1, K_2 ; $K_{1,*}$, $K_{2,*}$	this report indexes the two factors of a join by 0, 1 (G_0, G_1 , K_0, K_1 in section 8); the first factor, over which the annihilator is solved, plays the role of the coefficient field K_0 there

Here	Source 03	Reason
$\mathcal{K}_k(G)$ $\partial_\chi, \partial_i$ (Euler derivations); $\partial_c = \sum c_i \partial_i$ (the annihilator)	$\mathcal{H}_k(G)$ $D_\chi, D_i; D$	the same full Hahn field; this report's macro D is always a field in section 15.5 , and $D_{\mathbb{R}} = \mathbb{Z}$, $D_{\mathbb{C}} = \mathbb{Z}[i]$ are rings of integers; ∂_χ is the Euler derivation (16.2) with weight $\lambda = \chi$
δ , an ordinary relative derivation (section 37)	D	as above
Γ' , a subgroup in lemmas 30.1 and 30.3	Δ	Δ is the Euler derivation of section 22.3
$\Pi_k(G)$, purely infinite series; $\mathcal{A}_k(G) = D_k \oplus \Pi_k(G)$	$I_k(G); R_k(G)$	the same objects as in section 9 ; I is an index set and I_α an interval here, and R_j are omnific copies
$\mathcal{A}_k^D(G) = D \oplus \Pi_k(G)$ for a unital ring $D \subseteq k$, in particular $\mathcal{A}_k^k(G)$	$S; T$ (lemma 35.2)	S is a support set and the witness (8.3); T is kept for the local set in lemma 32.1
$\mathcal{A}_k(A_\kappa), \mathcal{A}_k(B_\kappa)$, $\mathcal{A}_k(A_\kappa + B_\kappa); \mathcal{A}_k(A_\kappa)[\mathcal{A}_k(B_\kappa)]$	$R_{A,k}, R_{B,k}, R_{E,k};$ S_k	as above; the ring generated by two rings is written as $R_0[R_1]$ in section 24
$\mathcal{T}$, a transcendence basis; $\varphi : \mathcal{T} \rightarrow E; dy, y \in \mathcal{T}$	$T; h; dt, t \in T$	as source 02 does (lemma 20.1); h, h_α are embeddings here; t is the Hahn variable
$\mathcal{I}_\kappa$, the index set of lemma 32.1	I_κ	I_α is an interval of section 7
$\mathfrak{B}_\eta$, the branch sets	B_η	B is the group $\text{span}_{\mathbb{Q}}\{\mu_n\}$ in the same theorem
Ξ_U , the omnific witnesses of Theorem E	Z_U	$Z_\alpha = (1+i)Y_\alpha$ are source 02's Gaussian witnesses (theorem 24.4)
Λ_α , the ordinal blocks of section 36	C_α	C_Θ is a localized coefficient field (lemma 23.1)
Theorems D, E, F, G	Theorems A, B, C, D	the lettered counter continues this report's Theorems A–C
Research questions 23–32	Questions 1–10	the question counter continues

Symbols kept, and tempting false readings.

- $H = \Gamma_0 \cap \Gamma_1$ is the common core of the two exponent groups, in the same role as the core H of [Theorem A](#); here it is an arbitrary subgroup of an abstract ordered group, and all examples have $H = 0$. $\Gamma = \Gamma_0 + \Gamma_1$ is the exponent group of the join E , as Γ is the outer group of $E = F((t^\Gamma))$ in source 02.
- $A, B; A_\kappa, B_\kappa; A_*, B_*$ are exponent *subgroups*, as A, B are in [lemma 8.1](#). They are not the rings $\mathcal{A}_k(H), \mathcal{B}_k(H)$ (calligraphic), not a parameter set A ([corollary 6.1](#)), and not the index set B or the labels b_i of [section 20](#).
- χ, χ_i are additive *characters* $\Gamma \rightarrow k$: the Euler weight λ of [definition 16.1](#) under another name. They are not the characters χ of the monomial maps $M_{\chi, \tau}$ of the omnific-preserving report.
- λ_n, μ_n, ρ_n ([Theorem F](#)) are positive *real numbers* used as exponents. λ_n is not the Euler weight λ of [definition 16.1](#) or a coset slice $\lambda_{g,H}$ of (5.1); μ_n is not $\mu = |F|$ of [theorem 21.1](#); ρ_n is not the embedding ρ of the omnific-preserving classification ([section 1.6](#)). p_n is the n th prime, not an evaluated partial p_{ij} of [theorem 19.2](#).
- $e_\alpha = \omega^{-\iota_0(\alpha)}$, $f_\alpha = \omega^{-\iota_1(\alpha)}$ and $r_\alpha = e_\alpha + f_\alpha$ are infinitesimal surreal *exponents* (for $\alpha \geq 1$; $e_0 = 1$). Here $\iota_0(\alpha) = 2 \cdot_{\text{ord}} \alpha$ is an *ordinal* product. *Tempting false reading*: “ $e_\alpha = \omega^{-2\alpha}$ with the

surreal product 2α .” For infinite α the two differ: $2 \cdot_{\text{ord}} \omega = \omega$, whereas the surreal $2\omega = \omega + \omega$. *Tempting false reading:* “ r_α is a real coefficient, as r_g in $\sum r_g \omega^g$.” It is not; r_α is itself a Conway exponent of the witnesses Ξ_U and W_α . Nor is e the generic exponent of [section 22](#).

- $\iota_0(\alpha) = 2 \cdot_{\text{ord}} \alpha$ and $\iota_1(\alpha)$ are ordinal-index maps, not the inner lifts ι_{σ_α} of the omnific-preserving report.
- F_U, F_η are mixed sums indexed by an incidence set or a branch; they are not the witness F of [Theorem B](#), the coefficient field F of source 02, or a polynomial. P_η is a prime witness indexed by a binary branch; P without index is a polynomial relation, and P_α are the partial sums of [section 10](#). W_α ([equation \(36.1\)](#)) is an omnific witness, not a leading polynomial $W_i(X, x)$ of [theorem 19.2](#).
- $\eta, \theta \in 2^{\mathbb{N}}$ are binary branches, as η is in [corollary 19.4](#); they are not the η_κ -orders of [section 11](#), and θ is not the inner support set Θ . s is a finite subset of κ or a binary word, and in [theorem 35.1](#) s_n are real t -exponents, not the coefficients $s_n = \omega^{g_n}$ of [\(8.3\)](#). δ is a derivation here, not the Taylor parameter of the omnific-preserving report; δ_{ih} is the Kronecker symbol. κ is an arbitrary infinite cardinal, not $\text{cf}(H)$, and λ without index is a cardinal in the proofs of [Theorem G](#) and [theorem 37.5](#).
- D is a unital subring of k in [section 35](#), in particular $D_{\mathbb{R}} = \mathbb{Z}$ or $D_{\mathbb{C}} = \mathbb{Z}[i]$ as in [section 9](#); it is never a derivation (source 02’s rule that D is a field applies to its own sections). q is an index of the generators u_q in [lemma 30.5](#) and a prime series in [section 35](#); g is a generic exponent in the proof of [proposition 37.2](#), and $\omega^{7/2}$ only in [remark 36.2](#).
- *Tempting false reading:* “the annihilator $\partial_{\mathcal{C}}$ is a derivation of E killing the compositum M .” It kills only a finitely generated subfield $K_1(u_1, \dots, u_m)$ chosen after a hypothetical relation is fixed; a nonzero strong derivation killing all of M does not exist ([proposition 37.2](#)).
- *Tempting false reading:* “the automorphisms of [theorem 37.5](#) are automorphisms of $\mathbf{No}(i)$ or of $\mathbf{Oz}$.” They are automorphisms of one set-sized complex Hahn join E fixing M , and nonidentity ones are not strong.
- *Strong* means preserving Hahn summability of set-indexed families, as in [section 1.5](#); ordinal operations are written $\cdot_{\text{ord}}, +_{\text{ord}}$ as in the source.

28.6 Place in the collection

[merge] Source 03 cites only this report from the collection (and the catalogue, at its pin). Checked against the collection at the placement commit `dec8d56`, the following reports bear on it; none is a dependency of its proofs.

1. *This report, source 01.* [Proposition 30.2](#) is the case $H = 0$ of [theorem 5.4](#) (the intersection is k and the fields are linearly disjoint over it), by the same coefficientwise argument; [lemma 30.1](#) is the elementary coset fact behind the slices [\(5.1\)](#). The definitions of $\Pi_k(G)$ and $\mathcal{A}_k(G)$ in [section 29](#) are those of [section 9](#). The source’s comparison with source 01 is accurate: [theorem 8.3](#) uses separated scales and an infinite independent coefficient list, and the source does not present that mechanism as new.
2. *This report, source 02* ([section 28.7](#)). Written from the same snapshot, source 02 answered Research question 3 in the separated case; the two sources overlap in no theorem.

3. *Tail spans and differential transcendence* (`surreal/tail-spans-and-differential-transcendence`, prefix `tail:`). The binary-branch sets (32.1) are the almost disjoint family `tail:eq:Aalpha` with the coding `tail:eq:binarycode` shifted by one: the source's code of a word s of length m is $2^m - 1 + \sum_j s_j 2^{m-1-j}$, that report's is $N(s) = 2^m + \sum_j s_j 2^{m-j}$ with 1-based indices, so the source's code is $N(s) - 1$. The same family serves [corollary 19.4](#). The linear independence of square roots of distinct primes, used for the exponents (34.1), is also the multiquadratic input of `tail:thm:surreal`, where the square roots are coefficients, not exponents. Source 03 cites neither.
4. *Omnific Diophantine geometry* (`surreal/omnific-diophantine-geometry`, prefix `odg:`). After `odg:prop:irreduciblect` that report cites the primality of $1 + \sum_{k \geq 1} \omega^{1/k}$ in $\mathbf{Oz}$ from the same sources (Pitteloud; Biljakovic–Kochetov–Kuhlmann; as reviewed in [3, Section 1.3]). [Proposition 35.3](#) is the same one-row input with the constant-term pullback written out, and the P_η are a new explicit continuum subfamily of such primes. A search of the collection found no other nonconstant prime element of $\mathbf{Oz}[i]$ (the set-sized-quotients report treats the ideals generated by ordinary Gaussian primes such as $1 + i$), so primality of the P_η in $\mathbf{Oz}[i]$ appears here first in the collection; this is not a priority claim. Every P_η has constant term 1, as `odg:prop:irreduciblect` requires of a nonconstant irreducible.
5. *Set-sized quotients of omnific integers* (`surreal/set-sized-quotients-of-omnific-integers`, prefix `osq:`). `osq:cor:primeexample` uses the prime $\omega^{\sqrt{2}} + \omega + 1$ (L'Innocente–Mantova's Theorem B) to obtain a proper-class domain $\mathbf{Oz}/(q)$ with no nonzero set-sized image. Its proof uses only primality and $\text{ct}(q) = 1$, so each P_η gives such a quotient; this is an observation of the merge, not a statement of either source.
6. *Bounded-support transcendence* (`surreal/transcendence-over-bounded-support`, prefix `bst:`). `bst:cor:exactcard` and `bst:cor:realgroup` compute transcendence degrees 2^κ and $\mathfrak{c}$ of one full Hahn field over its bounded-support fraction field. [Theorems E](#) and [F](#) compute the same cardinals for a join over the compositum of two full Hahn subfields. The bases differ, and the results are consistent: in each case the degree equals the cardinality of the ambient field. The Archimedean join $\mathcal{K}_k(A + B)$ of [Theorem F](#) has $\mathfrak{c}$ as its degree over both bases.
7. *Omnific-preserving automorphisms* (`opa:`). `opa:as:thm:main` shows that every ring automorphism of $\mathbf{Oz}$ extends to a strong automorphism of $\mathbf{No}$, and `opa:as:cor:setpairs` that every automorphism of a full real Hahn field on a nonzero divisible set group stabilizing its omnific ring is strong. The non-strong automorphisms of [theorem 37.5](#) are automorphisms of a complex Hahn join, not of $\mathbf{Oz}$, and are not claimed to preserve any omnific ring, so there is no conflict; [remark 37.6](#) combines the two results.

No `isc:hj:` label has a Lean mapping in `FORMALIZATION.md`.

28.7 Comparison with source 02

Remark 28.1 (Two answers to the degree question). [merge] Sources 02 and 03 were written from the same snapshot of this report and do not cite each other.

- (a) *Settings*. [Theorem 21.1](#) extends coefficients: it computes $\text{dtrdeg} = \text{trdeg} = |F|^{\aleph_0}$ for $F((t))$ over $Fk((t))$, with exponents in $\mathbb{Z}$ and the gap coming from the coefficient field F . [Theorem E](#) mixes exponents: both factors are full Hahn fields over the same k on interleaved groups, and the degree is 2^κ . Both degrees are maximal, equal to the cardinality of the ambient field; they

agree in value for $\kappa = \aleph_0$ and $|F| = \mathfrak{c}$.

- (b) *Mechanisms.* Source 03's finite-exclusive infinitude ([definition 31.1](#)) is word for word the finite private-index property [definition 19.1](#), and the private-label hypothesis of [theorem 20.2](#) is its label form. Source 02 kills a finitely generated coefficient field with coordinate *coefficient* derivations d_b ([lemma 20.1](#)); source 03 kills finitely many elements of the factor K_0 with a K_0 -linear combination of exponent-coordinate Euler derivations ∂_i ([lemma 30.5](#)), which plays the same role. In the second route of [remark 36.2](#) the ∂_i act on the outer coefficients $s_n = \omega^{g_n}$ of (8.3) as $s_n \partial / \partial s_n$, a logarithmic form of the coefficient derivations d_b .
- (c) *Scope.* Source 02 proves differential independence, tail resilience over every set of parameters and arithmetic nonintegrality, in the *separated* explicit join. Source 03 proves algebraic independence only, but without convex separation, with an exact set-sized degree, prime witnesses and a count of relative derivations and automorphisms.
- (d) *Class size.* [Theorem 22.2](#) and [corollary 23.4](#) work in the join of two actual copies of **No** ([section 7](#)); [Theorem G](#) works in explicit class *support subfields* $\mathcal{K}_k(A_*)$, $\mathcal{K}_k(B_*)$, which are not copies of **No**, as the source says.
- (e) *Coverage.* [Theorem D](#) as stated does not cover source 02's family (Y_α) : in the labels (α, n) its G_1 -exponents $b - ng$ do not depend on α , so hypothesis (ii) fails for the family, although each single Y_α satisfies it.

29 Conventions and foundational inputs of the third source

29.1 Hahn fields and summable families

Throughout the abstract argument, k is a set-sized field of characteristic zero and Γ is a set-sized divisible ordered abelian group. Divisibility makes Γ a $\mathbb{Q}$ -vector space, uniquely, because an ordered abelian group is torsion free. Its Hahn field consists of formal sums

$$k((t^\Gamma)) = \left\{ x = \sum_{\gamma \in \Gamma} x_\gamma t^\gamma : x_\gamma \in k, \text{ supp}(x) = \{\gamma : x_\gamma \neq 0\} \text{ is well ordered} \right\}.$$

Addition is coefficientwise and multiplication is convolution. The classical Hahn support theorem ensures that the sum of two well-ordered supports is well ordered and that each exponent has only finitely many representations from the two supports. It follows that every convolution coefficient is an ordinary finite sum. Inversion makes this a field. These are standard foundations, not results proposed as new here [[15](#), [1](#)].

A set-indexed family (x_j) is *Hahn summable* if its union of supports is well ordered and every exponent belongs to only finitely many of the supports. Its Hahn sum is then defined coefficientwise. No analytic convergence or passage to a topological limit is intended. In particular, the families of distinct monomials in [Theorem D](#) are summable because their total support is well ordered.

The natural valuation is $v(x) = \min \text{supp}(x)$ for $x \neq 0$. It is useful for orientation but does not enter the annihilator argument. In particular, no assumption about the cofinality of the valuation group is needed.

29.2 The Conway realization and the two coefficient roles

Conway normal form identifies $\mathbf{No}$ with real-coefficient, set-supported reverse-well-ordered sums $\sum r_g \omega^g$. Sending t^γ to $\omega^{-\gamma}$ identifies a Hahn field on a subgroup of $\mathbf{No}$ with a subfield of $\mathbf{No}$. Complex coefficients are handled through $\mathbf{No}(i) = \mathbf{No} \oplus i\mathbf{No}$. The union of the real and imaginary supports is again reverse well ordered, so this agrees with the formal complex Hahn presentation. The usual real-closedness theorem gives that $\mathcal{K}_{\mathbb{R}}(G)$ is real closed when G is divisible, and its complexification $\mathcal{K}_{\mathbb{C}}(G)$ is algebraically closed [1, 16].

An exponent that happens to be a real number is not a coefficient. For example, λ_n in Theorem F is fixed when it is regarded as an element of the coefficient field, while the monomial t^{λ_n} can have a nonzero Euler derivative. This distinction is essential. The derivations below are not assumed to respect a surreal exponential function, and are not the Berarducci–Mantova surreal derivation. They are additive-character derivations of a Hahn field.

29.3 Omnific and Gaussian omnific rings

For a subgroup $G \subseteq \mathbf{No}$ the additive group of purely infinite series is, as in section 9,

$$\Pi_k(G) = \left\{ \sum_{g \in S} c_g \omega^g \in \mathcal{K}_k(G) : S \subseteq G^{>0} \right\}.$$

The product of two such series again has strictly positive exponents. Thus $k \oplus \Pi_k(G)$ is a ring and its constant-term map to k is a ring homomorphism. For $k = \mathbb{R}$ and $k = \mathbb{C}$ respectively,

$$\mathcal{A}_{\mathbb{R}}(G) = \mathbb{Z} \oplus \Pi_{\mathbb{R}}(G) = \mathbf{Oz} \cap \mathcal{K}_{\mathbb{R}}(G), \quad \mathcal{A}_{\mathbb{C}}(G) = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}(G) = \mathbf{Oz}[i] \cap \mathcal{K}_{\mathbb{C}}(G).$$

These descriptions are exact. A real surreal is omnific precisely when it has no negative Conway exponent and its constant coefficient is an ordinary integer [14, 1].

We will not use the false blanket assertion $\text{Frac}(\mathcal{A}_k(G)) = \mathcal{K}_k(G)$ for arbitrary set-sized G . Even though every surreal is a quotient of full omnific integers, this identity need not remain true after imposing a fixed exponent group. The arithmetic corollary in section 33.3 only uses inclusions that follow directly from the definitions. [merge] This report proves the failure: for $H \neq 0$ the field $\text{Frac} \mathcal{A}_k(H)$ is $\text{Frac} \mathcal{B}_k(H)$ (proposition 9.3), over which $\mathcal{K}_k(H)$ is transcendental (theorem 10.2).

29.4 Set theory and class statements

Section 29 through section 35 use ordinary set mathematics with choice. The optional class results can be read in a class theory such as NBG, the framework of section 2. Every Hahn support is still a set. The class exponent groups used there have explicit finite-support coordinate descriptions, so no proper-class Hamel basis is selected. Finite linear algebra and finite polynomial identities over a class field have their usual meanings. Where a published theorem is stated for set groups, the application to a class ring will be localized to a set group containing all relevant supports.

30 Three algebraic lemmas

Whenever the data of Theorem D are used in this section, write $\Gamma = \Gamma_0 + \Gamma_1$, $K_j = k((t^{\Gamma_j}))$, $E = k((t^{\Gamma}))$, and $H = \Gamma_0 \cap \Gamma_1$.

30.1 Distinct exponent cosets cannot cancel

Lemma 30.1 (Coset independence). *Let $\Gamma' \subseteq \Gamma$ be an additive subgroup. If $\beta_1, \dots, \beta_r$ represent distinct cosets modulo Γ' , then $t^{\beta_1}, \dots, t^{\beta_r}$ are linearly independent over $k((t^{\Gamma'}))$ inside $k((t^\Gamma))$.*

Proof. If $c_j \in k((t^{\Gamma'}))$, then $\text{supp}(c_j t^{\beta_j}) \subseteq \Gamma' + \beta_j$. These cosets are pairwise disjoint. For a nonzero c_j , any exponent of $c_j t^{\beta_j}$ has a nonzero coefficient that cannot receive a contribution from another summand. Hence $\sum_j c_j t^{\beta_j} = 0$ forces every $c_j = 0$. $\square$

No convexity assumption appears in this argument. Even very closely interleaved cosets remain disjoint as sets of exponents. This is the replacement for a separated-scale coefficient expansion.

Proposition 30.2 (Intersection and linear disjointness). *Suppose $\Gamma_0 \cap \Gamma_1 = 0$, and put $K_j = k((t^{\Gamma_j}))$. Then $K_0 \cap K_1 = k$, and K_0, K_1 are linearly disjoint over k in $k((t^{\Gamma_0 + \Gamma_1}))$.*

Proof. An element of the intersection has support contained in both groups, hence in $\{0\}$, which proves the first assertion. To prove the second, let $y_1, \dots, y_r \in K_1$ be linearly independent over k , and assume $\sum_j x_j y_j = 0$ with $x_j \in K_0$. Every exponent in $\Gamma_0 + \Gamma_1$ has a unique expression $a + b$ with $a \in \Gamma_0$, $b \in \Gamma_1$. For a fixed a , the coefficients in the coset $a + \Gamma_1$ of the asserted relation therefore give

$$\sum_{j=1}^r (x_j)_a y_j = 0.$$

The independence of the y_j implies $(x_j)_a = 0$ for every j . As a was arbitrary, all x_j vanish. This is the linear-disjointness criterion. $\square$

[merge] This is the case $H = 0$ of [theorem 5.4 \(section 28.6\)](#).

30.2 Coordinate Euler derivations

Lemma 30.3 (An additive character gives a derivation). *Let $\chi : \Gamma \rightarrow k$ be an additive map. Then*

$$\partial_\chi \left(\sum_\gamma x_\gamma t^\gamma \right) = \sum_\gamma \chi(\gamma) x_\gamma t^\gamma \quad (30.1)$$

defines a k -linear derivation of $k((t^\Gamma))$. It preserves Hahn summability and commutes with Hahn sums. If χ vanishes on a subgroup Γ' , it annihilates $k((t^{\Gamma'}))$.

Proof. The support of the output is contained in that of the input, so the series is well defined. Additivity and k -linearity are immediate. For a product, its coefficient at γ is a finite convolution sum. Using $\chi(\alpha + \beta) = \chi(\alpha) + \chi(\beta)$ in that finite sum shows

$$\partial_\chi(xy) = \partial_\chi(x)y + x\partial_\chi(y).$$

For a summable family, its union of supports cannot increase under ∂_χ , and coefficientwise local finiteness cannot fail. The equality between differentiating the sum and summing the derivatives reduces to linearity in a finite coefficient sum. The last assertion follows from (30.1). $\square$

[merge] This is [lemma 16.2](#) for $F = k$, with the Euler weight $\lambda = \chi$.

Lemma 30.4 (Coordinate characters modulo the other group). *Under hypotheses (i) and (ii) of Theorem D, for each $i \in I$ there is a $\mathbb{Q}$ -linear map $\chi_i : \Gamma_0 + \Gamma_1 \rightarrow \mathbb{Q} \subseteq k$ such that*

$$\chi_i|_{\Gamma_1} = 0, \quad \chi_i(a_h) = \delta_{ih} \quad (h \in I).$$

Writing $\partial_i = \partial_{\chi_i}$, one has $\partial_i(K_0) \subseteq K_0$, $\partial_i(K_1) = 0$, and

$$\partial_i(F_U) = \begin{cases} t^{a_i+b_i} & i \in U, \\ 0 & i \notin U. \end{cases} \quad (30.2)$$

Proof. The classes of the a_h are linearly independent in $(\Gamma_0 + \Gamma_1)/\Gamma_1$: a finite rational combination lying in Γ_1 also lies in Γ_0 , hence in H , so all coefficients vanish. Extend these classes to a $\mathbb{Q}$ -basis of the quotient, assign value 1 to the class of a_i and 0 to the other basis vectors, and compose with the quotient map. Apply lemma 30.3. The value on $a_h + b_h$ is δ_{ih} , which gives (30.2). $\square$

Only hypothesis (i) is needed to construct these characters; hypothesis (ii) will ensure that a linear combination of the resulting nonzero monomial derivatives does not cancel.

30.3 Annihilating finitely many arbitrary Hahn series

Lemma 30.5 (Finite differential annihilator). *Let $u_1, \dots, u_m \in K_0$, and let $J \subseteq I$ consist of exactly $m + 1$ distinct indices. There are coefficients $(c_i)_{i \in J} \in K_0^J$, not all zero, such that the derivation*

$$\partial_{\mathbf{c}} = \sum_{i \in J} c_i \partial_i \quad (30.3)$$

annihilates $K_1(u_1, \dots, u_m)$. In addition,

$$\sum_{i \in J} c_i t^{a_i+b_i} \neq 0. \quad (30.4)$$

Proof. Consider the homogeneous equations

$$\sum_{i \in J} c_i \partial_i(u_q) = 0 \quad (1 \leq q \leq m).$$

There are m equations in $m + 1$ unknowns over the field K_0 , because $\partial_i(u_q) \in K_0$. Gaussian elimination gives a nonzero solution. A finite field-linear combination of derivations is again a derivation, so (30.3) is well defined. It annihilates every u_q and all of K_1 . The quotient rule then implies that it annihilates the field they generate.

For (30.4), write the left side as $\sum_{i \in J} (c_i t^{a_i}) t^{b_i}$. The cosets $b_i + \Gamma_0$ are pairwise distinct: a difference in Γ_0 would be in H , contrary to hypothesis (ii). The coefficients $c_i t^{a_i}$ lie in K_0 , and some are nonzero. Now apply lemma 30.1. $\square$

Remark 30.6 (Why arbitrary coefficient supports are harmless). The u_q may each have infinitely many, or uncountably many, support exponents. We do not choose coordinates outside all their supports. We choose one more differential direction than there are generators, and solve a finite homogeneous system over K_0 itself. This is the principal reason the proof applies to a compositum of *full* Hahn fields.

31 The mixed-support independence theorem

Definition 31.1. A family $\mathcal{U}$ of subsets of I has the *finite-exclusive infinitude property* if, for every finite list of distinct members $U_1, \dots, U_s$ and each j , the set

$$U_j \setminus \bigcup_{\ell \neq j} U_\ell$$

is infinite. This is a property of the incidence sets only; it says nothing about the order on the exponent group.

[merge] This is the finite private-index property of [definition 19.1](#), for families of subsets of an arbitrary index set.

Proof of Theorem D. Write $K_j = k((t^{\Gamma_j}))$, $M = K_0 K_1$, and $E = k((t^{\Gamma_0 + \Gamma_1}))$. The mixed exponents $a_i + b_i$ are pairwise distinct. Indeed, equality for two indices would give a nontrivial relation between the corresponding $a_i + H$. Their assumed well-ordering therefore makes every F_U a well-defined Hahn sum.

Suppose there is an algebraic relation among finitely many distinct $F_{U_1}, \dots, F_{U_s}$. Choose a nonzero polynomial

$$P \in M[X_1, \dots, X_s], \quad P(F_{U_1}, \dots, F_{U_s}) = 0,$$

of least possible total degree among nonzero relations on this tuple. It is not a constant polynomial. In characteristic zero, some partial $P_j = \partial P / \partial X_j$ is a nonzero polynomial. It has smaller total degree, so minimality gives

$$P_j(F_{U_1}, \dots, F_{U_s}) \neq 0. \tag{31.1}$$

There are only finitely many coefficients of P . Each is a rational expression in finitely many elements of K_0 and K_1 . Taking the union of the finite lists of K_0 -elements used in these expressions, there are $u_1, \dots, u_m \in K_0$ such that all coefficients of P lie in $K_1(u_1, \dots, u_m)$.

Choose $m + 1$ distinct indices in $U_j \setminus \bigcup_{\ell \neq j} U_\ell$, and call the resulting set J . Apply [lemma 30.5](#) to obtain $\partial_{\mathbf{c}} = \sum_{i \in J} c_i \partial_i$. This derivation annihilates the coefficients of P . By [\(30.2\)](#), it also annihilates each F_{U_ℓ} with $\ell \neq j$, whereas

$$\partial_{\mathbf{c}}(F_{U_j}) = \sum_{i \in J} c_i t^{a_i + b_i} \neq 0.$$

Differentiating the polynomial identity, using the ordinary finite polynomial chain rule, now yields

$$0 = \partial_{\mathbf{c}}(P(F_{U_1}, \dots, F_{U_s})) = P_j(F_{U_1}, \dots, F_{U_s}) \partial_{\mathbf{c}}(F_{U_j}).$$

Both factors on the right are nonzero by [\(31.1\)](#) and [lemma 30.5](#), contradicting that E is a field. Thus no algebraic relation exists. $\square$

31.1 A finite-generator formulation

The proof has a useful quantitative form even when the exclusive sets are not known to be infinite.

Proposition 31.2 (Generator-sensitive obstruction). *Fix $u_1, \dots, u_m \in K_0$ and finite subsets $U_1, \dots, U_s$ or arbitrary summable subsets of I . If each exclusive part $U_j \setminus \bigcup_{\ell \neq j} U_\ell$ has at least $m + 1$ elements, then the corresponding F_{U_j} are algebraically independent over $K_1(u_1, \dots, u_m)$.*

Proof. Choose a relation of minimal total degree over this fixed coefficient field. Its nonzero partial selects one j . The $m + 1$ available exclusive indices give the same annihilator and contradiction as above. $\square$

This statement is consistent with finite mixed sums lying in M : representing enough of them in a field $K_1(u_1, \dots, u_m)$ requires enough K_0 -generators. It does not give a bound on the expression length of an arbitrary Hahn series, because the choice of generators is itself unrestricted.

31.2 Stable variants

Corollary 31.3 (Weights and finite modifications). *In Theorem D, replace F_U by*

$$\tilde{F}_U = d_U + \sum_{i \in U} r_{U,i} t^{a_i + b_i}, \quad d_U \in M, \quad r_{U,i} \in k^\times.$$

Then the resulting family is still algebraically independent over M . Finite changes of the supports or coefficients also preserve the conclusion, provided the remaining nonzero terms satisfy the same exclusive infinitude property.

Proof. First take $d_U = 0$. On an exclusive set the derivative has the form $\sum c_i r_{U,i} t^{a_i + b_i}$; its coefficients in the distinct cosets remain nonzero whenever $c_i \neq 0$. The proof is unchanged. Translation by elements of the base field is an invertible polynomial change of variables. Finally, a finite sum of mixed monomials belongs to M , so finite modifications can also be absorbed into the translations. $\square$

Corollary 31.4 (Algebraic enlargement of the base). *Let M' be an algebraic extension of M contained in E . Every family proved independent over M above remains independent over M' . In particular this applies to the relative algebraic closure of M in E .*

Proof. For a finite tuple $\mathbf{F}$, the extension $M'(\mathbf{F})/M(\mathbf{F})$ is algebraic. The transcendence-degree formula, or the elementary exchange lemma for algebraic dependence, gives $\text{trdeg}(M'(\mathbf{F})/M') = \text{trdeg}(M(\mathbf{F})/M)$. Thus the degree equals the length of every finite tuple of distinct members, as required. $\square$

When E is real closed, the relative algebraic closure of M in E is real closed: positive square roots and roots of odd-degree polynomials in E are algebraic over M . When E is algebraically closed, the relative algebraic closure is algebraically closed for the same reason. Consequently passing from the compositum to its appropriate algebraic closure does not eliminate the transcendental gap. [merge] [Remark 22.4](#) is the same observation for the Euler jets of source 02.

32 Families of supports of the required size

The differential theorem separates the algebra from the combinatorics. This section supplies the support families. The constructions themselves are elementary standard independent-family and almost-disjoint-family arguments; no novelty is claimed for them.

32.1 Every infinite cardinal

Lemma 32.1 (A maximal independent incidence family). *For every infinite cardinal κ there is a set $\mathcal{I}_\kappa$ of cardinality κ and a family $(U_X)_{X \subseteq \kappa}$ of 2^κ subsets of $\mathcal{I}_\kappa$ such that every finite Boolean pattern on distinct members is realized at κ indices. In particular the family has finite-exclusive infinitude.*

Proof. Take

$$\mathcal{I}_\kappa = \{(s, T, \xi) : s \in [\kappa]^{<\omega}, T \subseteq \mathcal{P}(s), \xi < \kappa\}.$$

There are κ finite subsets s , only finitely many choices of T for each fixed finite s , and κ choices of ξ . Thus $|\mathcal{I}_\kappa| = \kappa$. For $X \subseteq \kappa$ define

$$U_X = \{(s, T, \xi) : X \cap s \in T\}.$$

Given distinct $X_1, \dots, X_r$, choose, for each pair, an element of their symmetric difference. The finite set s of the chosen elements makes all traces $X_j \cap s$ distinct. For any prescribed pattern $\varepsilon_j \in \{0, 1\}$, choose T to contain exactly those of these traces for which $\varepsilon_j = 1$. Then every (s, T, ξ) , $\xi < \kappa$, realizes the desired pattern. This gives at least κ witnesses, and there cannot be more because $|\mathcal{I}_\kappa| = \kappa$. Taking a pattern that distinguishes a pair also shows that all U_X are different. $\square$

Fixing any bijection between $\mathcal{I}_\kappa$ and κ transfers the family to subsets of an ordinal-indexed support. The exponent groups in the next section are explicit without this choice; the formula for the witnesses is explicit relative to the chosen bijection.

32.2 A concrete countable support

For a finite binary word $s = (s_0, \dots, s_{m-1})$, set

$$\text{code}(s) = 2^m - 1 + \sum_{j=0}^{m-1} s_j 2^{m-1-j}.$$

Words of length m have codes in $[2^m - 1, 2^{m+1} - 2]$, and words of different lengths have disjoint code intervals. For an infinite binary sequence $\eta \in 2^\mathbb{N}$ let

$$\mathfrak{B}_\eta = \{\text{code}(\eta|_m) : m \geq 1\}. \quad (32.1)$$

These are infinite subsets of $\mathbb{N}$.

[merge] With 1-based indices this is $N(s) - 1$ for the coding `tail:eq:binarycode` of the tail-spans report, and $\mathfrak{B}_\eta$ is its set `tail:eq:Aalpha` shifted down by one; the same kind of family serves [corollary 19.4](#).

Lemma 32.2 (Binary branches give exclusive tails). *Distinct $\mathfrak{B}_\eta, \mathfrak{B}_\theta$ have finite intersection. Hence the continuum sized family $(\mathfrak{B}_\eta)_{\eta \in 2^\mathbb{N}}$ has finite-exclusive infinitude.*

Proof. Once two branches first differ, their prefixes of every larger length differ, and therefore so do their codes. They can share only the finitely many earlier prefixes. For finitely many other branches, remove the finite union of these finite intersections from a fixed $\mathfrak{B}_\eta$; an infinite tail remains. $\square$

For example, the constantly zero and constantly one branches give

$$\mathfrak{B}_{000\dots} = \{1, 3, 7, 15, \dots\}, \quad \mathfrak{B}_{111\dots} = \{2, 6, 14, 30, \dots\}.$$

This fully specified family is used for the prime witnesses.

33 Maximal transcendence at every set cardinal

33.1 Two explicit independent exponent groups

It is important to distinguish ordinal operations used as indices from surreal arithmetic used inside the field. Define ordinal indices

$$\iota_0(\alpha) = 2 \cdot_{\text{ord}} \alpha, \quad \iota_1(\alpha) = (2 \cdot_{\text{ord}} \alpha) +_{\text{ord}} 1.$$

Here $\cdot_{\text{ord}}$ and $+_{\text{ord}}$ are ordinary ordinal multiplication and addition, not Hessenberg operations. They satisfy

$$\iota_0(\alpha) < \iota_1(\alpha) < \iota_0(\beta) \quad (\alpha < \beta).$$

Regard these ordinals as surreal numbers and put

$$e_\alpha = \omega^{-\iota_0(\alpha)}, \quad f_\alpha = \omega^{-\iota_1(\alpha)}, \quad r_\alpha = e_\alpha + f_\alpha. \quad (33.1)$$

Thus $e_0 = 1$, $f_0 = \omega^{-1}$, $e_1 = \omega^{-2}$, $f_1 = \omega^{-3}$, and so on. The symbols e_α, f_α are themselves surreal numbers that will be used as exponents in a second, outer Conway normal form.

Lemma 33.1. *For any infinite cardinal κ define*

$$A_\kappa = \text{span}_{\mathbb{Q}}\{e_\alpha : \alpha < \kappa\}, \quad B_\kappa = \text{span}_{\mathbb{Q}}\{f_\alpha : \alpha < \kappa\}.$$

Then $A_\kappa \cap B_\kappa = 0$, both groups have cardinality κ , and $(r_\alpha)_{\alpha < \kappa}$ is strictly decreasing and positive.

Proof. All the e_α and f_α are distinct Conway monomials. Uniqueness of finite Conway normal form implies that they are linearly independent over $\mathbb{R}$, hence over $\mathbb{Q}$. This proves the intersection claim. A finite rational combination of κ generators has only κ possibilities, and there are κ distinct generators, so the cardinalities are exactly κ .

If $\alpha < \beta$, the largest Conway exponent occurring in $r_\alpha - r_\beta$ is $-\iota_0(\alpha)$, with coefficient 1. Therefore the difference is positive. Each r_α is also positive. $\square$

33.2 Building the independent omnific integers

Transfer the family of [lemma 32.1](#) to a family $\mathcal{U}_\kappa \subseteq \mathcal{P}(\kappa)$ by a fixed bijection. Define

$$\Xi_U = \sum_{\alpha \in U} \omega^{r_\alpha} \quad (U \in \mathcal{U}_\kappa). \quad (33.2)$$

The decreasing support ensures that this is a Conway normal form. Its exponents are positive and its constant coefficient is zero, so it is an omnific integer. There are no analytic convergence conditions to check, even when κ is uncountable.

Proof of Theorem E. In [Theorem D](#), use

$$\Gamma_0 = A_\kappa, \quad \Gamma_1 = B_\kappa, \quad a_\alpha = -e_\alpha, \quad b_\alpha = -f_\alpha.$$

The classes of the a_α are independent modulo the zero intersection. The b_α are distinct. The mixed exponents $-r_\alpha$ are strictly increasing, with order type κ , so are well ordered. Since $t^{-r_\alpha} = \omega^{r_\alpha}$, the theorem proves that the 2^κ elements in (33.2) are algebraically independent over the ordinary compositum, for $k = \mathbb{R}$ and $k = \mathbb{C}$.

Let $\Gamma = A_\kappa + B_\kappa$. Its cardinality is κ . A Hahn series is determined by a function from Γ to k , so

$$|\mathcal{K}_k(\Gamma)| \leq |k|^\kappa = (2^{\aleph_0})^\kappa = 2^\kappa.$$

The independent family supplies the opposite bound and proves that the transcendence degree is exactly 2^κ . The factors are linearly disjoint by [proposition 30.2](#); their real or algebraic closedness follows from divisibility and the classical Hahn closedness theorem. The witnesses have real coefficients, so exactly the same family works inside the complexified fields. $\square$

Each factor field itself has cardinality 2^κ : arbitrary subsets of the decreasing support $(e_\alpha)_{\alpha < \kappa}$, or of the analogous f -support, give 2^κ different series, and the preceding upper bound applies to each factor. The ordinary compositum also has cardinality 2^κ , since it contains a factor and is contained in the join.

Corollary 33.2. *The relative real algebraic closure of $\mathcal{K}_\mathbb{R}(A_\kappa)\mathcal{K}_\mathbb{R}(B_\kappa)$ in the real Hahn join remains of transcendence codimension 2^κ . The relative algebraic closure of the complex compositum has the same property in the complex Hahn join.*

Proof. The independent family survives by [corollary 31.4](#); the upper bound is unchanged. $\square$

33.3 A fraction-field consequence that does not assume descent

Let $\mathcal{A}_k(A_\kappa)$, $\mathcal{A}_k(B_\kappa)$ and $\mathcal{A}_k(A_\kappa + B_\kappa)$ be as defined in [section 29](#). Denote by $\mathcal{A}_k(A_\kappa)[\mathcal{A}_k(B_\kappa)]$ the subring of the join generated by $\mathcal{A}_k(A_\kappa)$ and $\mathcal{A}_k(B_\kappa)$.

Corollary 33.3 (Maximal arithmetic-join gap). *For $k = \mathbb{R}, \mathbb{C}$ one has*

$$\text{trdeg}(\text{Frac } \mathcal{A}_k(A_\kappa + B_\kappa) / \text{Frac}(\mathcal{A}_k(A_\kappa)[\mathcal{A}_k(B_\kappa)])) = 2^\kappa.$$

The family $(\Xi_U)_{U \in \mathcal{U}_\kappa}$ witnesses this equality.

Proof. The ring $\mathcal{A}_k(A_\kappa + B_\kappa)$ contains both factor rings and every Ξ_U . Its fraction field is a subfield of the Hahn join. Also,

$$\text{Frac}(\mathcal{A}_k(A_\kappa)[\mathcal{A}_k(B_\kappa)]) \subseteq \mathcal{K}_k(A_\kappa)\mathcal{K}_k(B_\kappa).$$

Independence over the larger field implies independence over $\text{Frac}(\mathcal{A}_k(A_\kappa)[\mathcal{A}_k(B_\kappa)])$. This gives the lower bound 2^κ . The ambient Hahn join has cardinality 2^κ , which supplies the upper bound. $\square$

This is a failure of reconstructing the arithmetic Hahn join by finite ring operations and fractions from its two pieces. It is distinct from this report's failure of fraction-field descent for *intersections* ([Theorem B](#)). Neither argument authorizes replacing fixed-support omnific fraction fields by the corresponding full Hahn fields without proof. [merge] It is the set-sized, non-separated counterpart of [theorem 24.2](#), which obtains nonintegrality of the omnific join over the ring generated by two explicit separated copies, after any set adjunction.

34 A fully Archimedean, interleaved example

The nested monomial example has many distinct growth scales. We now remove that feature. All exponent groups in this section are subgroups of the ordinary real numbers.

34.1 Independent real exponents approaching zero

We use the elementary multiquadratic fact that the square roots of distinct positive primes are linearly independent over $\mathbb{Q}$. One way to see it is to work in the finite multiquadratic extension generated by any finite list of those roots. Unique prime factorization makes their square classes independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so adjoining the roots gives independent sign-changing automorphisms. Applying the automorphism that changes just one root's sign to a rational linear relation, and subtracting, forces that root's coefficient to vanish. Doing this for every root proves the assertion. This is ordinary finite Galois theory, not an assumption about infinite algebraic extensions.

Let $p_0 = 2, p_1 = 3, p_2 = 5, \dots$ and define

$$\lambda_n = \frac{\sqrt{p_{2n}}}{p_{2n}^2}, \quad \mu_n = \frac{\sqrt{p_{2n+1}}}{p_{2n+1}^2}, \quad \rho_n = \lambda_n + \mu_n. \quad (34.1)$$

Since $p^{-3/2} = \sqrt{p}/p^2$, these are exactly the quantities in [Theorem F](#). Set

$$A = \text{span}_{\mathbb{Q}}\{\lambda_n : n \geq 0\}, \quad B = \text{span}_{\mathbb{Q}}\{\mu_n : n \geq 0\}, \quad \Gamma = A + B.$$

Lemma 34.1. *The combined family of all λ_n, μ_n is $\mathbb{Q}$ -linearly independent. The groups A, B, Γ are countable and divisible, $A \cap B = 0$, and ρ_n strictly decreases to zero in $\mathbb{R}$. The group Γ is Archimedean and has no nonzero proper convex subgroup.*

Proof. Each number in the combined family is a nonzero rational multiple of the square root of a different prime, so the preceding independence fact applies. The claims about cardinalities, divisibility and intersection follow. As the primes strictly increase without bound, both (λ_n) and (μ_n) strictly decrease to zero; their sum does too. Every ordered subgroup of $\mathbb{R}$ is Archimedean. If a convex subgroup contains $g > 0$, then for any $h > 0$ in Γ there is a natural number m with $h < mg$. Convexity then puts h in that subgroup. Hence a nonzero convex subgroup is all of Γ . $\square$

Both A and B are dense in $\mathbb{R}$: each contains the rational multiples of a nonzero real number. Thus this is not a construction in which all nonzero exponents from one factor occupy a convex scale separated from the other. In particular, a proof depending on a nontrivial convex quotient of Γ cannot be used here. [merge] In particular the iterated presentation of [lemma 8.1](#), and with it the rank test [corollary 22.1](#), is not available for this pair.

34.2 Continuum many mixed sums

For the branch sets from [\(32.1\)](#), define

$$F_\eta = \sum_{n \in \mathfrak{B}_\eta} \omega^{\rho_n}, \quad P_\eta = 1 + F_\eta.$$

Every $\mathfrak{B}_\eta$ is an unbounded subset of $\mathbb{N}$, so its increasing enumeration selects a subsequence of (ρ_n) that still decreases to zero. These sums therefore have reverse-well-ordered support of type ω ; adjoining the constant term gives reverse support order type $\omega + 1$.

Proposition 34.2. *The family $(P_\eta)_{\eta \in 2^\mathbb{N}}$ is algebraically independent over $\mathcal{K}_k(A)\mathcal{K}_k(B)$, for both $k = \mathbb{R}$ and $k = \mathbb{C}$. In either case*

$$\text{trdeg}(\mathcal{K}_k(\Gamma)/\mathcal{K}_k(A)\mathcal{K}_k(B)) = \mathfrak{c}.$$

Proof. Use $a_n = -\lambda_n$, $b_n = -\mu_n$ in [Theorem D](#). The mixed t -exponents $-\rho_n$ strictly increase to zero, hence form a well-ordered set of order type ω . Apply [lemma 32.2](#) and then [corollary 31.3](#) to add the constant terms. The lower bound is $|2^{\mathbb{N}}| = \mathfrak{c}$. For the upper bound, Γ is countable and $|k| = \mathfrak{c}$, so the entire Hahn join has cardinality at most $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$. $\square$

For illustration, two members begin as

$$\begin{aligned} P_{000\dots} &= 1 + \omega^{\rho_1} + \omega^{\rho_3} + \omega^{\rho_7} + \omega^{\rho_{15}} + \dots, \\ P_{111\dots} &= 1 + \omega^{\rho_2} + \omega^{\rho_6} + \omega^{\rho_{14}} + \omega^{\rho_{30}} + \dots. \end{aligned}$$

Each displayed monomial already lies in the ordinary compositum because

$$\omega^{\rho_n} = \omega^{\lambda_n} \omega^{\mu_n}.$$

Nevertheless the full family of Hahn sums is algebraically independent over that compositum. This is the sense in which the obstruction is a mixed-summation obstruction rather than a missing-monomial obstruction.

35 Prime omnific and Gaussian omnific witnesses

35.1 The isolated external theorem

The following is the only specialized factorization theorem imported into the source. It is the one-row result of Pitteloud together with the transfer of Biljakovic–Kochetov–Kuhlmann, as stated in [\[3, §1.3\]](#); the latter paper also gives broader primality and transfer criteria in [Theorem F](#) and [Proposition 8.2.9](#).

Theorem 35.1 (Classical one-row prime theorem). *Let k be any characteristic-zero field and let G be a set-sized ordered abelian group containing the ordered additive group $\mathbb{R}$. If $s_0 < s_1 < \dots < 0$ are real numbers with supremum zero and all $c_n \in k$ are nonzero, then*

$$q = 1 + \sum_{n \geq 0} c_n t^{s_n}$$

is prime in the ring of Hahn series with coefficients in k and exponents in $G^{\leq 0}$.

No proof of [theorem 35.1](#) is claimed here; the references [\[17, 18, 3\]](#) supply this classical input. In particular, the independence argument does not reprove or replace the factorization theory behind it. Everything needed to pass from this input to the specific omnific and Gaussian statements is proved next. [\[merge\]](#) Checked in the merge against the arXiv text of [\[3\]](#) (version 5, Section 1.3): Pitteloud proved that $1 + b$, for b a sum over an ω -sequence of real exponents increasing to 0, is prime in $K((\mathbb{R}^{\leq 0}))$ for a field K of characteristic zero, and Biljakovic–Kochetov–Kuhlmann observed that it remains prime in $K((G^{\leq 0}))$ for every $G \supseteq \mathbb{R}$. The statement above agrees with that account; the original papers were not read in the merge.

35.2 Pulling primality through the constant term

Lemma 35.2 (Constant-term-one pullback). *Let k be a field, $D \subseteq k$ a unital subring, and G an ordered abelian group. In the ω convention consider the rings*

$$\mathcal{A}_k^k(G) = k \oplus \Pi_k(G), \quad \mathcal{A}_k^D(G) = D \oplus \Pi_k(G).$$

Suppose $q \in \mathcal{A}_k^D(G)$ is prime in $\mathcal{A}_k^k(G)$ and $\text{ct}(q) = 1$. Then q is prime in $\mathcal{A}_k^D(G)$. More precisely, divisibility by q in $\mathcal{A}_k^k(G)$ of an element of $\mathcal{A}_k^D(G)$ is already divisibility in $\mathcal{A}_k^D(G)$.

Proof. If $x = qu$ with $x \in \mathcal{A}_k^D(G)$, $u \in \mathcal{A}_k^k(G)$, apply the constant-term homomorphism. It gives $\text{ct}(u) = \text{ct}(x)/\text{ct}(q) = \text{ct}(x) \in D$, so $u \in \mathcal{A}_k^D(G)$. Now suppose q divides xy in $\mathcal{A}_k^D(G)$. It divides the same product in $\mathcal{A}_k^k(G)$, so it divides x or y in $\mathcal{A}_k^k(G)$. The preceding observation descends that divisibility to $\mathcal{A}_k^D(G)$. Also q cannot be a unit in $\mathcal{A}_k^D(G)$, since that would make it a unit in $\mathcal{A}_k^k(G)$. It is nonzero, as required for a prime element. $\square$

The constant-term condition is not cosmetic. Without it a quotient that exists after enlarging the constant ring need not lie in the smaller ring. The lemma uses exactly the feature supplied by the constant 1 in every P_η . [merge] For $D = D_k$ one has $\mathcal{A}_k^{D_k}(G) = \mathcal{A}_k(G)$; the ring $\mathcal{A}_\mathbb{R}^\mathbb{R}(G)$ is the omnific-preserving report's $\mathcal{A}_\Gamma = \mathbb{R} \oplus \Pi_\Gamma$ (section 1.5).

35.3 Localization handles the full proper-class rings

Proposition 35.3. *Let $r_0 > r_1 > \dots > 0$ be ordinary real numbers decreasing to zero. Then*

$$q = 1 + \sum_{n \geq 0} c_n \omega^{r_n}$$

is prime in $\mathbf{Oz}$ when $c_n \in \mathbb{R}^\times$, and is prime in $\mathbf{Oz}[i]$ when $c_n \in \mathbb{C}^\times$.

Proof. We give the complex case, with $D = \mathbb{Z}[i]$; the real case is identical with $k = \mathbb{R}$ and $D = \mathbb{Z}$. Suppose $xy = qz$ in $\mathbf{Oz}[i]$. The union of the Conway supports of x, y, z, q is a set. Let G be its $\mathbb{Q}$ -linear span together with the whole set $\mathbb{R}$ inside $\mathbf{No}$. This is a set-sized ordered divisible group containing $\mathbb{R}$. All four elements belong to $\mathcal{A}_k^D(G)$.

Under $t^g = \omega^{-g}$, the series q has real exponents $-r_n$ increasing to zero. By theorem 35.1, it is prime in $\mathcal{A}_k^k(G)$. By lemma 35.2 it is prime in $\mathcal{A}_k^D(G)$. Thus q divides x or y in that set-sized ring, and hence in the full Gaussian omnific ring. The series is nonzero and is not a unit: its largest Conway exponent is positive, and the largest exponent of a product of two nonzero elements of $\mathcal{A}_k^D(\mathbf{No})$ is the sum of their nonnegative largest exponents. Such a product cannot be 1 if one factor has positive largest exponent. This proves primality in the full class ring without applying a set theorem directly to a proper class. $\square$

Completion of the proof of Theorem F. The exponent and field assertions, and algebraic independence, are lemma 34.1 and proposition 34.2. For each branch η , enumerate $\mathfrak{B}_\eta$ increasingly. The resulting subsequence of ρ_n is strictly decreasing to zero. Apply proposition 35.3 with every coefficient equal to 1. This gives both primality assertions simultaneously. $\square$

Corollary 35.4 (Pairwise nonassociate primes). *The P_η form continuum many pairwise nonassociate primes in $\mathbf{Oz}$, and also in $\mathbf{Oz}[i]$, while remaining algebraically independent over the respective full Hahn composita.*

Proof. The only units of $\mathbf{Oz}$ are ± 1 , and those of $\mathbf{Oz}[i]$ are $\pm 1, \pm i$. Indeed, the leading-exponent argument in proposition 35.3 forces any unit and its inverse to have only constant terms, and the constants must be units of $\mathbb{Z}$ or $\mathbb{Z}[i]$. Since every P_η has constant term 1, two associates would have to be equal. Algebraic independence implies that distinct members are not equal. $\square$

The prime conclusion is arithmetic, whereas the independence conclusion is relative to a much larger field than $\mathbb{Q}$ or $\mathbb{Q}(i)$. One should not confuse the two: primality alone supplies no such relative transcendence statement. Conversely the mixed-support proof works for many series with constant term zero, which are not prime in $\mathbf{Oz}$ because they are divisible by every nonzero ordinary integer.

36 A fixed pair of class-sized Hahn subfields

36.1 Explicit coordinates, not a class Hamel basis

Use the exponents in (33.1) for all ordinals and define

$$A_* = \left\{ \sum_{\alpha \in F} q_\alpha e_\alpha : F \subseteq \mathbf{On} \text{ finite, } q_\alpha \in \mathbb{Q} \right\}, \quad B_* = \left\{ \sum_{\alpha \in F} q_\alpha f_\alpha : F \subseteq \mathbf{On} \text{ finite, } q_\alpha \in \mathbb{Q} \right\}.$$

These are proper-class divisible subgroups of $\mathbf{No}$ with zero intersection. Each exponent in $A_* + B_*$ has a unique *finite* coordinate expression in the e_α, f_α . Consequently its coordinate functions are explicitly defined class maps.

Let

$$K_{0,*} = \mathcal{K}_k(A_*), \quad K_{1,*} = \mathcal{K}_k(B_*), \quad M_* = K_{0,*}K_{1,*}, \quad E_* = \mathcal{K}_k(A_* + B_*).$$

The notation allows only set-sized Hahn supports. All field operations are well defined by localization: finitely many input series involve a set of exponent coordinates, and the relevant operations take place in a set-sized divisible-group Hahn field. The same observation proves real closedness for $k = \mathbb{R}$ and algebraic closedness for $k = \mathbb{C}$. The intersection and linear-disjointness proof in [proposition 30.2](#) is coefficientwise and applies without change to every finite relation.

For each ordinal α , the map extracting the negative of the e_α coordinate of a t -exponent is an additive character, vanishes on B_* , and takes value 1 at $-e_\alpha$. It therefore gives exactly the coordinate derivation used in the set-sized proof. No extension of a class-independent family to a class basis is required.

Lemma 36.1 (The mixed-support proof is local). *The conclusion of [Theorem D](#) applies to the class fields above for any set of the coordinate indices, with coefficients of a putative relation allowed anywhere in M_* . It also proves independence for a class of witnesses whenever each finite subfamily satisfies the exclusive infinitude condition.*

Proof. A polynomial relation uses finitely many witnesses and finitely many coefficients. Each coefficient belongs to $K_{1,*}(u_1, \dots, u_m)$ for finitely many $u_q \in K_{0,*}$. Choose $m + 1$ exclusive indices. The required characters are the explicit coordinate functions just described. A finite homogeneous system over the class field $K_{0,*}$ has a nonzero solution by the ordinary finite Gaussian-elimination proof; it involves only finitely many field operations. Coset independence is still valid, since the supports of each of the finitely many summands are sets in disjoint cosets. Every subsequent step is the finite polynomial calculation in [section 31](#). $\square$

A more general class group can be handled by set localization when suitable local characters exist. The explicit construction above is preferable here because it displays them and avoids additional foundational hypotheses about arbitrary class vector spaces.

36.2 An ordinal-indexed family of witnesses

For an ordinal α , let

$$\Lambda_\alpha = \{(\omega \cdot_{\text{ord}} \alpha) +_{\text{ord}} n : n < \omega\}.$$

These countably infinite ordinal blocks are pairwise disjoint. Define

$$W_\alpha = \sum_{n < \omega} \omega^{r(\omega \cdot_{\text{ord}} \alpha) +_{\text{ord}} n}. \quad (36.1)$$

Each W_α has countable, strictly decreasing positive support, so is an omnific integer belonging to E_* . [merge] The blocks are source 02's labels $\beta_{\alpha,n} = \omega \cdot \alpha + n$ of (22.6), here with $n \geq 0$ instead of $n \geq 1$.

Proof of Theorem G. The supports Λ_α are disjoint and infinite. Apply lemma 36.1 to any finite selection of the W_α . This proves algebraic independence of the ordinal-indexed family over M_* , for real or complex coefficients.

In addition, fix an infinite cardinal κ and use the family $\mathcal{U}_\kappa$ from section 33 on the first κ indices. The same local proof shows that its 2^κ witnesses remain independent over the *full* class compositum M_* , not merely over the small compositum from that section. The coefficients of a relation were never restricted to the first κ coordinates.

Finally, a set-sized transcendence basis of cardinality λ would bound every algebraically independent subset over M_* by λ (for a finite basis, by its finite size). This is the ordinary exchange argument for algebraic dependence; it uses only finite polynomial relations and remains valid over a class base field. The arbitrary cardinal lower bounds contradict such a bound. Hence no set-sized transcendence basis exists. $\square$

The theorem is deliberately not written as “ $\text{trdeg}(E_*/M_*) = \mathbf{On}$ ”: $\mathbf{On}$ is not a cardinal in the ordinary set-sized sense of transcendence degree. The precise assertions are an explicit independent class and independent sets of arbitrarily large cardinality. Also, no order-group isomorphism $A_* \cong \mathbf{No}$ or $B_* \cong \mathbf{No}$ has been proved or used. [merge] In the last proof λ is a cardinal, not the Euler weight.

Remark 36.2 (A second route to the transcendence of S). [merge] The class-local proof also applies to the explicit copies of section 7, and gives a second proof that the witness (8.3) is transcendental over $M = K_0 K_1$ in $E = \mathcal{K}_k(G_0 + G_1)$, for $k = \mathbb{R}$ and $\mathbb{C}$. In the t -convention, $S = \sum_{n \geq 1} t^{a_n + b_n}$ with

$$a_n = -g_n = -\omega^{\psi(n)} \in G_0, \quad b_n = ng \in G_1, \quad g = \omega^{7/2}.$$

Here $G_0 \cap G_1 = 0$; the a_n are distinct inner monomials, hence $\mathbb{Q}$ -linearly independent; the b_n are distinct; and the t -exponents increase, since consecutive differences are $g + g_n - g_{n+1} > 0$ by (8.1). The singleton family $\{\mathbb{N}_{>0}\}$ has finite-exclusive infinitude. The characters are explicit, as in lemma 36.1: for $\gamma = \gamma_0 + \gamma_1$ with $\gamma_j \in G_j$, let $\chi_n(\gamma)$ be minus the coefficient of the inner monomial $\omega^{\psi(n)}$ in γ_0 . It is additive with values in $\mathbb{R} \subseteq k$ (lemma 30.3 needs no more), vanishes on G_1 , and $\chi_n(a_m) = \delta_{nm}$. The proof of lemma 36.1, with the class fields K_0, K_1 in place of $K_{0,*}, K_{1,*}$, then gives transcendence of S over M . This route uses neither the iterated coordinates of lemma 8.1 nor the coefficient-field bound of lemma 8.2; unlike theorem 8.3 it needs characteristic zero and monomial coefficients $s_n = \omega^{g_n}$, on which ∂_{χ_n} acts as $s_n \partial / \partial s_n$. It does not give the differential transcendence of theorem 22.2. The observation is the merge's; the source does not apply its theorem to S .

37 Ordinary maps versus preservation of Hahn sums

The maximal gap has a further structural consequence. This section uses only the set-sized fields from [Theorem E](#) or [Theorem F](#). Put $E = \mathcal{K}_k(A + B)$ and $M = \mathcal{K}_k(A)\mathcal{K}_k(B)$ as before.

37.1 The strong relative structure is rigid

Definition 37.1. A field homomorphism on E is *strong* if it sends every Hahn summable family to a Hahn summable family and commutes with its sum. A derivation is strong under the analogous requirement. These are requirements on the specified Hahn presentation, not on an unspecified topology.

Proposition 37.2 (Strong relative rigidity). *Every strong field endomorphism of E fixing M pointwise is the identity. Every strong derivation of E annihilating M is zero.*

Proof. For $g = a + b \in A + B$, one has $\omega^g = \omega^a \omega^b \in M$, and $k \subseteq M$. Every $x \in E$ is the Hahn sum of its monomial terms $c_g \omega^g$, each of which belongs to M . A strong endomorphism fixing those terms fixes their sum. A strong derivation annihilating those terms annihilates their sum. $\square$

Notice that this is not automatic strongness of arbitrary field maps. It is rigidity *conditional* on preserving all the relevant Hahn sums. The next results show how large the distinction can be.

37.2 Ordinary relative derivations are abundant

Lemma 37.3 (Extending assignments on a transcendence basis). *Let E/M be a characteristic-zero extension and let $\mathcal{T}$ be a transcendence basis. Every function $\varphi : \mathcal{T} \rightarrow E$ extends uniquely to a derivation $\delta : E \rightarrow E$ annihilating M .*

Proof. Algebraic independence of $\mathcal{T}$ gives a derivation on the polynomial ring $M[\mathcal{T}]$ by the finite polynomial rule $\delta(y) = \varphi(y)$, $y \in \mathcal{T}$; the quotient rule extends it uniquely to $M(\mathcal{T})$. Since $E/M(\mathcal{T})$ is algebraic and characteristic zero, every minimal polynomial is separable. For an algebraic element z with minimal polynomial $p(X) = \sum_j a_j X^j$, any extension must satisfy

$$\delta(z) = -\frac{\sum_j \delta(a_j) z^j}{p'(z)}.$$

The denominator is nonzero. The standard quotient-ring calculation shows this formula defines a derivation on each simple algebraic extension. Iterating over finite algebraic extensions gives compatible maps by uniqueness, and their union defines δ on E . $\square$

Theorem 37.4 (Maximal ordinary differential freedom). *For the fields in [Theorem E](#),*

$$|\mathrm{Der}_M(E, E)| = 2^{2^\kappa}, \quad \mathrm{Der}_M^{\mathrm{strong}}(E, E) = \{0\}.$$

The vector space of relative Kähler differentials $\Omega_{E/M}$ has an E -basis of cardinality 2^κ . For the Archimedean example, these quantities become $2^{\mathfrak{c}}$ and $\mathfrak{c}$, respectively.

Proof. A transcendence basis has cardinality 2^κ by [Theorem E](#). Choosing values 0 or 1 independently at its members and applying [lemma 37.3](#) already gives 2^{2^κ} distinct derivations. The total number of functions $E \rightarrow E$ is $(2^\kappa)^{2^\kappa} = 2^{2^\kappa}$, so this is exact. The strong assertion is [proposition 37.2](#).

For completeness, the symbols dy , $y \in \mathcal{T}$, span $\Omega_{E/M}$: finite rational differentiation gives this for $M(\mathcal{T})$, and differentiating a separable minimal polynomial expresses each dz , $z \in E$, in that span. They are independent because the derivations that assign 1 to one basis member and 0 to all others extract their individual coefficients. Hence their cardinality is the dimension. The Archimedean calculation uses [proposition 34.2](#) instead. $\square$

Extend the explicit independent prime family of [Theorem F](#) to a transcendence basis. The lemma then gives, for each fixed η , a relative derivation with $\delta(P_\eta) = 1$ and $\delta(P_\theta) = 0$ for $\theta \neq \eta$. It annihilates every individual monomial in the expansion of P_η because those monomials lie in M . Thus it cannot commute with that Hahn sum. This concrete consequence is not a paradox: ordinary algebraic differentiation has no built-in rule for infinite formal sums.

The finite annihilators used to prove independence are a different object. They annihilate only a finitely generated portion of M , chosen after a hypothetical polynomial relation is given. Attempting to make a nonzero strong derivation annihilate *all* of M at once would contradict [proposition 37.2](#).

37.3 The complex field has many non-strong automorphisms

Theorem 37.5 (Hidden complex-field automorphisms). *For $k = \mathbb{C}$ in [Theorem E](#),*

$$|\text{Aut}(E/M)| = 2^{2^\kappa},$$

whereas its strong subgroup is trivial. In the Archimedean example, $|\text{Aut}(E/M)| = 2^\mathfrak{c}$.

Proof. Let M^{alg} be the relative algebraic closure of M in E . It is algebraically closed. Choose a transcendence basis $\mathcal{T}$ of E over M^{alg} . By [corollary 31.4](#), $|\mathcal{T}| = 2^\kappa$. Every permutation of $\mathcal{T}$ defines an automorphism of $M^{\text{alg}}(\mathcal{T})$ fixing M^{alg} . Such an automorphism extends to its algebraic closure E : the usual extension-by-maximality argument extends embeddings across algebraic elements by choosing a root of the transported minimal polynomial. The resulting image is algebraically closed and contains the base field, so it is all of E .

An infinite set of cardinality λ has 2^λ permutations. For the lower bound, partition it into λ disjoint pairs and independently swap or fix each pair; the upper bound is $\lambda^\lambda = 2^\lambda$. Thus the distinct permutations of $\mathcal{T}$ produce at least 2^{2^κ} distinct extensions. There are at most $|E|^{|E|} = 2^{2^\kappa}$ maps on E , giving equality. The strong subgroup is trivial by [proposition 37.2](#). The same proof with $|\mathcal{T}| = \mathfrak{c}$ handles the Archimedean example. $\square$

These automorphisms fix both entire complex Hahn factor fields, hence all mixed monomials, but nonidentity ones fail to preserve Hahn sums. They are not claimed to preserve the chosen real part, complex conjugation, omnific integers, order, or surreal exponentiation. In particular the theorem is not an assertion about ordered automorphisms of **No**, nor about the automorphism group of **Oz**. Those stronger compatibility conditions are genuinely additional.

Remark 37.6 (Omnific rings force the identity). [merge] Combined with the omnific-preserving report, the rigidity of [proposition 37.2](#) excludes one combination of the compatibilities just listed. Let $E_{\mathbb{R}} = \mathcal{K}_{\mathbb{R}}(A + B)$ and $M_{\mathbb{R}}$ be the real join and compositum of [Theorem E](#) or [Theorem F](#). By `opa:as:cor:setpairs`, applied with the nonzero divisible set group $\Gamma = A + B$ and $t^\gamma = \omega^{-\gamma}$, every field automorphism of $E_{\mathbb{R}}$ that stabilizes $\mathcal{A}_{\mathbb{R}}(A + B) = E_{\mathbb{R}} \cap \mathbf{Oz}$ is strongly $\mathbb{R}$ -linear; if it fixes $M_{\mathbb{R}}$, it is the identity by [proposition 37.2](#). Now let $\sigma \in \text{Aut}(E/M)$ for $k = \mathbb{C}$ commute with complex conjugation and stabilize $\mathcal{A}_{\mathbb{C}}(A + B) = E \cap \mathbf{Oz}[i]$. Then σ maps the fixed field $E_{\mathbb{R}}$ of conjugation onto itself and stabilizes $E_{\mathbb{R}} \cap \mathbf{Oz}$, so it is the identity on $E_{\mathbb{R}}$; since $i \in M$, σ is the identity. So a

nonidentity automorphism of [theorem 37.5](#) cannot both commute with conjugation and preserve the Gaussian omnific ring of E ; in the real joins, no nonidentity automorphism fixing $M_{\mathbb{R}}$ preserves the omnific ring. The argument depends on `opa:as:cor:setpairs`, an unrefereed result of that report, and was not in the source; it answers Research question [29](#) only for this combination of conditions.

38 Scope, comparison, and failure boundaries

38.1 What has been answered

At the pinned repository snapshot, Research question 3 of this report asks for an exact description of ordinary composita and for transcendence degrees of full Hahn joins over them. The results here answer its *degree component* as follows.

Setting	Answer	Witnesses and additional structure
Every infinite set cardinal κ	Exact degree 2^κ	2^κ omnific integers; real and complex versions
Countable Archimedean exponent group	Exact degree $\mathfrak{c}$	Explicit continuum family prime in both $\mathbf{Oz}$ and $\mathbf{Oz}[i]$
One fixed pair of class Hahn subfields	No set-sized transcendence basis	Explicit ordinal-indexed independent omnific family
Relative maps in the complex set examples	2^{2^κ} automorphisms	Only the identity preserves all Hahn sums

This report's one-witness separated-scale mechanism is not being represented as a new result. The new argument works without convex separation and supplies maximal-size independent families. The class result uses explicit finite-coordinate groups and does not invoke this report's independent full copies of **No**. [merge] Since the pin, source 02 has answered the degree component in the separated case and the Laurent cardinal cases, also with maximal-size families ([theorem 21.1](#)), and has given class-sized families in the join of two actual copies ([theorem 22.2](#)); the answers above are those for non-separated examples ([section 28.4](#) and [remark 28.1](#)).

The comparison is targeted, not a certification of originality against every file, pending archive, or paper. The repository catalogue was consulted to identify neighboring topics, and this report was inspected for its compositum result and research question. Published primality was checked separately. The asserted new results should still receive external mathematical review and a more exhaustive priority check before any publication claim.

38.2 What the proof does not imply

An arbitrary infinite mixed sum need not be new. If $u = t^a \in K_0$ and $v = t^b \in K_1$ with $a + b > 0$, then

$$\sum_{n \geq 0} (uv)^n = \frac{1}{1 - uv} \in M.$$

Here all a -directions are rationally dependent. The example explains why merely having infinitely many support terms cannot prove transcendence.

Distinct cosets really matter. If $b_i = 0$ for all i , an admissible sum of $t^{a_i+b_i}$ belongs to K_0 . More generally, a common b factors out and leaves a K_0 -series. The noncancellation conclusion (30.4) would fail without the distinct-coset hypothesis.

Finite support cannot work over the full compositum. A finite mixed sum already belongs to M . Finite-exclusive infinitude supplies arbitrarily many differential directions to overcome the unknown finite number of coefficient generators. A bound suited to only one fixed number of generators is not enough for the full compositum.

Not all incidence families give independence. For example, if $U_3 = U_1 \dot{\cup} U_2$, then $F_{U_3} = F_{U_1} + F_{U_2}$. The support hypothesis is therefore a substantive condition. Pairwise distinctness of supports alone is not sufficient.

A large independent set need not already be a basis. Even when its cardinality equals the transcendence degree, it may be properly extendible to a transcendence basis of the same cardinality. The prime family is not asserted to be a basis without extension.

Characteristic zero is used twice. It supplies nontrivial coordinate characters into k from a divisible $\mathbb{Q}$ -vector group and ensures a nonzero partial derivative for a nonconstant polynomial. In characteristic p , every additive map from a divisible group to $(k, +)$ vanishes, and inseparable polynomial relations can have all partials zero. The present proof does not extend verbatim.

Order and topology are separate structures. No conclusion about order-topological convergence of partial sums, no continuity classification of the non-strong maps, and no commutation with conjugation is proved. Hahn summability is the exact infinite-sum notion used here. Archimedean *exponent groups* do not make their Hahn *fields* Archimedean.

The full class of surreals is not the base field. Every witness belongs to $\mathbf{No}$, so it cannot be transcendental over $\mathbf{No}$. The base in each theorem is the displayed ordinary compositum. Likewise a prime in $\mathbf{Oz}[i]$ is a unit in the field $\mathbf{No}(i)$; primality is always a statement in the ring specified.

38.3 Logical independence of the ingredients

The cardinal-degree results do not depend on primality, and neither the set-sized nor the explicit class independence proof depends on this report's class-embedding theorems, which the source treats as unreviewed. The primality corollary requires the published one-row theorem in addition to the independence proof. The automorphism count requires algebraic closedness of the complex Hahn join but not preservation of its real structure. These dependency distinctions are useful both for checking the source and for reusing only the part needed in another project.

38.4 What the addition does not claim

[merge] This list collects every limitation stated by source 03, in its article and delivered README, with its location here, followed by those added in the merge.

- J1. The arguments are written out but have not been refereed, independently reviewed or checked by Lean; priority and novelty are not certified; no named longstanding conjecture is claimed solved (title page of the source; README).
- J2. The contribution proposed as new is the mixed-support criterion and its combined maximality and arithmetic applications, not the classical ingredients: Hahn fields, the support theorem, the closedness theorems and Conway normal forms are standard foundations (section 29; title page of the source).
- J3. Only the transcendence-degree part of Research question 3 is answered, and only for the examples constructed. No characterization of all ordinary composita and no membership test is given; the question is not settled in every case (sections 28 and 38; Research question 24; README).
- J4. Primality is imported. Theorem 35.1 is credited to Pitteloud, Biljakovic–Kochetov–Kuhlmann and L’Innocente–Mantova; no proof of it is claimed, no new one-row primality theorem is claimed, and the independence argument does not reprove or replace the factorization theory behind it. Its formalization is a separate project and must not be accepted as checked because the independence kernel is (section 35 and Theorem F; section 41.5).
- J5. Primality of a real element is not inferred to survive complexification; the Gaussian assertion is proved separately (section 41.1).
- J6. The class result concerns explicit support subfields $\mathcal{K}_k(A_*)$, $\mathcal{K}_k(B_*)$. It does not identify either with a full copy of **No**, uses none of this report’s copy constructions, and proves or uses no order-group isomorphism $A_* \cong \mathbf{No}$ or $B_* \cong \mathbf{No}$. It is not written as $\text{trdeg}(E_*/M_*) = \mathbf{On}$. No proper-class Hamel basis is selected; algebraic independence of a class means independence of every finite subfamily (Theorem G and section 36).
- J7. The derivations are additive-character derivations of a Hahn field. They are not the Berarducci–Mantova derivation and are not assumed to respect a surreal exponential (section 29).
- J8. The false blanket identity $\text{Frac}(\mathcal{A}_k(G)) = \mathcal{K}_k(G)$ is not used. Corollary 33.3 is a failure of reconstruction by finite ring operations and fractions, distinct from this report’s descent failure for intersections; neither argument authorizes replacing fixed-support omnific fraction fields by full Hahn fields (sections 29 and 33.3).
- J9. The independent-family and almost-disjoint-family constructions are standard; no novelty is claimed for them (section 32).
- J10. Proposition 31.2 gives no bound on the expression length of an arbitrary Hahn series (section 31).
- J11. Proposition 37.2 is rigidity conditional on preserving Hahn sums, not automatic strongness of arbitrary field maps. No single strong derivation annihilates the full compositum; the finite annihilators kill only a finitely generated part chosen after a relation is given (section 37; section 41.1).
- J12. The automorphisms of theorem 37.5 are not claimed to preserve the chosen real part, complex conjugation, omnific integers, order or surreal exponentiation. The theorem is not about ordered automorphisms of **No** or about $\text{Aut}(\mathbf{Oz})$ (section 37).
- J13. This report’s one-witness separated-scale mechanism is not represented as new. The repository comparison was targeted, not exhaustive; the catalogue was consulted for neighbouring topics

- only. External review and a more exhaustive priority check are needed before any publication claim ([section 38](#); README).
- J14. An arbitrary infinite mixed sum need not be new (a geometric series lies in M); the distinct-coset hypothesis is necessary; finite support cannot work over the full compositum; not every incidence family gives independence, and pairwise distinct supports are not enough ([section 38.2](#)).
 - J15. An independent set of the right cardinality need not be a transcendence basis; the prime family is not asserted to be one ([section 38.2](#)).
 - J16. Characteristic zero is used twice; the proof does not extend verbatim to characteristic p ([section 38.2](#); Research question 31).
 - J17. No order-topological convergence, continuity classification of the non-strong maps or commutation with conjugation is proved; Archimedean exponent groups do not make the Hahn fields Archimedean ([section 38.2](#)).
 - J18. **No** is not the base field; every witness lies in **No**. A prime of **Oz**[i] is a unit in **No**(i); primality is a statement in the ring specified ([section 38.2](#)).
 - J19. The cardinal results depend neither on primality nor on unreviewed class-embedding theorems; the primality corollary needs the published theorem; the automorphism count needs algebraic closedness of the complex join ([section 38](#)).
 - J20. Finite rational-rank factors are not covered: the proof needs infinitely many rationally independent directions, and the lower bound for $\mathcal{K}_k(\mathbb{Q} + \sqrt{2}\mathbb{Q})$ is open. Prime witnesses are obtained only for the countable Archimedean construction, not for uncountable κ . The algebraic theorem allows $H \neq 0$, but no example with $H \neq 0$ is constructed (Research questions 23, 28 and 26).
 - J21. The research questions are proposed by the arguments; they are not asserted to be previously published open problems. Research question 32 is an implementation task ([section 39](#)).
 - J22. The finite verification is not a proof of the infinite Hahn, cardinality or primality assertions and does not test primality in **Oz**. The floating-point check is orientation only; monotonicity and the limit are proved analytically. A numeric specialization typo found during verification was corrected before release ([section 41.4](#); README).
 - J23. No Lean source or kernel verification is claimed; the interfaces of [section 41.5](#) are proposals, not declarations of the collection ([section 41.5](#)).
 - J24. The user-supplied encyclopedia page served as orientation only; a publisher record for Pitteloud was consulted for bibliographic data only. The collection catalogue's inclusion of a report is not taken as evidence of correctness (Appendix A.9).
 - J25. Finite checks, typesetting checks and written proofs are the only evidence supplied; there has been no external review. The delivered README's PDF inspection (a contact sheet of rendered pages) does not verify the mathematics (Appendix A.9; README).
 - JM1. Research question 11 is answered only in its transcendence-defect part and only for the explicit non-separated examples; the image of the compositum and a denominator criterion stay open. Research question 14 is continued, not answered ([section 28.4](#)).

- JM2. Source 03 cites no report of the collection except this one. Its finite-exclusive infinitude is [definition 19.1](#); its branch coding is `tail:eq:binarycode` shifted by one; the square-root independence is also used in `tail:thm:surreal`; the one-row primality is cited in the Diophantine report; the degrees of `bst:cor:exactcard` and `bst:cor:realgroup` are over a different base ([section 28.6](#)).
- JM3. [Remarks 36.2](#) and [37.6](#) and the remarks comparing the sources are observations of the merge, read only in the merge. [Remark 37.6](#) depends on the unrefereed `opa:as:cor:setpairs`.
- JM4. The source was written against the pin, before source 02 was merged; its statements about this report are those of the pin. Where they are now incomplete (the degree question in the separated case, the exact membership test in the separated case, class families in actual copies), [merge] notes say so.
- JM5. No `isc:hj:` label has a Lean mapping in `FORMALIZATION.md`. The proofs were read when the addition was merged ([Appendix A.9](#)); that reading is not an independent review. The wording of [theorem 35.1](#) was compared with [3, Section 1.3]; the other new references were not checked.
- JM6. The recorded run was repeated in the merge on a copy; the shipped `code/03-hahn-joins-build.sh` does not run unchanged under the shipped names, and an adapted copy would overwrite a PDF and a verification record in its own directory ([Appendix A.10](#)).

39 Research questions of the third source

The questions below are proposed by the source's arguments. They are not asserted to be previously published open problems. The first two are especially direct next targets, and the final question is an implementation task rather than a conjecture about truth. [merge] The source's Question n is Research question $22 + n$ here.

Research question 23 (Finite rational-rank factors). Determine

$$\mathrm{trdeg}(\mathcal{K}_k(\mathbb{Q} + \sqrt{2}\mathbb{Q})/\mathcal{K}_k(\mathbb{Q})\mathcal{K}_k(\sqrt{2}\mathbb{Q})), \quad k = \mathbb{R}, \mathbb{C}.$$

Is it $\mathfrak{c}$? The ambient cardinality gives that upper bound, but the coordinate-annihilator proof has only one rational direction from each factor and therefore does not prove the lower bound. A new obstruction that survives finite differential rank is needed.

[merge] This is the smallest dense case of Research question 11. The rank test of [theorem 17.2](#) does not apply, since neither group is convex in the sum.

Research question 24 (Exact characterization of the ordinary compositum). Find a necessary and sufficient condition, expressed in coefficients or supports, for an element of $\mathcal{K}_k(A + B)$ to lie in $\mathcal{K}_k(A)\mathcal{K}_k(B)$. The mixed-support criterion supplies a large family outside even the relative algebraic closure, but not a membership test. The separated coefficient-field condition discussed in this report ([lemma 8.2](#)) is a useful comparison point, not a replacement for such a characterization in interleaved groups.

[merge] Since the pin, the separated case has an exact test, the finite coefficient-rank denominator criterion ([theorem 17.2](#) and [corollary 22.1](#)), and the coefficient-field condition is known not to be sufficient ([corollary 19.5](#)). For interleaved groups the question is the first part of Research question 11, and stays open.

Research question 25 (Weaker incidence hypotheses). Can finite-exclusive infinitude be replaced by an exact rank condition on the incidence functions 1_U modulo finitely supported functions? Finite linear identities among these functions already give polynomial relations among the series. Determine whether a suitable higher-order rank condition captures all algebraic relations for coordinate- independent mixed exponents.

Research question 26 (Prescribed shared Hahn cores). For $H \neq 0$, give sharp sufficient conditions on Γ_0/H and Γ_1/H for the existence of a well-ordered mixed support satisfying [Theorem D](#) with 2^κ incidence patterns. The algebraic theorem already allows a nontrivial intersection; the remaining problem is the ordered construction of enough suitable representatives and the corresponding optimal cardinal upper bound.

[merge] It continues Research questions 3 and 14; the latter asks for tail resilience over a prescribed core, which is not addressed here.

Research question 27 (Full surreal copies rather than support subfields). For pairs of full monomial-preserving copies of **No** over a prescribed Hahn core, determine when their exponent images contain the coordinate configurations of [Theorem D](#) at every set cardinal. A positive result would transfer the no-set-basis conclusion to those full-copy diagrams. The explicit class groups in [section 36](#) resolve a different case and should not be silently substituted for full copies.

[merge] For the explicit separated copies of [section 7](#), with trivial core, the no-set-basis conclusion is already known by another route: no set generates the join over the compositum even algebraically ([corollary 23.4](#)); and one configuration of [Theorem D](#) is exhibited there by [remark 36.2](#). Prescribed nonzero cores remain open (Research question 14).

Research question 28 (Uncountable prime witnesses). For $\kappa > \aleph_0$, can the 2^κ independent omnific integers in [Theorem E](#) be replaced by primes in **Oz**, or in both **Oz** and **Oz**[z]? The one-row theorem treats the countable real-exponent construction, not arbitrary uncountable nested supports. The question asks for a genuinely compatible primality mechanism, not merely an independent family with constant term one.

Research question 29 (Compatibility of the hidden complex automorphisms). Which automorphisms counted in [theorem 37.5](#) can commute with conjugation, preserve the real Hahn subfield, or preserve its omnific subring? The argument by permutations of a transcendence basis imposes none of these constraints. A classification would measure precisely how much additional rigidity each distinguished structure contributes.

[merge] *Partial status.* By [remark 37.6](#), which rests on `opa:as:cor:setpairs`, only the identity commutes with conjugation and preserves the Gaussian omnific ring of the join. The other combinations stay open.

Research question 30 (More than two factor fields). Develop a mixed-support criterion for $K_1 \cdots K_r$ inside the full Hahn field on $\Gamma_1 + \cdots + \Gamma_r$. One can annihilate all but one coefficient field and finitely many generators of the remaining field; what coset or tensor-independence hypothesis ensures that the surviving mixed terms do not cancel? This formulation isolates the exact place where the two-field proof would have to be strengthened.

[merge] Here the indices $1, \dots, r$ are the source's; they number r arbitrary factors. The question continues Research question 21, which asks the differential version for three or more separated copies with commuting Euler derivations.

Research question 31 (Positive characteristic). Find an analogue using higher or iterative derivations, or identify a sharp obstruction caused by divisible exponent groups and Frobenius. A meaningful statement must treat inseparable algebraic relations and cannot simply replace the ground field in [Theorem D](#).

Research question 32 (A small formal verification kernel). Formalize the support-coset lemma, additive-character Hahn derivations, and the finite annihilator argument in Lean. These three pieces should be separated from the normal-form identification with **No** and from published primality. A verified generic Hahn theorem would already cover the main new mathematical mechanism; [section 41.5](#) lists suggested interfaces.

[merge] It overlaps Research questions 10 and [22](#) in the coset and Euler-derivation kernel; none of the proposed interfaces exists in the collection’s Lean library.

40 Conclusion of the third source

An ordinary compositum has access to arbitrary series in each factor, but only finitely many such series in any one algebraic expression. The mixed-support theorem exploits that finiteness without making a finiteness assumption on their supports. A finite homogeneous system removes the coefficient dependence, and distinct exponent cosets keep the desired derivative nonzero.

This yields maximal transcendence gaps at every infinite set cardinal, a prime-witness version with interleaved real exponents, and a fixed class example with no set-sized transcendence basis. It also separates ordinary algebraic symmetries from symmetries of the Hahn presentation: in the complex examples the former are as numerous as possible while the relative strong subgroup is trivial. The finite-rank compositum problem and compatibility with full surreal-copy structures remain particularly clear next challenges.

41 Proof audit, finite verification and a formalization roadmap of the third source

[merge] This section prints the source’s Appendices A, B and C.

41.1 Dependency ledger

Result	Essential dependencies	Evidence in the source
Mixed-support independence	Hahn multiplication; set basis extension; finite linear algebra; characteristic zero	Full proof in sections 30 and 31
Maximal cardinal gap	Mixed theorem; independent incidence family; explicit monomial coordinates	Full proof in sections 32 and 33
Archimedean gap	Mixed theorem; square-root independence; binary branches	Full proof in section 34
Omnific primality	Classical one-row prime theorem; constant-term pullback; set localization	External theorem isolated; application proved in section 35
Class-sized example	Explicit class coordinates; finite local proof	Full argument in section 36 ; no class basis selection
Relative derivations and complex automorphisms	Transcendence-degree results; separability; algebraic-closure extension	Proofs in section 37

41.2 Potential failure points that have been checked

Support existence. The abstract theorem explicitly assumes a well-ordered mixed t -support. The nested example supplies a strictly increasing negative t -support indexed by κ , and the prime example supplies an increasing sequence with supremum zero. A supremum is not required to belong to a Hahn support.

Sign convention. Positive Conway exponents are negative t -exponents. The candidates Ξ_U and P_η therefore lie on the omnific side of the support cut, not on the infinitesimal side.

Coefficient-field membership. Each entry $\partial_i(u_q)$ lies in K_0 , not merely in the ambient join. This is why the homogeneous system can be solved with its coefficients c_i in K_0 and why the final coset-independence step is valid. Solving over the entire join instead would not establish the needed noncancellation.

Dependence of the annihilator on the relation. The annihilator is chosen after all the coefficients of one polynomial have been placed in a finitely generated field over K_1 . There is no claim that a single strong derivation annihilates the full compositum; [section 37](#) proves that it cannot do so nontrivially.

Nonzero polynomial partial. The relation is chosen of minimal total degree over a fixed coefficient field. A nonzero formal partial has smaller degree, so its evaluation cannot vanish. Merely choosing any polynomial relation would not justify this step.

Primality after complexification. Primality of a real element need not survive an arbitrary ring extension. We do not infer the Gaussian assertion from the real one. Instead we apply the characteristic-zero one-row theorem separately with complex coefficients and then descend along the constant-term map to $\mathbb{Z}[i]$.

Class localization. Any factorization identity in the full omnific ring uses a set of support exponents. Enlarging that set by $\mathbb{R}$ still yields a set group, so the external theorem applies within its stated set-sized scope. The class independence theorem uses explicit coordinate characters and finite systems rather than a proper-class choice of basis.

Cardinality and transcendence. The lower bound is supplied by proved algebraic independence, not by counting different series. The upper bound counts all coefficient functions on a group of cardinality κ . No continuum hypothesis or assumption that κ is regular is used.

41.3 Strengthening the multiquadratic justification

For clarity, the finite Galois fact used in [section 34](#) can be derived without appealing to an infinite construction. If F has characteristic not two and d is not a square in F , then an element $r \in F^\times$ becomes a square in $F(\sqrt{d})$ only if either r or r/d is a square in F . Indeed, writing a square root as $u + v\sqrt{d}$ gives $2uv = 0$; the two possibilities give exactly those alternatives. Induction shows that the kernel of

$$\mathbb{Q}^\times / (\mathbb{Q}^\times)^2 \longrightarrow \mathbb{Q}(\sqrt{p_1}, \dots, \sqrt{p_m})^\times / (\mathbb{Q}(\sqrt{p_1}, \dots, \sqrt{p_m})^\times)^2$$

is generated by the square classes of the p_j . Unique prime factorization says a new prime has a square class outside this subgroup, so every adjoining step doubles the degree. The finite extension has the expected 2^m monomial basis, and independently changing the signs of its generators preserves the multiplication relations. This supplies the sign-changing automorphisms used to prove linear independence of the square roots. [merge] In this paragraph F is a field of characteristic not two, r an element and $p_1, \dots, p_m$ distinct primes listed afresh, not the enumeration $p_0 < p_1 < \dots$ of [\(34.1\)](#).

41.4 A finite exact verification model

The source's program, shipped as `code/03-hahn-joins-verify.py`, checks finite algebraic and combinatorial instances. These checks are not a proof of the infinite Hahn assertions and do not test primality in **Oz**. Their purpose is to catch mistakes in the annihilator calculation, the incidence coding, and the explicit finite formulas.

For a concrete annihilator, let x_1, x_2, x_3 be independent formal variables and put

$$u_1 = x_1 + x_2 + x_3, \quad u_2 = x_1x_2 + x_3^2, \quad \partial_j = x_j \frac{\partial}{\partial x_j}.$$

The column vectors of the two derivative lists are

$$v = (x_1, x_2, x_3), \quad w = (x_1x_2, x_1x_2, 2x_3^2).$$

Their cross product gives

$$\begin{aligned} c_1 &= 2x_2x_3^2 - x_1x_2x_3, \\ c_2 &= x_1x_2x_3 - 2x_1x_3^2, \\ c_3 &= x_1x_2(x_1 - x_2). \end{aligned}$$

Thus $\partial_{\mathbf{c}} = \sum c_j \partial_j$ annihilates u_1 and u_2 exactly. If y_1, y_2, y_3 are additional independent variables and $F = x_1y_1 + x_2y_2 + x_3y_3$, then

$$\partial_{\mathbf{c}}(F) = c_1x_1y_1 + c_2x_2y_2 + c_3x_3y_3 \neq 0.$$

At $(x_1, x_2, x_3) = (1, 2, 3)$ this becomes $30y_1 - 24y_2 - 6y_3$, visibly nonzero. This is the finite polynomial shadow of the coset argument. [merge] In this model F is a polynomial, not a Hahn series.

The verification script uses exact symbolic arithmetic, enumerates small Boolean-pattern instances, checks exclusive binary tails for several distinct eventually periodic branches, and verifies the displayed cross-product identities and quotient-rule calculation. Numerical approximations to the first prime exponents are included only as an orientation check; the monotonicity and limit are proved analytically above. The actual output is shipped as `data/03-hahn-joins-verification.txt`. [merge] Its five PASS lines are: the exact cross-product annihilator for two generators and a quotient; the nonvanishing mixed derivative, with the specialization $30y_1 - 24y_2 - 6y_3$; all 34,112 Boolean patterns on up to four distinct subsets of a four-point universe (the finite analogue of lemma 32.1; the coordinate ξ is not modelled); eight distinct eventually periodic branches through depth 32, each with at least 29 exclusive nodes, and the codes $\{1, 3, 7, 15\}$, $\{2, 6, 14, 30\}$ of the two constant branches; and the decrease of the first ten exponents ρ_n in floating point. The rerun is in Appendix A.10.

41.5 A formalization roadmap

No Lean source or kernel verification is claimed for the new theorem. The following are proposed interfaces, not declarations alleged to exist in the repository.

Hahn support layer. Prove that multiplying a series supported in a subgroup Γ' by a monomial of exponent β places its support in $\Gamma' + \beta$. Deduce finite linear independence for monomials in distinct cosets. This part needs neither divisibility nor characteristic zero.

Character derivation layer. For an additive character $\chi : \Gamma \rightarrow k$, define the coefficient map $x_\gamma \mapsto \chi(\gamma)x_\gamma$ on Hahn series. Establish support containment, the coefficientwise Leibniz rule, and commutation with summable families. Then construct the coordinate characters from linear independence in Γ/Γ_1 using a set vector-space basis.

Finite annihilator layer. Package an $m \times (m + 1)$ matrix over K_0 and obtain a nonzero kernel vector. Form the finite linear combination of derivations and prove that it annihilates the intermediate field generated by K_1 and the u_q . Keep its coefficients in K_0 throughout; this is a useful type-level invariant that prevents an invalid noncancellation step.

Algebraic independence layer. Use minimal total degree, or an equivalent separable minimal-polynomial argument, to formalize the contradiction. The finite-exclusive support condition can be abstracted as the ability to select $m + 1$ indices for any finite relation. Cardinal amplification should be a separate theorem.

Surreal and arithmetic bridges. Only after the generic Hahn theorem is verified should one transport it through Conway normal form and add omnific support conditions. The one-row prime theorem is a distinct formalization project and must not be silently accepted as checked merely because the independence kernel is checked. The constant-term pullback lemma itself is an accessible independent commutative-algebra target.

[merge] The coset layer overlaps item (1) of section 13.2 and the support and base-change layer of section 27.2; the character derivation layer overlaps its derivation layer.

A Provenance of this report

A.1 One manuscript

This report is built from *one* manuscript: the archive `Surreal_Independent_Copies`, manuscript 06 of batch 31 of the collection, with main file `independent_surreal_copies.tex`, a 25-page PDF, dated 23 September 2026 and pinned to repository commit `bcac55ae6fd3f2b354e568ba1d2e94d496a2cc59`. It was placed in commit `9d28e28` as a new report. It answers no named question of the collection, and its subject, independence, disjointness and descent for full copies of **No**, is the subject of no existing report (section 1.6). There was no second source. Nothing was merged and nothing was selected away: every result, proof, example, question and limitation of the manuscript is printed here. [merge] This subsection describes the report as written in batch 31. In batch 34 a second manuscript was added as source 02 (Appendix A.7), and in batch 35 a third as source 03 (Appendix A.9); the report is now built from three manuscripts.

The package contained the article, its PDF and a README, and nothing else: no code, recorded data, audit or checksum list. The article was renamed `article.tex` and rewritten as described below; the README was replaced by the collection README; the delivered PDF is not shipped, and `article.pdf` is a build of this text.

The editorial changes are these.

- (1) Every label carries the prefix `isc:`. The source's 62 labels are kept, unchanged after the prefix. Twenty were added: `isc:sec:conventions`, `isc:sec:collection`, `isc:sec:nonclaims`, `isc:rem:base`, `isc:rem:floor`, `isc:warn:omega`, `isc:rem:join`, `isc:rem:scaling`, `isc:rem:infinite`, `isc:sub:model`, `isc:sub:bounded`, `isc:sub:formal`, `isc:sub:notmade`, `isc:app:report`, `isc:app:onesource`, `isc:app:pinned`, `isc:app:literature`, `isc:app:checked`, `isc:app:files` and `isc:app:numbers`. Labels on existing environments change no number.
- (2) The title page names the report as a single-source report of the collection, gives the batch and the pin, and extends the status paragraph. Sections 1.5 and 1.6, the ledger of section 14 and this appendix are new. Sentences marked [write] were added at the end of section 1.4; after proposition 4.4, corollary 6.1, theorem 6.2, (7.3), lemma 9.1, proposition 9.2 and proposition 9.3; at the start of section 10; after corollary 10.3; after research questions 1, 2, 6, 7, 8, 9 and 10; and in section 13.3.

No mathematical statement was changed and no symbol was renamed: the collisions are between reports, or between local letters of different sections, and section 1.5 fixes the meanings. Where the edit had to choose: the manuscript was placed as a new report, not as an addition to the omnific-preserving report, because it answers none of that report's questions and its spine (independence, exact intersections and descent) is not that report's (stabilizers, classification and coefficient motion). It was placed among the surreal reports because its fields are subfields of **No** and **No**(*i*). The title is kept.

A.2 Repository statements at the pin

The source compared itself with the catalogue, the omnific-preserving report and the bounded-support report at its pin, through links to that revision ([6, 7, 8]). Checked at the placement commit:

- At the pin the omnific-preserving report was the merged report of four manuscripts subtitled “Convex-scale stabilizers, definable constants, nondefinable monomials, and algebraic-parameter rigidity”. It had no part on embeddings and no `opa:as:` label. The source’s description of it as concerning stabilizers and rigidity was accurate. Its Part III, on automatic strongness and proper embeddings, was written in 20c4c9f, after the pin.
- The bounded-support report is unchanged between the pin and the placement commit (its last change, ce6715b, precedes the pin). The source’s description of it as proposing stronger independent families stands.
- The large-cardinal report was placed in 21375f8, after the pin, and is not cited by the source.
- No report of the collection supplies Theorem A, theorem 5.4, Theorem B for copies, or Theorem C; the overlaps are those of section 1.6.

A.3 References checked for this report

- *L’Innocente–Mantova* [3]. The arXiv page lists version 5 of 22 January 2024 and the journal DOI 10.1016/j.aim.2024.109513; the volume number was not checked. In the arXiv text, Fact 2.1.1 states that a Hahn field is algebraically closed, respectively real closed, if and only if its coefficient field is algebraically closed, respectively real closed, and its group is divisible. Section 2.3 introduces the surreal numbers as a Hahn field. Proposition 2.4.5 is stated for the Hahn field of series with supports of size below a cardinal κ : the fraction field of $\mathbb{Z} + K((G^{<0}))$ is the whole field if and only if $\text{cf}(G) \geq \kappa$ (or $G = 0$ and $\text{Frac } \mathbb{Z} = K$), and in particular $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$. For a set-sized G and $\kappa > |G|$ this is the full Hahn field, so its proof, with a lacunary series along a cofinal sequence, gives the non-membership obstruction cited in section 1.4 and section 10. It proves non-membership in the fraction field, not transcendence. The citations are therefore accurate.
- *Ehrlich–Kaplan* [5]. The author-hosted PDF (37 pages) has the stated title and authors; only its first page was read.
- Gonshor [1], van den Dries–Ehrlich [2], Marker [4] and the encyclopedia page [9] were not checked.

A.4 Proofs read when the report was written

The proofs of lemmas 3.1 and 3.2, theorem 3.4, theorem 4.1, proposition 4.2, proposition 4.4, section 5, Theorem A, theorems 6.2 and 6.3, corollary 6.5, sections 7 and 8, sections 9 and 10 and theorem 11.2 were read line by line when the report was written. No error was found. The explicit formulas were rechecked: $\Phi_0(\omega) = \omega^{\omega^{1/2}}$, $\Phi_1(\omega) = \omega^{\omega^{7/2}}$, $g = h_1(1) = \omega^{7/2}$, and the consecutive exponents of (8.3) differ by $g + g_n - g_{n+1} > 0$. One understatement is recorded after proposition 9.2. This reading is not an independent proof review.

A.5 Theorem numbers

Numbers are those of this build and of the delivered PDF, which agree: labels were added and no environment was inserted before an existing one. Remarks and warnings share the theorem counter, so the classification of monomial-preserving embeddings is Proposition 4.4 (after Remark 4.3), and the saturated-group theorem is Theorem 11.2 (after a definition).

A.6 Files

File	Role
<code>article.tex</code>	this report (delivered as <code>independent_surreal_copies.tex</code>)
<code>article.pdf</code>	a build of this text, not the delivered PDF
<code>README.md</code>	the collection README, replacing the delivered one

The source shipped no verification program, recorded run or audit, so there is nothing to rerun. The delivered README reports a build with pdfTeX 1.40.26 and a visual inspection of rendered pages and contact sheets; those renderings are not shipped. [merge] This table lists the files of source 01. The seven shipped files of source 02, with the prefix `02-composita-`, and their rerun are in Appendix A.8; the three of source 03, with the prefix `03-hahn-joins-`, in Appendix A.10.

A.7 Source 02 (batch 34)

[merge] Source 02 is manuscript 05 of batch 34 of the collection: the archive `Surreal_Composita_Research`, delivered in commit `a4611ad` and placed in commit `a7a435f` as an addition to this report with local number 02. Its main file `article.tex` (a standalone article with an internal bibliography) is titled *Beyond Composita of Surreal Copies: Exact rank tests, differential transcendence, and omnific witnesses*, dated 23 September 2026, with the author line “AI-assisted research draft prepared for Vladimir Reshetnikov”; its PDF has 29 pages. It is pinned to commit `7b256e0792df7718995b816ad12a77bf56731f8f`, at which this report’s `article.tex` was blob `7d5cfe644e7c902f2b594179c337a6c633350fc3`. The placement chose an addition rather than a new report because the manuscript answers this report’s Research question 3 in the separated case and extends its witness (8.3). Its manuscript, README and PDF are not shipped; its code, recorded data and two audits are shipped with the prefix `02-composita-` (Appendix A.8).

What it contributed. The finite coefficient-rank test for composita (theorem 17.2), the base-change lemmas of section 18, the rapid coefficient-degree theorem and its continuum family (theorem 19.2 and corollary 19.4), the failure of the finite coefficient-field criterion (corollary 19.5), the private-block theorem and tree amplification over the finite-generation envelope (theorem 20.2, corollaries 20.4 and 20.6, and proposition 20.5), the exact Laurent degrees $|F|^{\aleph_0}$ (theorem 21.1), the ordinal family Y_α of purely infinite omnific integers with independent Euler jets (theorem 22.2 and corollary 22.3), set-parameter tail resilience (theorem 23.3 and corollary 23.4), the arithmetic and Gaussian consequences (theorems 24.2 and 24.4), twelve research questions and a formalization decomposition.

Editorial changes.

- (1) Every label of the source carries the prefix `isc:cp:`, with the source’s name after it, except that its `sec:conventions` is `isc:cp:sec:foundations` and its appendix labels are replaced by `isc:cp:sec:verify` and `isc:cp:sub:formal`. The merge also labels the source’s unlabelled definition of coefficient rank, example, remarks and questions, and its new subsections and appendix subsections.
- (2) Symbols are renamed as in section 15.5; no normalization changed.
- (3) The source’s re-derivation of the explicit copies and of the re-expansion (its Subsections 8.1–8.2) is printed once, in sections 7 and 8, with the correspondence in section 22.1; the source’s added observations are kept there. Its duplicate of lemma 9.1 is kept inside lemma 24.1, where it serves a new statement.

- (4) The source’s citations of this report by section number are replaced by cross-references to the same sections. Its bibliography entries [10, 11, 12, 13] are added to this report’s; its entry for L’Innocente–Mantova is this report’s [3]. These four references were not checked in the merge.
- (5) Text marked [merge] was added in the merge: in section 15 (the first paragraph, the note on Y_0 , and sections 15.4 to 15.6), after theorems 17.2 and 22.2, lemmas 18.3 and 23.1, corollary 19.4, and example 19.6, in section 22.1, in the construction of the $a_{\alpha,n}$, in section 25.5, after Research questions 14, 17, 19 and 22, in section 27, and in this appendix. In source 01’s text, [merge] notes were added to the title page, to section 1.6 (the status of the research questions), after corollary 6.1 (with the reciprocal note for the dilation report), after lemma 8.2, after the proof of Theorem C, after Research question 3, at the start of section 14, in item W1, and in Appendices A.1 and A.6. No statement of source 01 was changed.
- (6) Cross-references to lemmas, propositions, corollaries, remarks and the warning were printed as “theorem” before the merge: the environments share the theorem counter, and cleveref named them by the counter. The merge adds cleveref’s type hint to the labels of these environments (26 of source 01 and 33 of source 02; for example the label of lemma 8.2 now carries the hint [lemma]) and a cleveref name for the warning. No label was renamed and no number changed; only the printed names of those references are corrected.

Where the merge chose. The addition is printed after the ledger of non-claims and before the appendix, so that no existing section, theorem, equation or question number changes; its sections keep the source’s numbering shifted by fourteen. The outer group is renamed Γ rather than keeping H , because H is this report’s common core and the source uses it in that sense too. The explicit copies are not reprinted.

Checks in the merge. The proofs of theorems 17.2, 19.2, 20.2, 21.1, 22.2, 23.3, 24.2 and 24.4, lemmas 18.2, 18.3, 20.1, 23.2 and 24.1, corollaries 18.4 and 20.4, and proposition 20.5 were read line by line when the addition was merged; no error was found. The following were rechecked: $g = \omega^{7/2}$ and $b = \omega^{15/4}$ lie in G_1 (inner exponents in (3, 4)), $\lambda(g) = 1$, $\lambda(b) = 0$, $\Delta\tau = \tau$ and $\Delta\omega^b = 0$ from (22.5); $\beta_{0,n} = n$, so $g_{0,n} = \omega^{\psi(n)}$ and $Y_0 = \omega^{\omega^{15/4}}S$ with S as in (8.3); the consecutive exponents of (22.8) differ by $g + g_{\alpha,n} - g_{\alpha,n+1} > 0$; and the source’s numbering (its Theorem 8.2 is the omnific theorem, its Theorem 10.4 the surcomplex one) agrees with its delivered PDF. This reading is not an independent proof review. The source’s statements about this report are those of its pin; the parts it used are unchanged since.

A.8 Files of source 02, and their rerun

[merge] The seven shipped files of source 02 are these.

Shipped path	Delivered name	Role
02-composita-SOURCE_AUDIT.md	SOURCE_AUDIT.md	pin, source scope, priority limits
02-composita-PROOF_AUDIT.md	PROOF_AUDIT.md	hypotheses, critical steps, non-claims
code/02-composita-verify.py	verify.py	exact finite checks (SymPy)
code/02-composita-build.sh	build.sh	runs the checks, then builds the source's PDF
data/02-composita-verification.json	verification.json	recorded run: 2,530 checks, 21 categories
data/02-composita-requirements.txt	requirements.txt	sympy==1.14.0
data/02-composita-BUILD_AUDIT.json	BUILD_AUDIT.json	build and PDF inspection record of the delivered PDF

All seven are byte-identical to the delivery. The manuscript, its README and its PDF are not shipped; the build audit therefore describes a PDF (29 pages, 499,044 bytes, pdfTeX 1.40.26) that is not in the collection.

The program writes a file only when given `-output`; otherwise it prints its JSON report. The delivered `build.sh` changes into its own directory and runs `python3 verify.py -output verification.json` before building `article.tex`. Under the shipped names it finds neither file in `code/`, and an adapted copy would write its record next to the script, not over the shipped data file. Rerun the checks on a copy, or with an explicit output path outside the report:

```
python code/02-composita-verify.py --output <scratch>/verification.json
```

In the merge the program was run on a copy with Python 3.14.4 and SymPy 1.14.0. It passed all 2,530 checks with the same 21 category counts as the recorded run; the only difference in the JSON record is the Python version (3.13.5 in the recorded run).

A.9 Source 03 (batch 35)

[merge] Source 03 is manuscript 01 of batch 35 of the collection: the archive `Surreal_Hahn_Joins_Research` (inner directory `maximal_transcendence_hahn_joins`), delivered in commit 267b910 and placed in commit dec8d56 as an addition to this report with local number 03. Its main file `article.tex` (a standalone article with an embedded bibliography) is titled *Maximal Transcendence in Hahn Joins: Mixed-support independence and prime omnific integers*, dated 23 September 2026, with the author line “AI-assisted research manuscript”; its PDF has 27 pages (cover, contents and 23 numbered pages). It is pinned to commit f388f950f44664f21f50f04b8ac74dab56c8ff2f, at which this report’s `article.tex` was blob 7d5cfe644e7c902f2b594179c337a6c633350fc3: the same snapshot as source 02’s pin, before source 02 was merged in b9f964c. The placement chose an addition rather than a new report because the manuscript takes Research question 3 as its target and answers the transcendence-defect part of Research question 11 for explicit examples. Its manuscript, README and PDF are not shipped; its program, build script and recorded run are shipped with the prefix `03-hahn-joins-` (Appendix A.10).

What it contributed. The mixed-support independence criterion (Theorem D) with its lemmas, generator-sensitive form and stable variants (sections 30 and 31); the incidence and branch families (section 32); exact degree 2^κ for interleaved real closed and algebraically closed factors with omnific witnesses and the arithmetic-join gap (Theorem E and corollaries 33.2 and 33.3); the Archimedean family P_η , independent and prime in $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ (Theorem F, propositions 34.2 and 35.3, lemma 35.2, and corollary 35.4); the class example (Theorem G and lemma 36.1); strong rigidity

and the counts of relative derivations and automorphisms (section 37); ten research questions; a dependency audit, a finite verification model and a formalization roadmap.

Sources and provenance, as the source states them (its Appendix D). The repository comparison is pinned to the commit above and was made on 23 September 2026. The primary target was this report’s `article.tex`, its question headed *An exact description of ordinary composita* (Research question 3) being the precise starting point. The collection catalogue `README.md` at the pin was consulted; it distinguishes drafts, review and formalization status, and the source does not infer correctness from inclusion in it or use its unreviewed embedding theorems as black boxes. For the published prime input the source checked L’Innocente–Mantova’s arXiv version 5 of 22 January 2024, especially its Section 1.3 and its references to Pitteloud and Biljakovic–Kochetov–Kuhlmann, which are distinct from the proposed new arguments; a short publisher record for Pitteloud was consulted to verify bibliographic data. A user-supplied Wikipedia page served as orientation only, not as a proof source. The source gives the build commands (three `pdflatex` passes, then `python verify.py`) and says that finite checks, typesetting checks, mathematical proofs and external review are separate forms of evidence, of which only the first three are provided.

Editorial changes.

- (1) Every label of the source carries the prefix `isc:hj:`, with the source’s name after it, except that `sec:arithmetic` is kept as `isc:hj:sec:arithmetic` on the same subsection, the appendix labels `app:audit`, `app:verification` and `app:formal` become `isc:hj:sub:audit`, `isc:hj:sub:verification` and `isc:hj:sub:formal`, `app:sources` becomes this appendix `isc:hj:app:source`, and `main:mixed`, `main:cardinal`, `main:prime`, `main:class` become `isc:hj:main:mixed` etc. The merge also labels the source’s unlabelled definitions, questions, conclusion and new subsections, and gives every lemma, proposition, corollary, definition and remark label a `cleveref` type hint.
- (2) Symbols are renamed as in section 28.5; no normalization changed. The source’s Theorems A–D print as Theorems D–G, and its Questions 1–10 as Research questions 23–32.
- (3) The source’s citations of this report are replaced by cross-references to the same places; its hard-coded section numbers in the dependency ledger and the audit are replaced by cross-references. Its bibliography entries [14, 15, 16, 17, 18] are added to this report’s; its entries for Gonshor and L’Innocente–Mantova are this report’s [1, 3]; its entries for this report and for the catalogue at its pin are replaced by cross-references and by `README.md`. Only the wording of theorem 35.1 was compared with [3] in the merge; the new entries were not checked.
- (4) Text marked [merge] was added in the merge: in section 28 (the first paragraph and sections 28.4 to 28.7), after the fraction-field sentence of section 29, after proposition 30.2, lemma 30.3, and definition 31.1, after corollary 31.4, after (32.1), after corollary 33.3, after lemma 34.1, after theorem 35.1 and lemma 35.2, after (36.1), after the proof of Theorem G, remarks 36.2 and 37.6, in section 38 (after the table, and section 38.4), at the head of section 39 and after Research questions 23, 24, 26, 27, 29, 30 and 32, in section 41, and in this appendix. In the text of sources 01 and 02, batch-35 [merge] notes were added to the title page, to section 1.6 (status of the research questions), after the proof of Theorem C, after Research questions 3, 11, 14, 21 and 22, in item 1 of section 15.4, at the start of section 14, in items W1 and M3, and in Appendices A.1 and A.6. The [merge] tag marks text of both merges; the notes naming source 03 are those of batch 35. No statement of sources 01 or 02 was changed.

Where the merge chose. The addition is printed after source 02’s sections and before the appendix, so that no existing section, theorem, equation or question number changes; its sections keep the

source’s numbering shifted by twenty-seven, its lettered theorems continue the lettered counter, and its questions continue the question counter. The compositum is renamed M and the factors are indexed 0, 1, so that the second route of [remark 36.2](#) reads with $\Gamma_0 = G_0$, $\Gamma_1 = G_1$ and K_0, K_1 as in [section 8](#). The source’s $\mathcal{K}_k(G)$ definitions of [sections 28](#) and [29](#) are printed, not replaced by references, because they state the source’s exactness claims; they agree with [sections 1](#) and [9](#).

Checks in the merge. The proofs of [lemmas 30.1, 30.3 to 30.5, 32.1, 32.2, 33.1, 34.1, 35.2, 36.1 and 37.3, propositions 30.2, 31.2, 34.2, 35.3 and 37.2, Theorems D, E and G, corollaries 31.3, 31.4, 33.3 and 35.4, and theorems 37.4 and 37.5](#) were read line by line when the addition was merged; no error was found. The following were rechecked: the mixed exponents are pairwise distinct under hypothesis (i); the cosets $b_i + \Gamma_0$ are distinct under (ii); $r_\alpha - r_\beta$ has leading exponent $-\iota_0(\alpha)$ for $\alpha < \beta$, and $2 \cdot_{\text{ord}} \alpha$ is strictly increasing in α ; the codes of words of length m fill $[2^m - 1, 2^{m+1} - 2]$, with $\mathfrak{B}_{000\dots} = \{1, 3, 7, 15, \dots\}$ and $\mathfrak{B}_{111\dots} = \{2, 6, 14, 30, \dots\}$; $\rho_0 = 2^{-3/2} + 3^{-3/2} \approx 0.5460$; the cross product and the specialization $30y_1 - 24y_2 - 6y_3$ of [section 41.4](#); the count $\sum_{r=1}^4 \binom{16}{r} 2^r = 34,112$ of the recorded Boolean patterns; and the source’s numbering (Lemma 3.1 to Theorem 10.5, Question 1) against a build of its text, which agrees with its delivered PDF. The interleaving used in [section 28.4](#) was checked: $e_0 = 1 > f_0 = \omega^{-1} > e_1 = \omega^{-2}$, so neither of A_κ, B_κ is infinitesimal relative to the other. This reading is not an independent proof review. The source’s statements about this report are those of its pin; the sections it used ([sections 8](#) and [12](#)) are unchanged since, and the later source 02 is taken into account by the [merge] notes.

A.10 Files of source 03, and their rerun

[merge] The three shipped files of source 03 are these.

Shipped path	Delivered name	Role
<code>code/03-hahn-joins-verify.py</code>	<code>verify.py</code>	exact finite checks (SymPy); prints its report, writes no file
<code>code/03-hahn-joins-build.sh</code>	<code>build.sh</code>	three <code>pdflatex</code> passes, then <code>python3 verify.py > verification.txt</code>
<code>data/03-hahn-joins-verification.txt</code>	<code>verification.txt</code>	recorded run: five PASS lines, Python 3.13.5, SymPy 1.14.0

All three are byte-identical to the delivery. The manuscript, its README and its PDF are not shipped. No requirements file was delivered; the program needs Python 3 and SymPy (the recorded run used SymPy 1.14.0).

The program prints to standard output and writes no file. The delivered `build.sh` changes into its own directory, runs `pdflatex` three times on `article.tex`, overwriting `article.pdf` there, and then runs `python3 verify.py > verification.txt`, rewriting that record. Under the shipped names it stops at once, as a test on a copy confirmed: there is no `article.tex` in `code/`, the first `pdflatex` run fails (leaving a `texput.log`), and the script runs with `set -e`, so the checks are never reached. An adapted copy would write `verification.txt` next to the script, not over the shipped data file. Rerun the checks on a copy, or redirect the output outside the report:

```
python code/03-hahn-joins-verify.py > <scratch>/verification.txt
```

In the merge the program was run on a copy with Python 3.14.4 and SymPy 1.14.0. All five checks passed; apart from line endings, the output differs from the recorded run only in its second line, the Python version (3.13.5 recorded).

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